Cash That Never Sleeps

A small manual of commercial deposit liquidity at regional banks, worked with frontier AI models

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ABSTRACT

A regional bank holds commercial cash it cannot insure, and that cash runs on two clocks. On the run clock, a balance above the insured limit leaves first, in hours, when the bank is doubted. On the settlement clock, a tokenized deposit can move to another bank at any hour, while the bank's assets, its repo market and the large-value settlement rail keep business hours. This manual asks whether a regional bank can keep such cash, and at what price, by pricing nine routes on one list in four terms, against five screens and a rule written before any number was read. Read strictly, the champion for the run clock is a pooled Treasury-backed vehicle sized to a 99 percent standard at 37.94 basis points a year; with each open legal question resolved in the route's favor, a member-owned mutual with a Treasury-backed tail wins at 21.03, and the deciding question is whether a national bank may contribute entry capital to a state-licensed mutual. For the settlement clock none of the nine priced routes qualifies strictly, with every open legal question counted as a failure, though an unpriced route on the instant-payment rails might (Section 6.14). With each open legal question resolved in the route's favor, a collateralized weekend netting facility wins at 10 to 34 basis points for a fifth of the balance, provided its Sunday settlement cash is Reserve Bank credit against loans already pledged there, with a prefunded joint account on the model of the live instant-payment network as challenger at 25.35. At the assumed weekend peak net debit the facility cannot keep the balance nine tenths lendable and hold a member's cap to a rare event at once: a cap that binds in 1 percent of a member's weekends needs backing of a third of the balance, and at the assumed dispersion the two hold together only if the median weekend peak net debit is at most 1.56 percent of the balance, against the 5 percent assumed, a number no bank reports. Holding today's bind rate, under the announced Sunday-to-Friday Fedwire schedule the facility misses the lendable screen by 0.004 if the peak net debit scales with the hours (0.044 under square-root scaling); under seven-day operation it passes only if weekend Reserve Bank credit carries a net debtor's debits to Monday's securities opening, and fails if it holds reserves against the outflow instead. The clocks apply by kind of deposit: a plain uninsured balance runs on the run clock alone, an insured tokenized balance on the settlement clock alone, and an uninsured tokenized balance on both, at about twice the mutual's price for the run clock and with a lendable share 0.20 below the mutual's. The work was an experiment with frontier AI models, and Part 5 records what they built, what they got wrong, what an outside model family found that same-family checks did not, and what only a practitioner caught.

Keywords: uninsured deposits, bank runs, tokenized deposits, settlement hours, regional banks, deposit insurance, mutual insurance, AI-assisted research.

Part 1. The problem

Chapter 1. Two clocks

A corporation that keeps $100 million in an operating account at a regional bank holds, above the insured limit, an unsecured claim on one bank, repayable on demand, with no coupon that explicitly compensates it for the bank's failure risk. The bank holds, in the same balance, the funding it lends against and the funding that leaves first when the bank is doubted. That is the whole of the first problem, and nothing about it is new. What is newer is that the same balance, once recorded on a shared ledger, can leave at an hour when the bank has nothing to settle with, which is the second problem.

1.1 The run clock

The run clock is measured in hours. The FDIC's staff study of the spring 2023 failures gives the shape of it. The top 0.5 percent of depositors by tax identification number, with balances above roughly $49 million at Silicon Valley Bank, $16 million at Signature and $6 million at First Republic, held about 38 to 39 percent of Silicon Valley Bank's deposits, 62 percent of Signature's and 50 percent of First Republic's. Of that cohort, 74, 65 and 74 percent had withdrawn at least three quarters of their 6 March balance by 17 March (Chu et al. 2026). On Thursday 9 March one bank received $54 billion of outbound wire requests, of which $41 billion completed and $12.1 billion were queued for Friday and then cancelled when the bank was closed; its net deposit loss that day was about $30.2 billion, a fifth of its deposits, and it lost half of its deposits in three business days. Wires carried 87 percent of that bank's net loss, and among depositors who ran with more than $1.5 million of business deposits, accounts drawn down by more than 90 percent held 85 percent of those deposits, which is the behavior of a depositor closing an account rather than trimming a balance (Chu et al. 2026).

Runs pass through to failure mainly at banks with weak fundamentals. Correia, Luck and Verner (2026), working from 3,984 US bank runs between 1863 and 1934, find a bank in the lowest decile of fundamentals failing after a run with a probability of 59 percent and the strongest decile seldom failing, so a run is far from a sufficient condition for the loss a protection contract covers, and every failure-contingent route in this manual triggers on the receiver's appointment and not on the run. Full coverage announced after a failure did not retain depositors who had already decided to leave, either. The Silicon Valley bridge bank lost 14.4 percent of deposits on its first day, after the 12 March systemic risk exception had covered every dollar in it, and 40 percent of the 6 March balance over its first week (Chu et al. 2026). Protection arranged in advance, which the depositor holds before the stress starts, is a different object, and whether it would change what a depositor does in a run is the untested behavioral question that runs through this manual to the pilot in Chapter 10.

1.2 What the record shows about where the money went

The record of where those balances went is narrower than either a banker or a depositor would like. Read at the level of totals the episode healed, since the small domestically chartered banks in the Federal Reserve's H.8 release regained their 1 March 2023 deposit level on 10 April 2024 (Board of Governors 2026a); read by composition, the uninsured share fell. The uninsured share of domestic deposits at the $10 billion to $250 billion banks, the group whose depositors ran, fell from 41.7 percent at the end of 2022 to 33.4 percent by mid-2023 and stood at 36.5 percent in mid-2026, while the system's share went from 42.3 to 42.9 percent (FDIC 2026b). About 3.3 points of the group's drop is arithmetic, the three failed banks leaving the group, against about 4.0 points of the own decline of the banks still filing in mid-2023, a wider set than Table 1's. The survivors on their own are in Table 1. They lost about three and a half points of uninsured share by mid-2023, have regained all but about one of them, and hold more uninsured dollars than before the run.

129 surviving banks	2022 Q4	2023 Q2	2024 Q4	2026 Q2
Uninsured share of domestic deposits	38.61%	34.97%	35.47%	37.47%
Uninsured deposits, $ billion	1,851.0	1,723.5	1,837.6	2,177.4

Table 1. Fixed panel of the $10 billion to $250 billion group's 2022 Q4 members that survived: 149 banks, less the three that failed in 2023, less 17 that merged away or stopped filing by 2026 Q2, each bank's own figures at each date whether or not it stayed in the size band. Dropping the 35 banks whose domestic deposits grew more than 30 percent, a screen for acquirers, leaves 94 banks at 39.08 percent in 2022 Q4 and 37.90 in 2026 Q2. Source: author's computation from FDIC bank-level data (FDIC 2026c); the panel script and its results are in the repository (Appendix A).

The large uninsured balances that ran were mostly wholesale money, and the data carry no depositor type, so they cannot say whether the uninsured money the surviving regionals hold today belongs to the depositors who left, or which share of what left was passive corporate cash. The documentation comes in pieces. The run banks bridged with Federal Home Loan Bank advances and discount-window borrowing and replaced the uninsured deposits with insured deposits at higher rates through the third quarter of 2023, some in reciprocal form. They sent 30.5 percent of their Friday 10 March payments to the three custody banks against a typical 14.2, which the authors read as, to a considerable degree, a flight out of the banking system into money funds and securities held through custodians (Cipriani, Eisenbach and Kovner 2024). The extra outflow was concentrated at the super-regionals of $50 billion to $250 billion of domestic assets (Luck, Plosser and Younger 2023). Home Loan Bank advances system-wide, $1,045 billion on 31 March 2023, were back at $810 billion by the end of 2023 (Federal Home Loan Banks Office of Finance 2023, 2026). The reciprocal networks grew from about $157 billion at the end of 2022 to $462.8 billion on 31 March 2026 (Ryfe and Saretto 2023; FDIC 2026a). Regional banks have since relied more on reciprocal and, to a lesser degree, brokered deposits, which cost more than core insured deposits and may not be as stable under stress (Board of Governors 2026b). Most of the growth in money fund holdings since 2022 was household cash, with about one dollar in eight belonging to nonfinancial companies (Board of

Governors 2023a, 2023b, 2026b, 2026c; Investment Company Institute 2026). The exit the run banks' depositors used is open for the next run, and the regionals now carry a funding mix that leans on costlier insured money.

1.3 The settlement clock

The second clock is the newer one. A tokenized deposit is a bank deposit recorded on a shared ledger, a liability of the issuing bank that the bank lends on, and above the insured limit it carries every risk the first problem carries (Garratt and Shin 2023); what the ledger adds is the hour. A holder, or software acting for the holder, can instruct a transfer at 2 a.m. on a Saturday and see it recorded in seconds. If the transfer is to another holder at the same bank nothing further happens, because the bank has moved a balance from one account to another on its own books. If the transfer is to another bank, the first bank now owes the second, and the question is what the two banks settle in.

The Federal Reserve is partly always on. The FedNow Service settles around the clock in master accounts, every day of the year, with more than 1,800 institutions live by July 2026 and a network transaction limit of $10 million, and it offers a liquidity management transfer of up to $500 million under the operating procedures of October 2026, final when recorded. The published schedule processes that transfer on the same 24-hour cycle as instant payments, while the procedures refer to a window for it without stating its hours (Federal Reserve Financial Services 2025, 2026b, 2026c, 2026d, 2026e). What confines that transfer is a purpose rule, whatever the hours of that window, since the operating circular allows a participant to send one only to support liquidity needs related to payment activity in the FedNow Service or in another instant payment system backed by a joint account at a Reserve Bank, so a large interbank claim arising from a weekend transfer of tokenized deposits is outside it as written. A transfer of tokenized deposits to a holder at another bank is itself a customer payment, and an arrangement could route transfers of up to $10 million each through FedNow as credit transfers at any hour; the gap is for larger transfers and for the netted claims between banks. The settlement asset for large-value interbank claims, a balance at a Federal Reserve Bank moved through the Fedwire Funds Service, is available from 9:00 p.m. Eastern time on the preceding calendar day to 7:00 p.m. on five days and observes every Saturday and Sunday as a holiday, and the National Settlement Service, through which clearing arrangements settle net positions, keeps similar hours (Federal Reserve Financial Services 2026a). From Friday at 7:00 p.m. to Sunday at 9:00 p.m., about 50 hours, there is no large-value central bank settlement in which an interbank claim above the FedNow limit, or a netted position between banks, can settle. The Board announced on 9 October 2025 that Fedwire Funds and the National Settlement Service will run Sunday through Friday, weekday holidays included, in 2028 or 2029 and on a voluntary basis for participants, which cuts the weekend window to 26 hours, Friday 7:00 p.m. to Saturday 9:00 p.m., and Chapter 6 prices what that schedule and a seven-day one would do (Board of Governors 2025a, 2025b). The expansion covers Fedwire Funds and the National Settlement Service only; the Board's

notice says it is not considering the Fedwire Securities Service, through which Treasury collateral moves, so a bank that wants cash against its bills on a Sunday evening will still have to have pledged them before the weekend.

What keeps business hours is the funding side of the bank. Loans pay on business days and securities settle on business days through the Fedwire Securities Service, which keeps hours of its own; the discount window, the Home Loan Banks, the tri-party and bilateral repo market and the money funds keep business hours; and an asset sale needs a market that is open. So a bank facing a weekend outflow of tokenized deposits has nothing to sell and no one to borrow from until Monday, and a bank receiving the inflow holds a claim on the first bank that it cannot convert either. Throughout this manual the phrase “outside Fedwire hours” means that window, and never that the Federal Reserve is closed, which it is not.

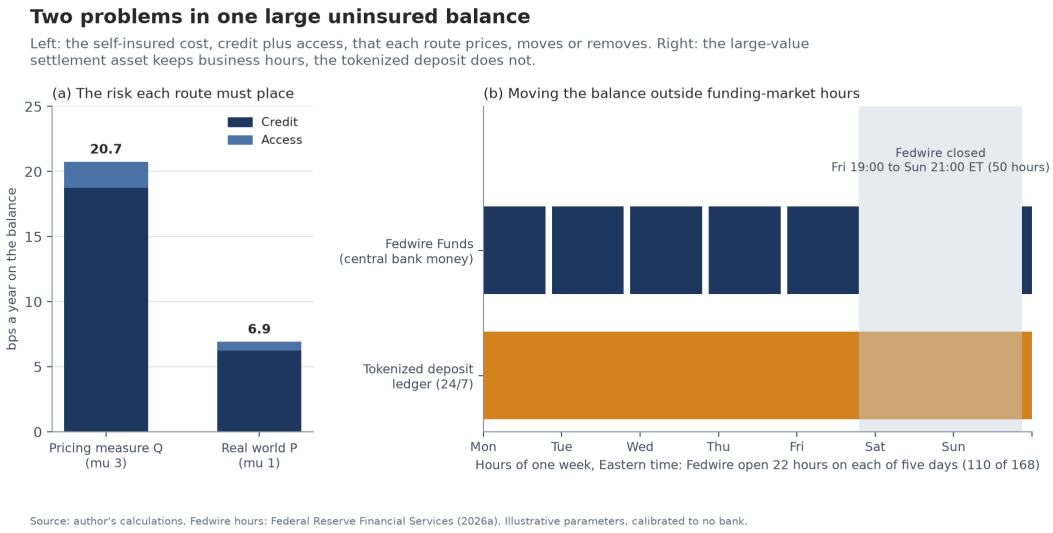


Figure 1. Two problems in one large uninsured balance. (a), left: the self-insured cost, credit plus access, that each route in Chapter 4 prices, moves or removes. (b), right: the large-value settlement asset keeps business hours and the tokenized deposit does not; the Fedwire Funds Service is closed from Friday 19:00 to Sunday 21:00 Eastern, about 50 hours. Source: author’s calculations; Fedwire hours from Federal Reserve Financial Services (2026a); illustrative parameters, calibrated to no bank.

1.4 The governing question, and how to read the manual

These are two problems with one toolkit. The tools that keep a balance stable through a failure and the tools that let it move outside Fedwire hours draw on the same stock of Treasury bills, the same membership of banks and the same depositor’s willingness to pay, so the two have to be priced together. Not every deposit carries both risks, though. A plain uninsured balance is on the run clock alone, an insured tokenized balance on the settlement clock alone, and only an uninsured tokenized balance on both (Section 2.6). The question this manual works is whether a regional bank can keep commercial cash it cannot insure and cannot always settle, and at what price.

Every number in this manual was computed by frontier AI models under my direction, in seven stages over about two and a half weeks, and every chapter through Chapter 9 ends with a short box, "What the models did here", recording what the models built for that chapter and what they got wrong; Chapter 9 gives the full account, and Chapter 10 the questions I most want human experts to challenge.

The numbers are illustrative and calibrated to no bank, mutual, vehicle, facility or investor, and Appendix B lists every parameter with its label. The legal readings are a practitioner's and not counsel's. Every model number traces to one of the engines in the repository, which the repository README maps chapter by chapter, so that a reader who prefers other conventions can rerun them in a few minutes.

> **What the models did here.** The original outside draft, in stage 1, used the $54 billion of wire requests to argue that credit protection cannot stop runs, misreading a pre-failure Thursday as a post-failure Monday, and a Claude red-team review restated it against the FDIC staff study. Claude then barred several correct figures from that study as unverifiable because its checks used search extracts while the study sat in my own library; the outside models kept asserting them and a primary-source check restored them. In stage 2 the first drafts said the uninsured corporate money left the regionals and did not come back. A fact check of 501 items found every table cell right; a later check aimed at the premise found it wrong, since the data carry no depositor type and the surviving banks hold more uninsured dollars than before the run; Table 1 was then replicated from the FDIC's bank-level API, and a scoped check of the rewritten passages cut "largely" back to the authors' "to a considerable degree".

Chapter 2. What a dollar of commercial cash carries, and the rule

2.1 Two risks in one claim

A deposit above the insured limit carries two distinct risks that the deposit contract bundles into one claim: the eventual credit loss, the shortfall between the balance and what the receivership ultimately pays, and the loss of access, the period between the bank's closure and the date the cash is available again, during which the depositor cannot run payroll or settle with vendors from that account. Depositor preference sets the first. Under 12 U.S.C. 1821(d)(11) the receiver pays its own administrative expenses first, then domestic deposit liabilities (insured and uninsured pari passu, with the FDIC as subrogee for the insured depositors it has paid), then other general and senior liabilities, then subordinated obligations, then shareholders. Depositor preference does not, however, put the deposit above secured creditors, and 12 U.S.C. 1430(e) gives a Federal Home Loan Bank's security interest priority over any other party including the receiver. A bank that funds a run with Home Loan Bank advances, as all 22 run banks of March 2023 did to some degree (Cipriani,

Eisenbach and Kovner 2024), therefore encumbers assets that would otherwise support deposit recoveries, so the loss on a deposit claim depends on how the run was funded as well as on the loan book, and it is likely to be worst in the years when several banks fail together.

Resolution practice sets the second. The FDIC has paid advance dividends to uninsured depositors since 1984 (FDIC 1984), sized on the estimated recovery value of the failed bank's assets, with a receivership certificate for the remainder; the advance was 50 percent at IndyMac in 2008 (FDIC 2008), and one was announced for Silicon Valley Bank before the systemic risk exception overtook it (FDIC 2023b). The stated practice is case by case and usually within 30 days of closing, and the rest of the claim arrives as the receivership liquidates, over years, in certificates that are neither cash nor easily financed (FDIC n.d.). The public answers are not routes a bank can choose. The systemic risk exception under 12 U.S.C. 1823(c)(4)(G), which since Dodd-Frank can be invoked only for an institution already in receivership and needs two thirds of two boards and the Treasury Secretary, was used in March 2023 for two banks with uninsured shares of 93.8 percent on the Call Report at the end of 2022 and a reported 90 percent (FDIC 2026c, from the panel script of Appendix A; FDIC 2023c). Statutory coverage, whose unlimited form the FDIC's 2023 options paper priced at a 70 to 80 percent rise in insured deposits and a cut of more than 40 percent in the fund's reserve ratio, with targeted coverage of business payment accounts needing funding the report calls hard to estimate (FDIC 2023a), has not been enacted.

2.2 The four terms

Pricing protection on the claim means pricing those two risks plus the cost of being ready to pay. The annual clearing cost, in basis points of the covered balance, is the sum of four terms, credit, access, standing capacity and frictions:

$$c^{*} \;=\; \lambda_{Q}\,\mathrm{LGD}_{dep} \;+\; \lambda_{Q}\,\max(0,\;1-a-\mathrm{LGD}_{dep})\,(r+s_{b})\,D \;+\; k \;+\; f, \qquad \lambda_{Q}=\mu\,\lambda_{P}$$

Here λ_{P} is the physical one-year default intensity of the bank, μ a risk-premium multiplier so that λ_{Q} is the risk-neutral intensity a market price would embed, LGD_{dep} the loss given default on a deposit claim under depositor preference, a the advance-dividend fraction, D the years from failure to final dividends, r the risk-free rate, s_{b} the provider's funding spread, k the standing-capacity cost and f the frictions. The quantity $1-a-\mathrm{LGD}_{dep}$ is the slice of the claim that will eventually be recovered but has to be bridged in the meantime, and the max operator keeps it from going negative if an advance dividend were ever to exceed ultimate recovery. The first two terms are the risk itself, what the depositor bears unprotected; the last two are the cost of moving it, and they are paid only if someone else takes it. The credit term can be priced under the pricing measure Q, with an intensity that already contains a risk premium, or under the real-world measure P, with the historical intensity plus an explicit charge for the capital a seller holds against the exposure. Stacking them, a risk-neutral premium plus a capital charge plus a separate wrong-way charge, is how a modest number becomes a large one without anyone having added information, and the rule throughout is one measure, never stacked.

The base case takes λ_P at 0.25 percent, μ at 3, deposit LGD at 25 percent, an advance dividend of 50 percent, two years to final dividends, a risk-free rate of 4.3 percent (the 3-month Treasury bill as of 15 October 2026, rounded; Federal Reserve Bank of St. Louis 2026a) and a funding spread of 100 basis points, and Appendix B lists every parameter with its label (Table 18). At those inputs the risk the balance carries, with nothing for moving it, is about 21 basis points a year under the pricing measure and about 7 under the real-world one (Table 2).

Measure	Credit	Access	Self-insured (credit + access)
Q (mu 3), pricing	18.75	1.99	20.74
P (mu 1), real world	6.25	0.66	6.91

Table 2. The self-insured cost of the uninsured balance, basis points a year on a $100 million balance above the insured limit: the credit and access terms of the four-term formula with no transfer cost. Source: the routes engine's results, Section 3.

The access term is small. The problem that feels most acute, being unable to reach the cash, costs about 2 basis points a year in expectation, and only about 9 with no advance dividend and three years to final payment, because it is a financing cost on a partial balance for a limited period, multiplied by a small probability. The credit term carries the weight, and it scales with the default intensity, which is why that one input dominates every sensitivity in this manual.

That cost is the object every route must put somewhere. A route can price it, as a funded vehicle or a mutual does when private capital is paid to hold it and as reciprocal placement does when the FDIC holds it against the assessments the banks pay. It can move it to a party who does not charge for it, as a Home Loan Bank letter of credit does through the super-lien and a pledge does through the bank's own encumbered collateral. Or it can remove the balance, as letting it go does, and pay the replacement spread instead. Where it lands is my reading throughout, not a finding of law.

2.3 Franchise value and lendable value

A bank values the balance twice over. The franchise value is the relationship, the margin, fees and cross-sell on the balance, which Chapter 3 values over five years and finds worth 1.0 to 30.5 basis points a year across its sweep. The lendable value is the dollar as funding. A deposit that stays on the balance sheet and funds a loan is worth, to a lender, whatever the next dollar of funding would cost over the deposit's all-in cost, and a deposit that has been encumbered by the route that protects it, or that has funded a bill the bank cannot plan to lend, is worth less than its face as funding however safe it has been made. The lendable share, the share of each deposit dollar that can still fund a loan after whatever the route encumbers on the bank's own balance sheet, is therefore the second axis of every comparison in Chapter 4, beside the price, and it is the axis a fee comparison alone leaves out. The lending that funding supports has to be stated more narrowly, as an availability claim at community and smaller regional banks, which lend on soft information that large banks at a distance do not collect (Berger, Miller, Petersen, Rajan and Stein 2005; Stein 2002; Petersen and Rajan 1994; Nguyen 2019; FDIC 2018, 2024; Gruenberg 2022; Federal Reserve Banks

2025). The link from the regionals' 2023 deposit losses to small-business credit is unmeasured (Caglio, Dlugosz and Rezende 2024; Cortés and Strahan 2017; Gilje, Loutskina and Strahan 2016), so the credit point is a qualitative reason for the designs and stays out of every price.

2.4 The five screens

Routes are scored before they are priced. The five screens in Table 3 are the conditions a route must meet for a private corporate depositor at a regional or community bank that wants to keep lending. Each verdict is pass, fail, open (a question the sources do not settle) or not applicable. S5 is computed by the engines for the routes with a failure payoff, passes by construction for a route the FDIC insures, and does not apply where there is no failure payoff to price.

Screen	Definition
S1	Open to a private corporate depositor at a national bank
S2	Lendable share at least 0.9
S3	Cover continues, and can grow, below well capitalized or at CAMELS 4 or 5
S4	No hidden public subsidy
S5	Pays in full with probability at least 99 percent in the year the depositor's own bank fails (Gaussian copula, correlation 0.30)

Table 3. The five screens. The lendable share is the share of each deposit dollar that can still fund a loan after whatever the route encumbers on the bank's own balance sheet; S5 is the reliability standard Chapter 3 derives, measured conditional on the depositor's own bank failing. Source: the routes engine's results, Section 8.

S1 is there because the Supreme Court held in Texas and Pacific Railway Co. v. Pottorff, 291 U.S. 245 (1934), that national banks lack the power to pledge assets to secure a private deposit, and a route that works only for public units or only under a state charter is not an answer for the depositor this manual is about. A route that makes the dollar safe by turning it into a bill has solved the depositor's problem and abolished the bank's, and S2 screens it out. S3 is there because cover that stops growing in the quarter the bank weakens is cover that stops in the very quarters the depositor most wants it. A route whose cost falls on the deposit insurance fund is subsidized, and S4 screens it out. S5 is there because a pooled design that pays par in an average year and falls short in the year the depositor's own bank fails has priced the wrong year, and Chapter 3 shows why the conditional standard is the one that belongs in the design.

2.5 The rule, stated before any result

Among routes passing every applicable screen, the champion is the route with the cheapest all-in price, taken at the base of its range where the price is a range, and the challenger is the next cheapest route, variants of one route competing inside the route first; an open verdict is not a pass. The applicable screens are S1 to S5 for the run-clock problem and for both problems together, and S1, S3 and S4 for the settlement-clock problem, whose objective is movement rather than lendable funding or failure cover; the result with S2 also applied to it is reported. Because several verdicts

turn on legal questions no source settles, the rule is run twice, once strictly and once with every open item resolved in the route's favor, so that the reader can see which open question decides the answer. The strict reading is the one the rule names; the settled reading is the one a reader who believes the open questions will resolve favorably would act on; and the gap between them is the price of the legal uncertainty.

Beside every champion and challenger the same design is re-priced under a Student-t(5) copula with its shortfall under that copula, because the Gaussian standard in S5 is a convention and Chapter 3 finds the two copulas disagree by a factor of about twenty on how often a given capacity falls short, and nothing in the record chooses between them. A reader who prefers other screens, another bar for the lendable share, a probability on each open question instead of a fail, or a weighted scorecard instead of a cheapest-price rule, can rerun the routes engine with those conventions; the rule here was chosen before any number was read, and that is the only claim made for it.

2.6 Which deposit runs on which clock

The two clocks are two different risks, and not every deposit runs on both. A plain insured balance runs on neither, since the FDIC carries the credit loss and the loss of access and the balance moves in business hours like every other plain deposit. Above the limit the plain balance runs on the run clock alone, because it leaves first when the bank is doubted and whatever it sends outside Fedwire hours goes by FedNow, under that rail's limit, and settles as it goes. Tokenizing a balance puts it on the settlement clock whether it is insured or not, because the mismatch of Section 1.3 is between the hours the deposit can move and the hours the bank's assets and the rail keep, and insurance does not remove that mismatch, though it changes its form. An insured holder's transfer fits under FedNow's $10 million limit and could settle on its own at any hour, but only from reserves the bank cannot replenish before Monday; where transfers are netted between banks, the claim waits for Fedwire, which is the case the insured rows of Table 17 price. An uninsured tokenized balance, the deposit in which the manual's two problems meet, runs on both.

The encumbrances add on the absolute convention, under which the lendable share is the share of each deposit dollar that can still fund a loan after whatever the route encumbers (Section 2.3), so a deposit that needs a route for each clock gives up what each route encumbers, the mutual's capital contribution on one side and the always-on bills or prefunded reserves on the other. The lendable value of an uninsured tokenized balance, in the sense of Section 2.3, is therefore lower than a plain uninsured balance's, and its price higher, because the routes stack where the risks do. Section 7.3 prices each kind of deposit on the list and says what each should be charged, and Section 8.4 records the one case the stacking does not price, a run that starts on a weekend in an uninsured tokenized balance.

> **What the models did here.** The four-term formula came out of the first review of the stage-1 outside draft, which had compared a 175 and an 85 basis point price built by stacking a risk-neutral premium, a capital charge and a wrong-way charge on top of

one another, and mixing four capital regimes in one “35 percent effective risk weight”; the review’s rule, one measure and never stacked, became the formula. The Claude model that built the engines then flagged twelve specification issues in its first brief instead of silently fixing them, among them a capital hurdle charged on capital that itself earns the risk-free rate, and I decided each. The risk-free rate was wrong for most of a day: a 3.9 percent figure from a search snapshot carried a “verify before relying on it” flag from the first brief on and was caught only by a direct check against the Federal Reserve’s data, at 4.28 percent on 15 October 2026. In stage 4 the rule and the screens were chosen in one round before the routes engine ran, and the engine’s one addition to them was the verdict that decides Chapter 5, the mutual’s S1 question in its national-bank form, which stage 2 had asked only for a Texas state bank.

Part 2. The run clock

Chapter 3. The price of keeping a dollar

3.1 Who could sell protection, and what each would charge

The clearing cost is the annual charge at which a provider with no informational edge would be indifferent to writing protection that pays par on the covered balance at the FDIC's appointment as receiver, and it is the shadow cost the bank's uninsured funding carries whether or not anyone charges it. Three kinds of provider could write it, and they differ in the capacity term and the pricing measure. A funded investor vehicle holds Treasury bills, so it has no negative carry, and its capacity cost is the liquidity premium investors want for locking up funds, 10 basis points within a 5 to 15 range. A bank selling a standby letter of credit incurs a capacity cost through its liquidity coverage ratio; on my reading of 12 CFR 249.32(e) the ratio assigns a 30 percent outflow to a liquidity facility extended to a non-financial corporate (10 percent for a credit facility; a commitment to buy a deposit claim fits neither cleanly), which at a net carry on high-quality liquid assets of 40 basis points is 12 basis points, plus a capital cost. A regional mutual pool that prefunds a reserve of 10 percent of its coverage at an opportunity cost of 100 basis points carries 10 basis points of capacity, an illustrative choice within a 5 to 20 percent range. Frictions are 15 basis points for each. Table 4 gives the base case.

Provider	Credit	Access	Capital	Capacity k	Frictions f	c*	Range
Funded investor vehicle	19	2.0	0	10	15	46	36 to 56
Bank standby letter of credit seller	6	0.7	15	12	15	49	37 to 49
Regional mutual pool	19	2.0	0	10	15	46	41 to 56

Table 4. Base-case clearing cost by provider type, basis points a year on the covered balance. The bank row is priced under the real-world measure (expected loss plus capital); the other two under the pricing measure. The range reflects the capacity and frictions ranges. Rounded components may not add to the rounded total; the unrounded funded-vehicle price is 45.7. Source: the clearing-cost engine's results, Section 2.

The transfer cost, capacity plus frictions at 25 basis points, exceeds the risk itself, credit plus access at 21, so at a low default probability more than half of what a depositor would pay for protection is the cost of moving the risk rather than the risk. At base the bank seller's price is set by capital rather than by the market. Its capital cost of about 15 basis points (a 100 percent credit conversion factor, a 20 percent risk weight, a 13.5 percent common equity ratio and a 10 percent pre-tax hurdle net of the 4.3 percent the capital earns on its own) makes the real-world price of 49 bind over the pricing-

measure price of 48 below a default probability of about 0.28 percent. The seller's own credit needs the same care, since a bank or a pool of banks is more likely to fail when the reference bank fails. The depositor bears that wrong-way risk, so I treat it as a reduction in what the protection is worth to the depositor, equal to λ_Q times the probability that the seller fails given that the reference bank fails, times the deposit loss, times the loss on the claim against the seller. That haircut is 3.8 basis points at base for a conditional joint-failure probability of 40 percent and 2.8 to 4.7 across 30 to 50 percent, small in basis points because it scales with λ_Q, and it sits entirely in the tail years.

Among the sensitivities, the default probability is the driver that matters. Across λ_P of 0.10, 0.25, 0.50, 1.00 and 2.00 percent the funded vehicle's price moves from 33 to 46, 66, 108 and 191 basis points. The risk-premium multiplier is next, moving the price from 39 at μ of 2 to 60 at μ of 5, and deposit LGD third, from 39 at 15 percent to 56 at 40 percent, while the advance dividend and the time to final dividends each move the price by 4 basis points or less across their full ranges.

Each of the three provider types has a history, and the mutual's is the least reassuring. In Ohio in March 1985 Home State Savings Bank lost about $145 million through ESM Government Securities; the loss exhausted the privately run Ohio Deposit Guarantee Fund and the state declared a holiday for the fund's member thrifts (Federal Reserve Bank of Cleveland 1986; Rose 2024). Two months later a run on Old Court Savings and Loan spread through the privately insured Maryland Savings-Share Insurance Corporation system, the governor capped withdrawals at $1,000 per account per 30 days on 14 May (Hughes 1985), and the corporation was replaced by a state fund on 17 to 18 May (Maryland Comptroller 1990). Rhode Island's fund followed in 1991 (Pulkkinen and Rosengren 1993; Todd 1994). A pool that prefunds a reserve of 10 percent of its coverage can pay par on a failed member only if that member's covered balances are at most 10 percent of the total; beyond that it draws post-event assessments from members who are, in the state of the world where a member has failed, themselves under stress. The Massachusetts Depositors Insurance Fund is the working counterexample, with a limit: created in 1934 as a private, industry-sponsored fund, it has covered deposits above the FDIC limit since all its members joined the FDIC by 1986, after the Ohio and Maryland failures (FDIC 1997; Gallagher 2020), so what has lasted is private excess cover on top of FDIC insurance and not private primary cover. Those histories are why the pooled design below is funded by outside investors and capped per bank, and why the mutual of Chapter 5 sits on an FDIC-insured base and carries a tail it does not own.

3.2 The wedge, and the three clocks a failure runs on

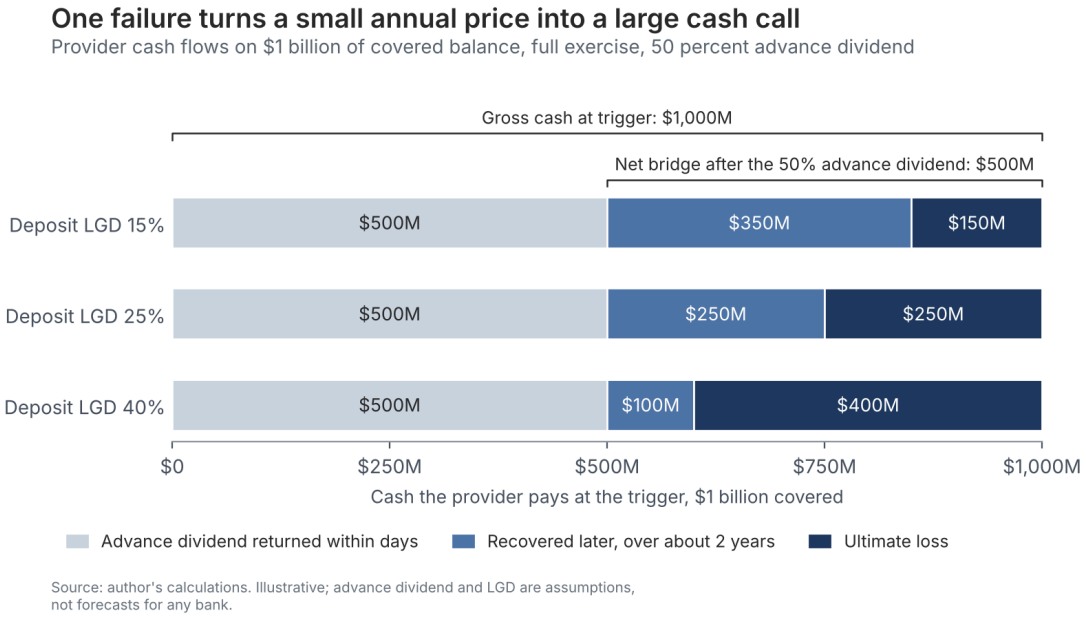


Figure 2. Provider cash flows on $1 billion of covered balance at full exercise with a 50 percent advance dividend. The provider pays $1,000 million at the trigger, receives $500 million back within days, recovers $100 million to $350 million over about two years and loses $150 million to $400 million depending on deposit LGD. The expected cost is small only because the probability is. Source: author's calculations from the clearing-cost engine; illustrative parameters, calibrated to no bank.

The wedge between what the protection costs each year and what it pays at the trigger decides whether anyone sells it. Against the cash flows in Figure 2 the annual premium at the base-case price is about $4.6 million, so one event at 25 percent LGD costs the provider roughly 60 years of premium, and the cash call at the trigger is the whole balance, which whoever sells the protection needs to be able to find on a Friday afternoon. The three cash flows arrive on three different clocks, par at the trigger, the advance dividend within days and the final recovery over years, and separating them is the idea the pooled design is built on.

3.3 Pooling: what a cluster costs

A vehicle writing on many banks changes the shape of the risk without changing its size. Take 50 reference banks, each capped at 2 percent of the covered notional, each at the base default probability of 0.25 percent and a 25 percent deposit LGD, so that one failure is a par payment of 2 percent of the pool and a credit loss of 0.5 percent. Failures are driven by a common factor at an illustrative asset correlation ρ (a one-factor Gaussian copula, in which a low value of the common factor is a bad year for every bank at once). Expected failures are 0.125 a year and the expected loss is 0.0625 percent of the pool at every ρ; what correlation does is move probability out of the one-failure years into the zero-failure years and the many-failure years. The probability of three or more failures in a year, the 2023 pattern, is 0.03 percent under independence, 0.19 percent at ρ of 0.10,

0.51 percent at 0.20 and 0.84 percent at 0.30. The cash a cluster calls for is smaller than the single-name arithmetic suggests, since three failures among 50 banks at 2 percent each need 6 percent of the pool at the trigger, half of it back within days as advance dividends and 1.5 percent lost at a 25 percent LGD, while a single-name vehicle holds Treasury collateral equal to the whole covered stock.

The reference design holds one source of cash against those flows, a funded buffer θ of 20 percent of covered notional invested in Treasury bills. The buffer pays par at the trigger, is the first loss, and carries the bridge between the advance dividend and the final recovery. With no facility there is nothing to draw, nothing to term out and no provider who can fail to perform, and since the term need after the advance dividend is never more than the cash paid at the trigger, the buffer that paid par can always carry the bridge. If the cash needed at the trigger exceeds the buffer, the vehicle cannot pay par; that is the protection failure, and I assume it is shared pro rata across the failed names, with the unpaid depositor keeping its deposit claim and bearing LGD on the unpaid share. A hybrid variant replaces half the buffer with a committed liquidity facility of 10 percent, drawn at the trigger only for the par payments above the buffer and repaid from the advance dividends within the week, and since the expected drawn cost is a few thousandths of a basis point the facility earns its place only when its commitment fee is below the lock-up premium investors charge on the buffer. At the base spreads, a 10 basis point premium and a 20 basis point fee, the all-funded buffer is cheaper, which is why it is the reference and the hybrid a variant. A bank providing the facility would carry, on my reading of the liquidity and capital rules, a cost floor of 55.4 to 117.0 basis points a year on the undrawn amount, so if a facility is ever written it belongs with a non-bank that holds the Treasuries anyway.

The timing standard is simultaneous. All of a year’s failures are assumed to land inside one advance-dividend window, so the cash needed at the trigger is the number of failures times 2 percent; spring 2023 had two failures over one weekend. LGD is allowed to depend on the state of the world, 15 percent in a normal year rising to 40 percent when the common factor is three standard deviations below its mean, because fire-sale asset values and heavier secured Home Loan Bank borrowing push losses up in exactly the years when failures bunch, and the advance dividend falls with it, at two thirds of estimated recovery. Under that mapping the average failed bank has an LGD of 23.8, 26.7 and 29.0 percent at correlations of 0.10, 0.20 and 0.30. Failure probabilities are computed on the real-world measure at the 0.25 percent default probability, while prices use the same copula run at three times that probability per bank, which with a fixed LGD reproduces the single-name credit and access terms exactly.

The reference design has the same credit and access terms as the single-name vehicle, 18.8 and 2.0 basis points, and a capacity term of 2.0, the lock-up premium of 10 basis points on a buffer of 20 percent of notional; with frictions of 15 its price is 37.7 basis points against 45.7 for the single-name vehicle. The saving over the single-name vehicle is the capacity term only, so pooling at this capacity saves 8 of the 10 basis points of lock-up premium. How low the price can go is a question about

frictions, since the reference design's price is 22.7 basis points plus frictions, and fifty names, rating-agency review and legal opinions in several states seem to me more likely to cost more than 15 basis points than less.

3.4 When it fails, measured in the year that matters

The price buys a less certain par. A fully funded single-name vehicle cannot run short of cash, by construction; the pooled vehicle fails when more banks fail in one week than its capacity covers. At 20 percent capacity and a correlation of 0.30 the real-world probability of being unable to pay par is 0.0097 percent a year, about one year in 10,000, under a Gaussian copula, and under a Student-t copula with five degrees of freedom, which has tail dependence the Gaussian lacks, the same capacity fails in 0.21 percent of years, about one in 500. The tail model moves the answer by a factor of about twenty.

Those probabilities are per year, averaged over the years in which nothing happens, and they are the wrong number for the depositor. A depositor collects only in the year its own bank fails, and that selects the bad years, in which the common factor is low and the other 49 banks are more likely to fail too. The standard that belongs in the design is therefore conditional, the probability that the vehicle falls short in a year in which the depositor's own bank has failed, together with the severity, the share of par left unpaid when it does. Figure 3 draws it.

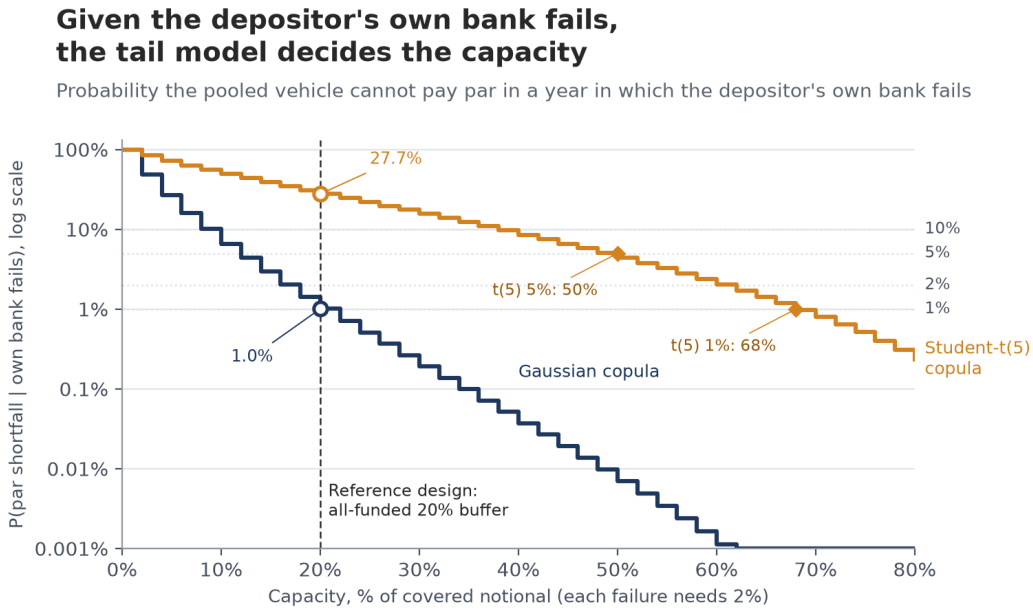


Figure 3. Probability that the pooled vehicle cannot pay par in a year in which the depositor’s own bank fails, against capacity, under a Gaussian copula and a Student-t copula with five degrees of freedom, both at an asset correlation of 0.30 and a 0.25 percent default probability. At the reference design’s 20 percent the Gaussian figure is 1.0 percent and the Student-t figure 27.7 percent; the diamonds mark the capacity the Student-t copula needs for 5 and 1 percent conditional standards, 50 and 68 percent. Source: author’s calculations from the clearing-cost engine; illustrative parameters, calibrated to no bank.

At the reference design a depositor whose own bank has failed is paid in full in 99.0 percent of cases under the Gaussian copula and 72.3 percent under the Student-t. When it is not, it is left holding about a quarter of its balance as a deposit claim under the first and about 40 percent under the second.

The way to read the price is then to fix a standard and ask what each copula needs to meet it (Table 5). Under the Gaussian copula a 1 percent conditional standard takes 22 percent of notional and the all-funded design prices at 37.9; under the Student-t it takes 68 percent and 42.5, against 45.7 for the single-name vehicle that never falls short. At the 1 percent standard the pool’s saving over the single-name vehicle shrinks from about 8 basis points under the Gaussian copula to about 3 under the t, so part of what pooling appears to save is a saving on tail coverage whose size depends on a copula nobody can calibrate from the handful of US bank-failure clusters on record. Note that the reference design’s 20 percent sits just above the Gaussian 1 percent standard, at 1.008 percent short, which is why Chapter 4 carries a variant of the vehicle sized to the standard at 37.94.

Copula	Conditional standard	Capacity needed (failures)	Achieved	All-funded price	Saving against single-name
Gaussian	5%	12% (6)	4.37%	36.9	8.8
Gaussian	1%	22% (11)	0.72%	37.9	7.8
Student-t(5)	5%	50% (25)	4.38%	40.7	5.0
Student-t(5)	1%	68% (34)	0.98%	42.5	3.2

Table 5. Iso-reliability: the smallest capacity at which the conditional shortfall probability meets a round standard, and the price at that capacity, by copula; correlation 0.30, default probability 0.25 percent, 50 banks at 2 percent each, prices on the base spreads (lock-up premium 10, frictions 15). The single-name vehicle is 45.7 and never falls short. Source: the clearing-cost engine's results, Section 39.

3.5 The settlement rule at the trigger

A 27.7 percent conditional shortfall under the Student-t copula is a defect of the settlement rule as much as of the tail model. The reference design exhausts its 20 percent buffer after ten failures because the vehicle pays 100 cents on the dollar at the trigger and so advances the share of par that the FDIC itself returns as an advance dividend, 50 percent at base assumptions. If the vehicle is built for the credit loss and the carry rather than to advance the FDIC's dividend out of first-loss capital, a rule under which the depositor remains the claimant at the cutoff, collects the advance dividend directly from the receiver, and at day seven assigns the receivership certificate to the vehicle against par less the dividend actually paid, stretches the same buffer across more failures. The 20 percent buffer then covers 20 simultaneous failures at a 50 percent dividend and 16 at 40 percent, the conditional shortfall falls from 1.008 to 0.137 percent under the Gaussian copula and from 27.7 to 10.5 percent under the Student-t at the same 37.7 basis point price, and the 1 percent Student-t standard needs a 39.0 percent buffer at 39.6 basis points instead of 68.0 percent at 42.5. The provider's expected cost, and so the price, is the same under either rule. What changes is the depositor's position: under gross par it is paid within the stated business days of the trigger; under the day-seven rule it holds the dividend when the FDIC pays it and the rest at day seven, so a depositor who needs the whole balance in the first week still has an access reason to run, and the size and timing of the dividend are decided by the FDIC, outside the contract. Chapter 5 uses this day-seven rule as the fallback behind a settlement rule drawn from Massachusetts practice.

3.6 Who pays, and the three-party condition

The clearing cost becomes useful when it is set against what the depositor earns for bearing the risk, and that depends on what the balance would earn elsewhere. Call the difference the depositor's advantage Δ, in basis points, measured as a yield spread for passive cash and as a gap in total account value for an operating balance. Passive excess cash, the reserves a company holds beyond what its payment cycle needs, can sit in a government money market fund or be spread across banks through a reciprocal network, and for that cash Δ is the spread the bank pays over the government fund, usually small for a large balance and sometimes negative. An operating balance

cannot sit in a fund, because payroll, ACH debits, lockbox receipts and the balance a credit line's covenants call for all have to clear through a bank account during the business day, and cash in a fund neither makes payments nor reduces a loan's cost; its outside option is an operating account at a large bank. That account typically pays little or no interest and earns an earnings credit instead, which offsets the bank's fees and is not cash interest, so for an operating balance Δ is the gap in total account value between the two accounts, and the reference price of 37.7 basis points is an implied hurdle on that gap, not a yield the depositor should expect to earn.

In the passive-cash regime, with Δ between 0 and 50 basis points, single-name protection clears only where Δ exceeds 25 basis points, and at the top of the band only up to a default probability of 0.30 percent; at 25 the depositor is paid for bearing the risk up to 0.30 percent while capacity and frictions alone consume a single-name buyer's whole advantage. A depositor can be fairly compensated for the risk and still rationally decline to buy protection, and for passive cash I think that is the usual case, which is why Chapter 1's record of where the money went reads as it does. In the operating-balance regime, at a 100 basis point advantage over a large bank's operating account, protection from any provider clears up to a default probability near 0.9 to 1.0 percent, about four times the base assumption. Who pays the premium does not change the cost, since in every protected case the depositor's net yield plus the provider's income equals the bank's all-in cost, and the provider's income is the clearing cost whoever pays it; who pays changes the posted rate, the accounting and the tax line, and leaves the cost where it was. Who pays the reciprocal network's fee can also change with the state of the cycle, which Section 5.7 takes up.

Whether anyone will pay is a separate condition: the provider needs at least the clearing cost, the depositor will give up at most its advantage Δ, and the reference bank will contribute at most what retaining the balance is worth to it. I write that retention value as the cut in the depositor's annual probability of leaving in a run that protection buys, times the present value of the relationship (margin, fees and cross-sell on the covered balance over five years at the risk-free rate plus 3 percent, an annuity factor of 4.067). Protection clears when the price is at or below the sum of what the depositor and the bank will give up. With no retention value the depositor alone needs Δ of at least 37.7 basis points at the reference price. A relationship worth 100 basis points a year with a one-point cut in run probability is worth 4.1 to the bank and leaves the depositor needing 33.7; at 150 and a five-point cut, the largest values in the sweep, the bank's 30.5 leaves 7.2. In no cell does the bank's retention value alone cover the price. Chapter 7 adds the bank's replacement cost to the condition and shows what that does.

3.7 The replacement spread

Uninsured deposits fund loans, and a bank that lost them to a fund or a reserve claim would replace them with Home Loan Bank advances, brokered deposits or term debt, or shrink. The shadow cost adds to the deposit's posted rate what it would cost to make the balance whole; the replacement cost compares the deposit with the funding that would take its place, and for a lender the second comparison is the one that moves loan pricing. At a 20 to 40 percent uninsured share and a 50 to

300 basis point replacement spread, where the deposit's cost counts the earnings credit and the cost of the services as well as any interest, holding the margin would raise the bank's funding cost by 10 to 120 basis points per dollar of funding, before any shrinkage; the figures are round, chosen to bracket rather than to describe any bank. Against that, protection at the clearing cost is cheaper than replacement on each dollar that would otherwise leave whenever the replacement spread exceeds about 38 basis points, and pure cost on each dollar that would have stayed, which is the behavioral question again. The replacement spread is the ceiling on what a bank would rationally pay for stability on a dollar that would otherwise leave, and the spread times the chance of leaving is the ceiling on a dollar covered; that is a second reason the price belongs in funds transfer pricing before anyone sells the product, and Chapter 6 adds a third.

> **What the models did here.** This chapter is where the models erred toward the confident number. The first outside version in stage 1 carried a 31 basis point headline resting on an unsourced 1.67 basis point repo fee and 7.5 basis points of frictions, and a claim that a 10 percent buffer "covers up to six simultaneous failures" when six need 12 percent at the trigger; the Claude model that built the engines replicated the 31 and showed what it rested on. The second outside version led with 99 percent reliability that held under one copula while its own table showed a 27.7 percent shortfall under the other, set a 1.01 percent target equal to its design's own result, and dropped from its write-up the fat-tailed answer its own code had computed. Claude's errors ran the other way. A dearer lead design at 38.6 sat in the project's own price table for four versions beside an all-funded buffer at 37.6 until the outside version drew the conclusion, and the conditional reliability the design now stands on was computed from the fifth model version but was not the standard until the outside version made it one. A gate that traced about 1,450 numbers in the final-version draft found eight wrong.

Chapter 4. Nine routes on one price list

Prices in this chapter are per $100 million of corporate balance above the insured limit. A reported figure is a published fee or disclosure with its date, an author assumption is a round illustrative range, and a model number comes from one of the engines of Appendix A. Each route is described by what it does with the 21 basis points of Table 2, what it consumes on the bank's balance sheet, where it stops, and what it costs.

4.1 Pledged securities

The bank pledges securities worth at least 110 percent of the deposit, the California local-agency template (Cal. Gov. Code 53651 and 53652, as reproduced by a secondary source), so that the depositor takes no principal loss at failure. The risk the balance carries stays on the bank's own encumbered collateral, and because the deposit has funded securities rather than loans the lendable share is about zero, the narrow-bank arrangement by another name, with $110 million of high-

quality liquid assets tied up for each $100 million covered. For a national bank the charter ends the route under Pottorff, the statutory exceptions covering public money (12 U.S.C. 90) and trust funds (12 U.S.C. 92a(d)), and most states mirror the bar, Colorado's letter-of-credit provision and Illinois's 2025 clause "(h) to secure deposits." being the exceptions found. Author assumptions put the cost at 10 to 30 basis points a year if the securities are already held and must be replaced as liquid assets, and at 165 to 330 if they are bought for the purpose, a 150 to 300 basis point loan-to-security spread on 110 percent of the balance.

4.2 Federal Home Loan Bank letter of credit

The member's Home Loan Bank issues a standby letter of credit in favor of the depositor, fully secured by collateral the member has pledged under its advance agreement (12 CFR 1269.2(b)). The deposit stays on the balance sheet and funds loans, so the lendable share is about one; what the bank gives up is $100 million of borrowing capacity and the collateral behind it, which it will want on the Thursday of a run. The published public-unit schedules price the standard structures at 10.5 to 18.0 basis points a year by district, with the variants in Table 18 (FHLBank Des Moines 2026; FHLBank San Francisco 2025; FHLBank Boston 2022; FHLBank Topeka 2023). The fee is small because the risk has been moved rather than removed. The Home Loan Bank's security interest outranks the receiver (12 U.S.C. 1430(e)), so a letter drawn after a failure is paid from assets that would otherwise have supported the deposit insurance fund's recovery, and on my reading the 21 basis points land on the fund; fee plus that risk is 31.2 to 38.7 basis points under the pricing measure. Whether a private corporation can be the beneficiary is the open point. The regulation's list of permitted purposes does not restrict it (12 CFR 1269.2(a)), but the Home Loan Banks' product terms for public-unit letters do, one district requiring the member to warrant that the beneficiary is a public instrumentality, and a letter whose cover is the member's pledged assets raises the same Pottorff question that a direct pledge would. The route is open for public units and an open question for private corporations.

4.3 Reciprocal placement

The network places the depositor's balance across member banks in amounts under the insured limit and the placing bank receives reciprocal deposits of equal size, so its funding is unchanged, the depositor is insured in full, and the lendable share is about one. Here the deposit insurance fund carries the 21 basis points against the assessments the banks pay, a priced public cover rather than a hidden one. The placing bank keeps its customer, but the equal balance that comes back is other banks' customers' money, recorded in the network custodian's name, so the inflow carries no relationship of its own and shrinks when the bank's own customer draws the placed balance down (IntraFi 2022; FDIC 2026a). The scarce resource is the non-brokered allowance, under which a well-capitalized agent institution rated CAMELS 1 to 3 may treat as non-brokered 50 percent of the first $1 billion of its liabilities, 40 percent of the next $9 billion and 30 percent above that, up to $30 billion (12 U.S.C. 1831f(i); 12 CFR 337.6(e); FDIC 2026a). That is a cap on how much of the bank's reciprocal funding escapes brokered treatment, not on how much a depositor may place. Outside

the statute the placing bank sets its own program caps, and the depositor's investment policy has to accept that a $200 million balance becomes at least 800 insured deposits at banks the depositor did not choose. A bank below well capitalized or at CAMELS 4 or 5 may keep what it holds up to a special cap but may not grow past it without every reciprocal deposit becoming brokered, and, if below well capitalized, may not pay above the applicable rate cap (12 CFR 337.6; 12 CFR 337.7). And placement is end-of-day, so it does nothing for the balance that has to sit at the operating bank intraday. The only public fee I have found is one bank holding company's disclosure of an annualized 12.5 basis points at the end of 2025 (Chain Bridge Bancorp 2026). With a rate uplift the depositor demands of 0, 10 or 25 basis points, author assumptions, the route prices at 12.5, 22.5 or 37.5, with 22.5 as the base.

4.4 Letting the balance go

The bank declines the other routes and the depositor sweeps the excess to a government money market fund, a tokenized Treasury fund that transfers between approved holders at any hour, or a stablecoin, and the bank replaces the funding with Home Loan Bank advances, brokered deposits or term debt, or shrinks the loan book. Nothing stops this route, nothing is consumed by it, and the lendable share is zero because the dollar has left. The 21 basis points leave with the balance. The bank pays the replacement spread instead, borne by borrowers through the loan rate or by shareholders through the smaller balance sheet, 50 to 300 basis points on each dollar replaced (Chapter 3), base 175. A stablecoin issuer that keeps reserves in bank deposits is itself a large uninsured depositor, as the $3.3 billion of USDC reserves at Silicon Valley Bank showed in March 2023 (Circle Internet Financial 2023).

4.5 Third-party funded vehicle

Outside investors fund a special-purpose entity up front, the proceeds sit in Treasury bills, and the vehicle writes protection against that collateral, paying par on the covered balance at the FDIC's appointment as receiver against assignment of the deposit claim and collecting the advance dividend and the later dividends as assignee. Because the buffer is the provider's, the lendable share is one and nothing is pushed onto the fund or the bank's contingent liquidity; the premium pays private capital to hold the 21 basis points. The clearing-cost engine of Chapter 3 prices three designs. A single-name vehicle locks up collateral equal to the whole covered balance and never falls short, at 45.74 basis points, 21 of it the risk and 25 the lock-up premium and frictions. The pooled reference design writes on 50 banks, each capped at 2 percent of covered notional, behind an all-funded buffer of 20 percent that pays gross par at the trigger, at 37.74, and a third design sizes the buffer to the S5 standard rather than to a round number, 22 percent or eleven failures of 2 percent, at 37.94. The wrapper and the tail limit the route. A guarantee sold to the party holding the exposure can be financial guaranty insurance under state law, the route around it is a security-based swap with an eligible contract participant, and a vehicle writing many such contracts may cross the dealer de minimis threshold (Chapter 8). No such vehicle exists that I can verify, and above the stated cluster the depositor holds its deposit claim on no public commitment.

4.6 Member-owned mutual with a Treasury-backed tail over a reciprocal base

The stack has three layers, set out in Chapter 5. The base is reciprocal placement (route 4.3), insured by the FDIC, where the depositor's policy and the bank's program cap allow it and while the member stays well capitalized and rated CAMELS 1 to 3. The middle is a member-owned excess-deposit mutual in the form of the Ohio credit-union share guaranty corporation (Ohio Revised Code Chapter 1761), funded by a 3 percent capital contribution paid in at entry and held in Treasury bills and by a special assessment capped at two times the annual assessment. The top is the funded vehicle of route 4.5 moved up to attach where the mutual's resources run out, at 3.42 percent of covered excess, and to pay up to an exhaustion point of 10.0 percent, above which the loss is public; where the mutual pays the FDIC at closure, the tail also advances the special-assessment slice that the mutual collects from survivors in the week after, so that the cash is there on the day it is due. Most of the 21 basis points sit with the members, the tail's buffer, the fund and the depositor carrying the rest. The lendable share is 0.97, since the capital contribution sits in bills at the mutual. Nothing is pledged, so Pottorff does not bite, but whether a national bank may make the entry capital contribution to a state-licensed mutual insurer is a question for counsel that no stage of this work answered, and the rule counts it open. The mutual engine prices the reference design at 21.03 basis points a year on covered excess, 14.54 for the mutual's own layer and 6.50 for the tail, and an alternative design that meets a 5 percent standard under the Student-t(5) copula, exhausting at 21.8 percent, at 31.01.

4.7 Always-on collateralized netting facility

A consortium of banks signs standard repurchase agreements with each other, positions Treasury bills at a tri-party custodian before Friday's close, and lets a calculation and collateral agent net every weekend transfer of tokenized deposits between members, each net debit secured at all times by bills worth, after a 2 percent haircut, at least the debit and settled in central bank money at Sunday's opening; Chapter 6 sets out the mechanics. Each member's net debit is capped at the lesser of its pledged bills after the haircut and a disclosed share of its tokenized balances, 20 percent at the illustrative inputs, and transfers past the cap are queued to the opening. This is a liquidity product, so the 21 basis points stay with the depositor, and the facility does nothing for the balance if the bank fails except deliver the net creditors their collateral. Per $100 million of balance it ties up $20.4 million of bills positioned each weekend, and the lendable share is 0.80, because each always-on dollar is matched by a bill the bank cannot plan to lend. At a 20 percent share the facility costs 10.0 to 34.0 basis points a year on the whole tokenized balance, the sum of author-assumed term ranges, base 22.0; the usage charge between net debtors and net creditors nets to zero and is not in the price. Chapter 6 then asks what backing the cap needs if it is to bind rarely, and finds that, at the weekend outflow the manual assumes, the lendable screen and a rare cap cannot both hold.

4.8 One ledger with a credit tranche and a liquidity tranche

One membership, agent and collateral ledger can carry both a liquidity tranche, the facility of route 4.7 collateralized in the members' own bills, and a credit tranche, the mutual with its tail of route 4.6, because the money that moves on the weekend is the money that would otherwise be a claim in Monday's receivership. The two tranches share a ledger and a membership and never a reserve, since the weekend on which many members are net debtors is the weekend before the Monday on which one of them is closed (Chapter 6). Who bears what is unchanged from the two routes it joins; the saving is the second membership, agent and collateral ledger, about one basis point on the engine's reading, in the access and friction terms. The lendable share is 0.77, the route inherits the mutual's open question on the capital contribution and the facility's on the safe-harbored repo, and it costs 29.7 to 55.0 basis points a year, base 41.8, a model number for the credit tranche plus an author assumption for the liquidity tranche less the saving.

4.9 Prefunded joint-account settlement

The ninth route is the one the live instant-payment network already runs. An operator, which need not be a bank, holds a joint account at a Reserve Bank whose rights and liabilities are shared among the member banks. Each member prefunds a position equal to a disclosed share of its tokenized balance during Fedwire hours, and weekend transfers between members settle instantly and finally against those positions in central bank money. A position can be topped up or drawn down at any hour by a FedNow liquidity management transfer from the member's master account, which the purpose rule of Chapter 1 permits for an instant payment system backed by a joint account and which the network's rules effective 1 November 2026 allow for weekend funding and, for the first time, weekend defunding (Board of Governors 2017; Federal Reserve Financial Services 2026b, 2026d; The Clearing House 2026). A payment beyond the sender's position is rejected rather than queued, the network's practice and not my choice. This is the second liquidity product on the list, so the 21 basis points stay with the depositor. What it consumes is reserves: each prefunded dollar sits in the joint account and cannot fund a loan, so at a 20 percent share, chosen to match the facility, the lendable share is 0.80, and the slice is not counted as a high-quality liquid asset in the base, since no public statement I have found settles how a participant's share of a joint-account balance is treated under the liquidity coverage ratio. The price is 13.6 to 37.1 basis points a year on the whole balance, base 25.35, from credit of 0 to 0.5, access of 2 to 6, frictions of 2 to 6 and capacity of 9.6 to 24.6, author assumptions except the rates inside the capacity term: the joint account's balances earn the rate on reserve balances, 3.90 percent, under a Board order specific to the live network's account, and a 3-month bill yielded 4.13 percent the same day, so the prefunded slice forgoes the loan spread over bills plus the 23 basis points the bill earns over reserves (Board of Governors 2023c, 2026d; U.S. Department of the Treasury 2026). That second item is why the route costs more than the facility at the same share, and at equal shares the joint account is cheaper only below a prefunded share of 5.4 percent. That comparison takes the facility's Sunday settlement cash as Reserve Bank credit against loans the bank already holds there, within its net debit cap headroom (Section 6.2). Where the facility's Sunday cash must instead be new reserves or newly bought securities, its price

at a 20 percent share is 39.10 or 34.50 with a lendable share of 0.60, and the joint account at 25.35 is cheaper at every share, winning at a 20 percent share once 19.6 percent of the Sunday capacity comes from new reserves or 26.8 percent from bought securities (the routes engine's results, Section 7). When the route runs on the live network's RTP rails, or on FedNow, each payment above $10 million would have to be split under that network's per-payment limit, if its rules allow a split (Section 6.14). The limit is $10 million on RTP since 9 February 2025 and on FedNow (The Clearing House 2024, 2026, Rule II.C.2; Federal Reserve Financial Services 2025), and the joint account as specified here, with rules of its own, does not carry it. A new joint account carries no interest order of its own; if its balances earned nothing the price would be 91.6 to 115.1, base 103.35, below the one-ledger structure in the ranking. Without a weekend top-up a member's position is exhausted, and payments rejected, in 4.156 percent of weekends at a 20 percent share, 0.3769 percent of the balance rejected on average, and a top-up lowers that rate only by drawing master-account reserves the bank cannot lend either. The route is open on the first screen, because a new joint account needs the account-holding Reserve Bank's approval under the Board's 2017 guidelines and none has been sought, and it fails the lendable screen at a fifth of the balance as the facility does, passing at a tenth or less. Section 6.2 says why a pool needs no entity of its own.

4.10 The price list

Table 6 is the price list, one row per priced variant, with the basis of each figure stated, because a model number and a reported fee are not the same kind of number and the rule ranks across both, and with the lendable share beside the price, because the fee comparison alone leaves out the measure that decides whether the bank can lend the dollar.

Route	Variant	Price (bps a year)	t(5) price	Lendable share	Basis
R1	Pledged from securities already held	10.0 to 30.0 (base 20.0)	n/a	0.00	author assumption
R1	Securities bought for the purpose (150 to 300 spread on 110%)	165.0 to 330.0 (base 247.5)	n/a	0.00	author assumption
R2	Public-unit letter fee by district, standard structures	10.5 to 18.0 (base 14.3)	n/a	1.00	reported figure, FHLBank schedules 2022 to 2026
R3	Placement fee plus rate uplift (base uplift 10.0)	12.5 to 37.5 (base 22.5)	n/a (FDIC-insured)	1.00	reported fee (Chain Bridge Bancorp 2026); assumed uplift
R4	Replacement spread on each dollar replaced	50.0 to 300.0 (base 175.0)	n/a	0.00	author assumption
R5	Single-name funded vehicle	45.74	45.74	1.00	model number, clearing-cost engine
R5	Pooled all-funded reference design (theta 20%, 50 banks)	37.74	37.74	1.00	model number, clearing-cost engine
R5	Pooled, capacity sized to the S5 standard (11 failures, theta 22%)	37.94	37.94	1.00	model number, clearing-cost engine
R6	Reference design (X 10.0%, Gaussian 1% standard, q_acq 0.5)	21.03	23.02	0.97	model number, mutual engine

Route	Variant	Price (bps a year)	t(5) price	Lendable share	Basis
R6	Alternative design (X 21.8%, Student-t(5) 5% standard)	31.01	33.73	0.97	model number, mutual engine
R7	Facility at a 20% always-on share, 2% haircut	10.0 to 34.0 (base 22.0)	n/a	0.80	author assumption (illustrative)
R8	R6 reference design + R7 less the shared-infrastructure saving	29.7 to 55.0 (base 41.8)	43.83	0.77	model number + assumption
R9	Joint-account position at a 20% prefunded share, balance earning interest	13.6 to 37.1 (base 25.4)	n/a	0.80	author assumption (illustrative); reported rates

Table 6. The price list, basis points a year per unit of a $100 million corporate balance above the insured limit, with the lendable share. The base of a range is its midpoint except for R3 (the 10 basis point uplift; midpoint 25.0) and R8 (R6 plus R7's base less the 1.19 saving), and the rule ranks on base prices; a t(5) price is shown only where the route has a failure payoff. R7 and R9 are a different product (liquidity outside Fedwire hours) shown on the same scale for comparison only. Source: the routes engine's results, Sections 4 and 5.

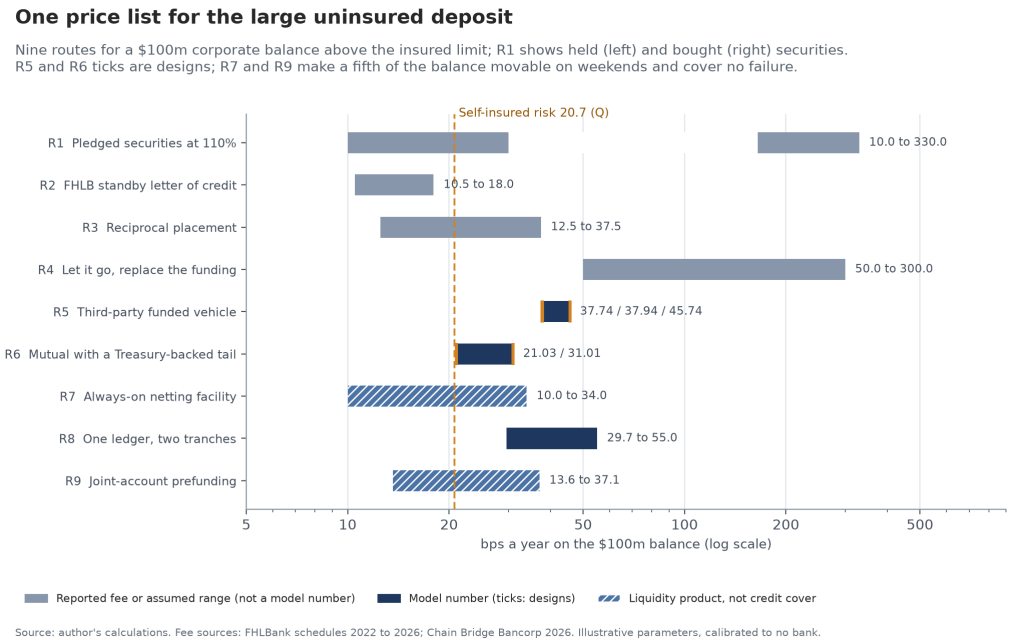


Figure 4. One price list for the large uninsured deposit: nine routes for a $100 million corporate balance above the insured limit, R1 shown for held and for bought securities, R5 and R6 ticks at the priced designs, R7 and R9 liquidity products that make a fifth of the balance movable on weekends and cover no failure. Source: author’s calculations; fee sources as in Table 6; illustrative parameters, calibrated to no bank.

The letter of credit’s fee leaves the risk unpriced, and fee plus the risk the super-lien moves to the fund is 31.2 to 38.7 basis points under the pricing measure, which puts it beside the funded vehicle rather than below it. Letting the balance go is priced on each dollar replaced, and above about 55 basis points it is dearer than every route on the list except pledging securities bought for the purpose. On the lendable-share axis the menu collapses into two groups, the routes that keep the dollar on the balance sheet and the routes that do not, and the choice within the first group is between a subsidized price with capacity consumed and an unsubsidized price with none (Figure 5).

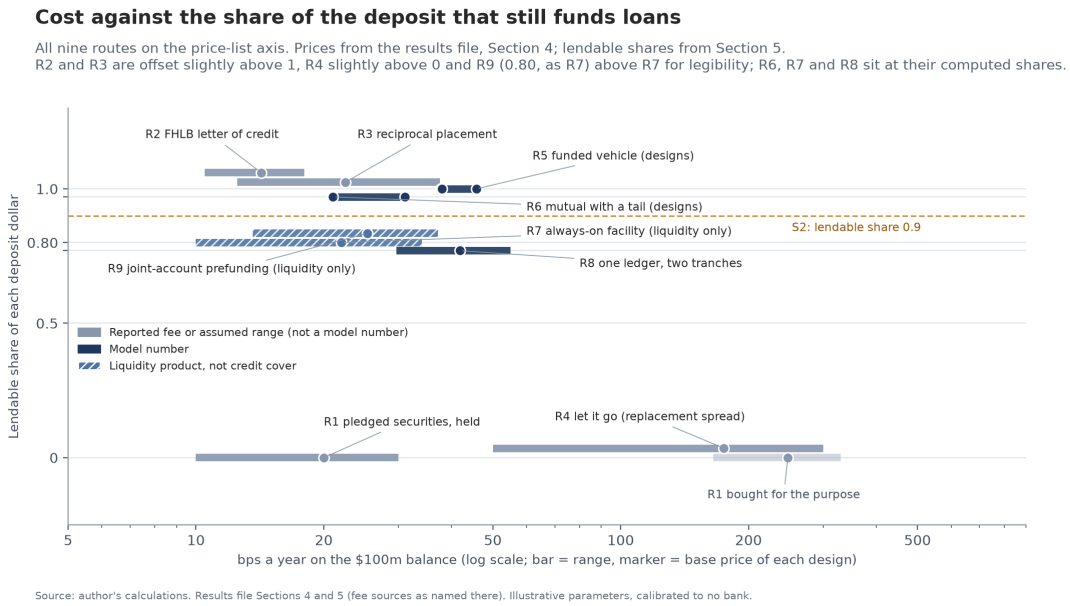


Figure 5. The share of the deposit that still funds loans against cost, for the nine routes of Table 6 at their base prices on the cost axis of Figure 4: the letter of credit, reciprocal placement, the mutual and the vehicle along the top at lendable shares of 0.97 to 1, the facility (0.80), the joint account (0.80, drawn slightly above the facility for legibility) and the one-ledger structure (0.77) a step below them, and pledged collateral and letting the balance go along the bottom at a share of about zero; the dashed line is the S2 threshold, a lendable share of 0.9, and for legibility the letter of credit and reciprocal placement are drawn slightly above one and the let-it-go route slightly above zero. Source: author's construction from Table 6; illustrative parameters, calibrated to no bank.

Figure 6 shows where the 21 basis points of Table 2 land, and every bar sums to the self-insured cost, because the risk is neither created nor destroyed by the route, only placed. The pledge puts it on the bank's own encumbered collateral, the letter on the fund through the super-lien, reciprocal placement on the fund against assessments and the vehicle on the private buffer. The mutual puts most of it on its members (19.19 of the 20.74 under the pricing measure, with 0.55 on the tail, 0.99 on the fund and 0.01 on the depositor), the facility and the joint account leave it on the depositor, and letting the balance go sends it away. The facility moves one further thing the figure does not show. The safe-harbored weekend repo puts net creditors ahead of the receiver for the capped outflow, worth up to 1.38 basis points a year under the real-world measure and 4.15 under the pricing measure, an upper bound that assumes every failure follows a weekend in which the capped outflow left (Chapter 6).

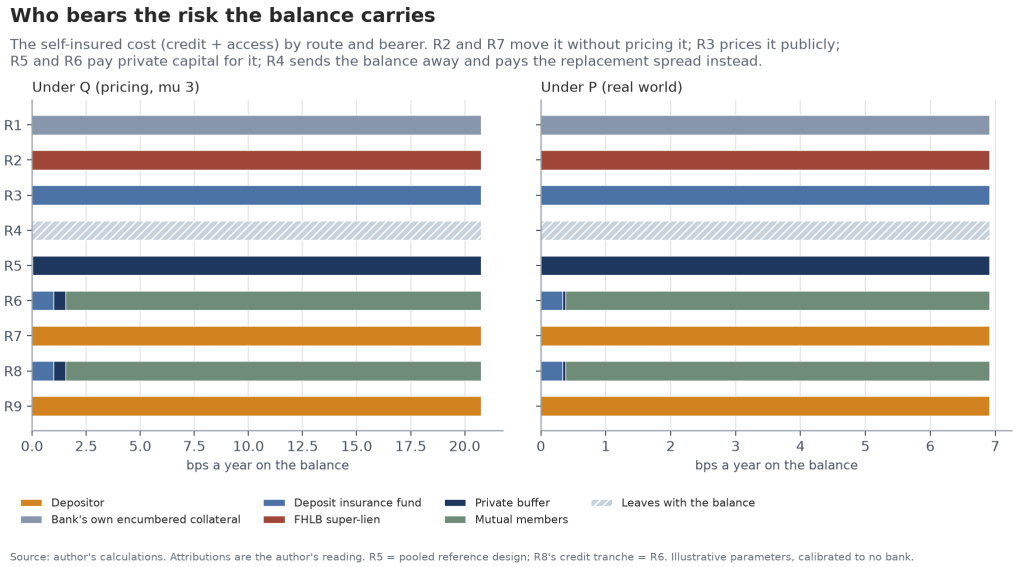


Figure 6. Who bears the risk the balance carries: the self-insured cost (credit plus access) by route and bearer, under the pricing measure and the real-world measure. Source: author's calculations; attributions are the author's reading; R5 is the pooled reference design and R8's credit tranche is R6; illustrative parameters, calibrated to no bank.

4.11 The screens

Table 7 is the screens grid, with the reasons behind each verdict, and their sources, in the routes engine's results. The verdicts that carry the result are the letter of credit's failure on S4 (the super-lien) and its open S1 and S3, and the pooled reference design's failure on S5 by a hair, 1.01 percent short against a 1 percent standard, which is why the sized variant exists. The mutual's open S1 is its only non-pass. The facility fails S2 at a 20 percent share and is open on S4, the one-ledger structure inherits both open questions with a lendable share of 0.77, and the joint account fails S2 at the same share and is open on S1 for the unsought Reserve Bank approval, scored as the letter's and the mutual's unsettled approvals are scored, while passing S3, since the position is the member's own prefunded money with no capital condition, and S4, since it is prefunded and final with no secured priority over the receiver and no Reserve Bank credit, on my reading.

Variant	S1	S2	S3	S4	S5
R1 held	fail	fail	pass	open	n/a
R2	open	pass	open	fail	n/a
R3	pass	pass	fail	pass	pass
R4	pass	fail	n/a	pass	n/a
R5 single	pass	pass	pass	pass	pass
R5 pooled	pass	pass	pass	pass	fail
R5 sized	pass	pass	pass	pass	pass
R6	open	pass	pass	pass	pass
R6 alt	open	pass	pass	pass	pass
R7	pass	fail	pass	open	n/a
R8	open	fail	pass	open	pass
R9	open	fail	pass	pass	n/a

Table 7. The screens grid: pass, fail, open (a question the sources do not settle) or not applicable; R1 bought shares R1 held's verdicts; S5 is computed for R5, R6 and R8, passes by construction for R3 and does not apply where the route has no failure payoff. Source: the routes engine's results, Section 8.

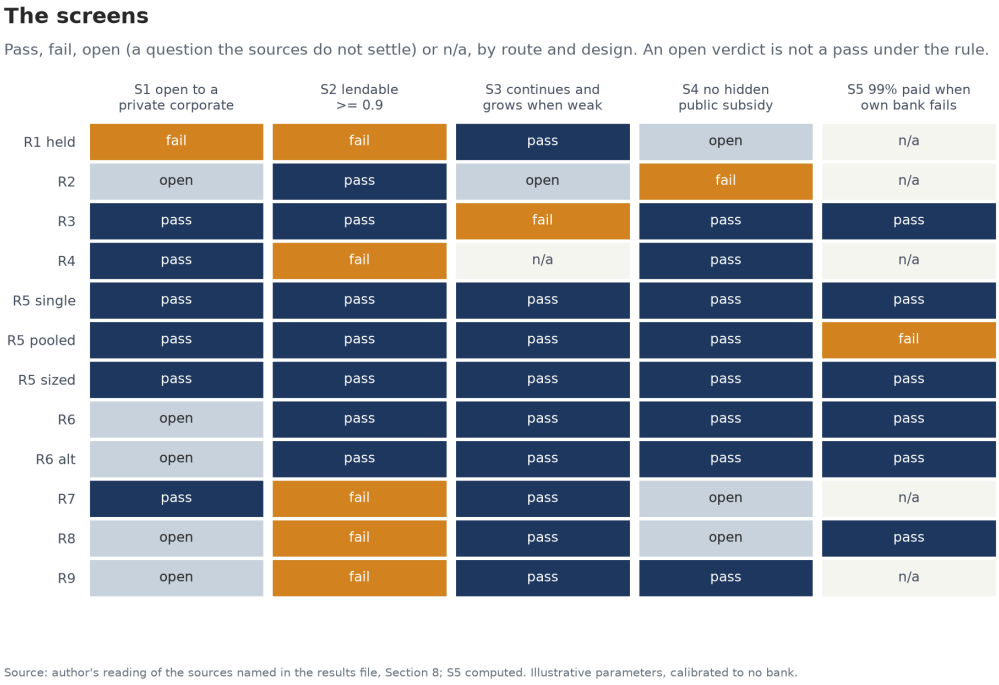


Figure 7. The screens: pass, fail, open or not applicable, by route and design; an open verdict is not a pass under the rule. Source: author's reading of the sources named in the routes engine's results, Section 8; S5 computed; illustrative parameters, calibrated to no bank.

Table 8 re-prices every computed design under the Student-t(5) copula and sets the result beside the Gaussian price, with the shortfall in the year the depositor's own bank fails under each copula. The price barely moves and the reliability moves by a factor of about twenty to thirty-five for the designs sized to the Gaussian standard. The reference design misses the Gaussian standard by 0.0084 percentage points, and one more failure of capacity passes at 37.94. Under the fat-tailed copula the same 22 percent is short one year in four of the years that matter, and the design that meets a 5 percent standard under that copula needs 25 failures of capacity, half the pool, at 40.74.

Design	Price	t(5) price, same design	Gaussian: short	Gaussian: paid in full	t(5): short
R5 single-name	45.74	45.74	0 (fully funded)	100.00%	0 (fully funded)
R5 pooled reference (20%)	37.74	37.74	1.01%	98.99%	27.73%
R5 pooled sized to S5 (22%)	37.94	37.94	0.72%	99.28%	24.75%
Memo: R5 pooled sized to t(5) 5% (50%)	40.74	40.74	0.007%	99.99%	4.38%
R6 reference (X 10.0%)	21.03	23.02	0.99%	99.01%	20.19%
R6 alternative (X 21.8%)	31.01	33.73	0.032%	99.97%	4.95%
R8 (credit tranche = R6 reference)	41.84	43.83	0.99%	99.01%	20.19%

Table 8. Reliability of every computed design: the probability that the depositor is not paid in full in the year its own bank fails, correlation 0.30, state-dependent loss, under the Gaussian copula (the S5 column) and the Student-t(5) copula, beside the price under each. Source: the routes engine's results, Section 9.

The lendable bar of 0.90 in S2 is the rule of record, set before the run, and Table 19 in Appendix B shows what moves if it is lowered. Two orientations bracket it. Loan-to-deposit ratios have fallen since 2008: the median regional bank of $10 billion to $100 billion in assets lent 1.05 of its deposits at the end of 2007 and 0.85 at the end of 2025, and the median bank under $1 billion lent 0.84 and then 0.79 (FDIC 2026d), whole-book ratios that include insured retail deposits and so overstate what is prudent against runnable uninsured commercial money. The liquidity coverage ratio's outflow rates for the same money are 25 percent for operational deposits and 40 percent for non-operational unsecured wholesale funding from a provider that is not a financial-sector entity (12 CFR 249.32(h)(4) and (h)(1)), that is, lendable shares of 0.75 and 0.60, and the rule binds only the largest banking organizations, so that too is an orientation and not a bar for the banks of this manual. A lower bar is fair only if the route's encumbered bills or reserves can also be the bank's own liquidity buffer against the same deposit, the open encumbrance question of Section 8.4; if they

cannot, the encumbrance stacks on the buffer and a lower bar overstates what the bank keeps lending. At 0.85 nothing changes. At 0.80 the facility and the joint account pass S2 at a 20 percent share, so with S2 applied to the settlement problem the answer with the open items resolved is the facility at 22.00 with the joint account as challenger at 25.35, on the Sunday-cash condition of Section 4.9, since where that cash must be new reserves or bought securities the facility lends 0.60 and fails S2 at every bar. At 0.75 the one-ledger structure passes as well and becomes the answer, with the open items resolved, for both problems together at 41.84. The run-clock champions do not move at any bar, and the cap test of Chapter 6 fails at the assumed 5 percent median at every bar at a dispersion of 0.8, the boundary running from 1.52 to 3.81 percent of the balance counting the grossed-up bills. The routes engine and the settlement engine both take the bar as a setting (`--s2-min`), so a reader can rerun them at any bar (the routes engine's results, Section 9b; the settlement engine's results, Table A6).

> **What the models did here.** The routes engine was built in stage 4 in about 47 minutes by the Claude model that built the engines: it imports the clearing-cost engine and the mutual engine by path, copies no code from either, reads fourteen headline figures from them in memory before any new result, and prices, screens and ranks the routes, eight of them at that stage and the ninth added in stage 7 after the outside review of Chapter 9 asked why a prefunded pool had been dismissed. Its three findings that changed the work were not errors: the 0.0084-point miss of the reference design, the mutual's S1 question in its national-bank form, and the facility's failure on S2 at a fifth of the balance. The drafting errors were in the prose around it. The first draft called letting the balance go the dearest route at any spread, when bought securities and the top of the one-ledger range are dearer. It put two Home Loan Bank districts' prices "at one district" and applied the rate cap to a bank rated CAMELS 4 that is still well capitalized, where the rule applies below well capitalized, and wrote "the base of a range is its midpoint" for two routes where it is not. It captioned Figure 5 as eight routes when the chart drew five, a fail the fact checker found only by opening the image.

Chapter 5. The champion

Table 9 is what the rule produced, run strictly and then with every open item resolved in the route's favor, for each problem alone and for both together.

Problem	Reading	Champion	Challenger (next route)	Next-cheapest passing variant
(a) stable lendable funding through a failure	strict (open is not a pass)	R5 pooled vehicle sized to the S5 standard (11 failures, theta 22%), 37.94 bps	none	R5 single-name vehicle, 45.74 bps
(a)	open items resolved in the route's favor	R6 mutual with tail, reference design (X 10.0%, q_acq 0.5), 21.03 bps	R5 sized, 37.94 bps	R6 alternative design (X 21.8%), 31.01 bps
(b) move the balance outside Fedwire hours	strict	none	none	none
(b)	open items resolved	R7 facility at a 20% always-on share, 22.00 bps (10.0 to 34.0)	R9 joint account at a 20% prefunded share, 25.35 bps	R9, 25.35 bps
(a) and (b) together	strict	none	none	none
(a) and (b) together	open items resolved	none	none	none

Table 9. The rule's result. With S2 also applied to problem (b), no route passes under either reading; with the joint account's balances earning no interest, R8 at 41.84 is again the challenger on problem (b). R7's 22.00 takes its Sunday settlement cash as Reserve Bank credit against loans already pledged at the Reserve Bank; where that cash must be new reserves or bought securities, R9 at 25.35 is the champion on problem (b) with the open items resolved (Section 4.9). The "none" rows for both problems together are the rule's verdict at the 0.90 bar; what such a balance should be charged regardless is Table 17 (Section 7.3). Source: the routes engine's results, Sections 7 and 9.

5.1 The two readings and my position

Read as written, the rule names the pooled Treasury-backed vehicle with its buffer sized to the standard, eleven failures of 2 percent among fifty banks, at 37.94 basis points a year, and names no challenger, because no other route passes every screen. The champion is a vehicle nobody has built, a fifth of a basis point above the reference design of Chapter 3, which it beats only because that design falls short of the 1 percent standard by 0.0084 percentage points (1.0084 percent short, so 98.99 percent paid in full when the depositor's own bank fails). Under the fat-tailed copula it is short in 24.75 percent of the years that matter against 27.73 for the reference design, at the same price, because with a fixed loss given default the price does not move with the copula and the shortfall at a given capacity does. The strict rule picks the one route whose open legal questions the engine scores as general rather than charter-specific, and pays about 17 basis points a year for that.

Resolve each open verdict in the route's favor and the answer changes once. The mutual with a Treasury-backed tail becomes the champion at 21.03 basis points and the sized vehicle the challenger at 37.94, while the letter of credit still fails S4 and the facility still fails S2. The one verdict that decides between the two readings is the mutual's S1, whether a national bank may make the 3

percent entry capital contribution to a state-licensed excess-deposit mutual insurer. Nothing is pledged, so the Pottorff bar does not bite, but a capital contribution to a state-licensed insurer is an investment, and whether a national bank's powers reach it is a question I have not resolved and which is for counsel.

The rule as written names the vehicle. In my judgment the mutual is the better design, by about 17 basis points a year on every dollar of covered excess, with a lendable share of 0.97 against 1.00 and a reliability under either copula within a few points of the vehicle's, if one legal question resolves and the FDIC accepts the Massachusetts settlement rule and the receivership carry it entails. Without that rule the mutual still prices at 21.39 at the same exhaustion point and 22.73 re-sized to the standard, and the legal question is the first item of the ask in Chapter 10. The Student-t(5) copula gives the same picture. The mutual stays about 15 basis points cheaper under it and falls short a little less often than the sized vehicle in the year the depositor's own bank fails (Table 8), and the designs that meet a fat-tailed standard (the mutual's alternative at 31.01, the vehicle sized to 50 percent at 40.74) keep the same ordering.

5.2 The three layers

The champion's stack has three layers, each doing what it does cheapest (Figure 8). The base is reciprocal placement, insured by the FDIC, where the depositor's investment policy and the bank's program cap allow it and while the member stays well capitalized and rated CAMELS 1 to 3; it carries whatever part of the balance those conditions admit and is outside the model (Section 5.7 treats that layer as an option across the cycle). The middle is a member-owned excess-deposit mutual for regional and community banks, covering the uninsured excess that is not placed, which I call covered excess. The top is the Treasury-backed vehicle of Chapter 3, moved up from first loss to attach above the mutual's own resources and cover the cluster tail up to a stated exhaustion point, above which the loss is public.

The mutual is a private licensed insurer in the Ohio form: a member-owned guaranty corporation under state insurance law, with an insurance subsidiary writing the excess cover and the tail layer in place of reinsurance. Members are insured depository institutions of $1 billion to $250 billion in assets, 100 to 200 of them in the sizing, admitted on a grid of public Call Report and FR Y-9C metrics, with no member above 5 percent of covered excess and each member's covered excess capped by a tiered schedule on its liabilities whose shape mirrors the reciprocal rule. There is no per-depositor cap, because the depositor's claim stays a plain deposit at its own bank and the mutual's promise is to the member's excess depositors as a class. The mutual's resources are a reserve and a call. The reserve target is 3 percent of covered excess, paid in by members as a capital contribution at entry (as the Ohio statute requires of credit unions that join a share guaranty corporation) and held in Treasury bills, with the carry priced as the members' opportunity cost on that capital, 100 basis points over bills. After a loss year the mutual may levy a special assessment on surviving members, capped at two times the annual assessment and collectable within the week, 0.42 percent of covered excess at the base, so the mutual's own resources meet cash calls up to 3.42 percent, with the week's

lag bridged by the tail where the cash is due at closure (Section 5.3). The annual assessment is risk-classed on the same public-metric grid as admission (three classes at 0.15, 0.35 and 0.75 percent annual default probability in the sizing).

The tail layer attaches where the reserve and the collectable special assessment run out, at 3.42 percent of covered excess, and is 6.58 percent thick, so the stack pays up to an exhaustion point of 10.0 percent of covered excess, about 60 failures of average weight under the Massachusetts rule and 30 under the day-seven rule. The tail's funded commitment is larger than its layer, 7.00 percent of covered excess, because under the Massachusetts rule it also advances the special-assessment slice at closure and is repaid when the assessment is collected, so that an assessment still receivable on the day the FDIC is paid is not counted as cash. The exhaustion point is the design's standard, and the stack as built stops a little short of it. The tail's layer is fixed at the exhaustion point less the mutual's attachment, and a failed member pays no special assessment, so when members holding a share S of covered excess fail the collectable call shrinks by that share and the stack pays at most the exhaustion point less 0.42 percent times S, about 9.9 percent of covered excess in the state where it first exhausts. The slice of the tail's funded commitment that would have filled that gap goes unused, and under the day-seven rule, where no advance is needed, the whole 0.42 percent advance slice is unused though its lock-up and frictions are charged. A contract under which the tail pays up to the exhaustion point would use that slice, and Section 8.4 carries it as an open item. The exhaustion point meets the reliability standard of Chapter 3, a 1 percent probability that the stack falls short in the year the median member fails, under the Gaussian copula; the alternative design meets a 5 percent standard under the Student-t(5) copula and exhausts at 21.8 percent. Above the exhaustion point the depositor holds its deposit claim and the layer is public on no public commitment.

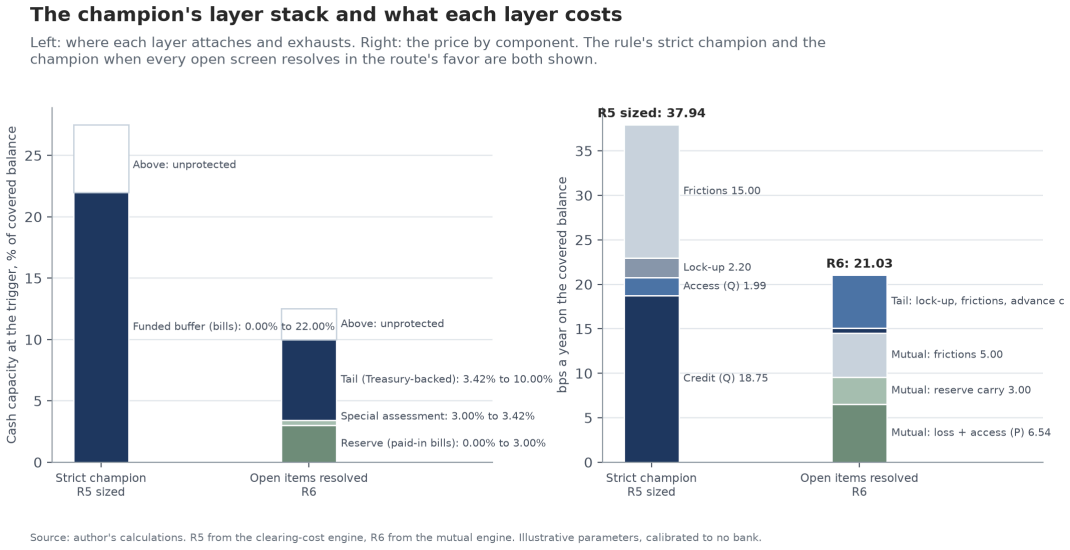


Figure 8. The champion's layer stack and what each layer costs. Left, where each layer attaches and exhausts, as cash capacity at the trigger in percent of the covered balance; right, the price by component in basis points a year. The rule's strict champion (R5 sized) and the champion when every open screen resolves in the route's favor (R6) are both shown. Source: author's calculations; R5 from the clearing-cost engine, R6 from the mutual engine; illustrative parameters, calibrated to no bank.

The mutual and the tail are structured to a stated financial-strength rating target, because corporate investment policies are written to ratings and a depositor whose policy will not take 800 bank names can hold one rated claim. As far as I have found, no nationally recognized statistical rating organization rates any US deposit or share insurer today; Morningstar DBRS extended its government-related-entities methodology to deposit insurance and guarantee funds in June 2025 and names no US fund (Morningstar DBRS 2025), so a rated excess-deposit mutual would be a first.

5.3 How the layers settle

Two settlement rules are priced and a mixture of them is the base. Under the Massachusetts rule (rule M, the practice of that state's excess-deposit fund in the failures of 1989 to 1994), the stack pays the FDIC the uninsured loss share at closure so that an acquirer assumes every deposit whole on day one; the cash needed is the loss, LGD times covered excess, and the stack holds no claim afterward. Because the special assessment is collected from survivors in the week after closure, the cash on the day is the reserve plus the tail, and the tail advances the assessment slice and is repaid when it is collected, at the risk-free rate plus the funding spread for up to seven days; in a year in which members holding 10 percent of covered excess fail at a 40 percent loss, the gap at closure would be 0.3786 percent of covered excess with the assessment counted as a receivable and is zero with the advance, and across the conditional distribution the tail advances in 4.35 percent of the years that matter, its largest cash call at closure being 6.896 percent against a commitment of 7.000. The expected carry on the advance rounds to zero; what the fix costs is the larger standing commitment, on which the tail's lock-up and frictions are charged, and the advance carries no charge for a survivor that fails to pay its assessment, which I assume away and record in Chapter 8. Under the day-seven rule of Chapter 3 (rule B), the depositor collects the FDIC's advance dividend on its own claim and the layers pay the remainder of par on day seven, own the receivership certificate and fund the still-recoverable slice until the receivership pays out. Whether an acquirer appears is a market state, so the sizing draws rule M or rule B once for all of a year's failures with an acquirer probability of one half, an orientation rather than a calibration. Rule M needs half the cash of rule B at the base loss and two thirds at a 40 percent loss, which is why the settlement rule is a design lever; every failure under rule B at the same exhaustion point prices the stack at 21.39 basis points. Rule M places a cost of its own, since the receivership carries the recovery of the excess from closure until its assets are liquidated, a timing cost the FDIC carries in any all-deposit purchase and assumption, booked at 0.99 basis points a year under the pricing measure to the deposit insurance fund. The manual counts the mutual as passing the no-hidden-subsidy screen on the reading that this is ordinary resolution practice, and lists that reading among the open items in Chapter 8. Rule M also depends on the FDIC accepting a private fund's cash for the uninsured loss share, which it did in Massachusetts for small savings banks in the early 1990s (FDIC 1989, 1990, 1991, 1992a, 1992b, 1994); whether it would do so for a large regional, and in a cluster, is a resolution-policy question.

5.4 The price by layer and what moves it

The champion's 21.03 basis points a year are 14.54 for the mutual's own layer, priced under the real-world measure because its members hold the risk at cost, and 6.50 for the tail premium, priced under the pricing measure because the tail's investors are outsiders (Table 10). The alternative design's 31.01 differs almost entirely in the tail, whose lock-up and frictions scale with its commitment.

Component (bps a year on covered excess)	Reference, mixed rule	Alternative, mixed rule	Reference, rule B only
Mutual: expected loss on its retention (P)	6.21	6.21	6.17
Mutual: access (P)	0.33	0.33	0.65
Mutual: reserve carry	3.00	3.00	3.00
Mutual: frictions	5.00	5.00	5.00
Mutual, own layer	14.54	14.54	14.83
Tail: expected loss in the layer (Q)	0.50	0.45	0.89
Tail: access (Q)	0.05	0.04	0.09
Tail: lock-up on the commitment	0.70	1.88	0.66
Tail: frictions on the commitment	5.25	14.10	4.93
Tail: carry on the rule-M assessment advance	0.00	0.00	0.00
Tail premium	6.50	16.47	6.57
All-in assessment	21.03	31.01	21.39

Table 10. Price by layer. The mutual prices its own layer under the real-world measure; the tail's investors price under the pricing measure (three times the physical default probability); the tail's lock-up and frictions are charged on its funded commitment, the layer plus the rule-M advance capacity. Source: the mutual engine's results, Section 6.

About 7.1 of the 21.03 is the risk itself and about 13.9 is running cost, the carry on the members' paid-in reserve, the mutual's frictions, and the tail's lock-up premium and frictions, so the design lives or dies on frictions, and the two live excess insurers say frictions can be low. The Massachusetts fund ran at about 1.6 basis points of net expenses on its insured excess deposits in fiscal 2025 (Depositors Insurance Fund 2025). The Ohio credit-union excess insurer's whole premium was about 11 basis points on its limits (American Mutual Share Insurance Corporation and Excess Share Insurance Corporation 2026), so the mutual's 5 sits between the two, and the sweep from 2 to 10 moves the price from 18.05 to 26.00. Against the vehicle's 37.7 the saving of

about 17 basis points comes from the mutual retaining the first loss at the real-world measure, from a tail that is a thin layer whose frictions scale with its funded commitment, and from a settlement rule that halves the cash a failure draws half the time.

The sensitivities in Table 11 move one input at a time from the base and re-size each design’s capacity to the Gaussian standard in each case, so that the price carries the cost of staying at the standard; Figure 16 in Chapter 7 draws them. The default intensity dominates both tornados, as it did in Chapter 3. The mutual’s next inputs are its own frictions and the loss given default, the vehicle’s the risk-premium multiplier and the loss. The pricing correlation does not move the vehicle’s price at all, since with a fixed loss and no facility its credit and access terms are expectations the correlation does not change. The member count, the size dispersion, the concentration cap and the covered excess sought each move the mutual’s price by less than 0.2 basis points.

Input	R6: low	high	spread	R5 sized: low	high	spread
Default intensity lambda_P, 0.10% to 1.00%	14.10	53.50	39.40	24.89	101.35	76.46
Loss given default, 15% to 40%	18.29	25.52	7.22	31.23	47.99	16.76
Risk-premium multiplier mu, 2 to 5	20.71	22.11	1.40	31.02	51.76	20.74
Pricing correlation rho, 0.10 to 0.30	20.63	21.70	1.06	37.94	37.94	0.00
Mutual frictions f_m, 2 to 10 bps	18.05	26.00	7.95			
Reserve opportunity cost, 50 to 150 bps	19.54	22.52	2.98			
Vehicle frictions f, 10 to 20 bps				32.94	42.94	10.00
Lock-up premium s_lock, 5 to 15 bps				36.84	39.04	2.20

Table 11. Range of the all-in price across one-at-a-time sweeps, basis points a year, capacity re-sized to the Gaussian 1 percent standard at correlation 0.30 in each case; the Gaussian shortfall stays at 0.98 to 0.99 percent for the mutual and 0.708 to 0.963 for the vehicle across every row. Source: the routes engine’s results, Section 11.

5.5 Reliability under both copulas, and the years after a loss

Reliability is measured as Chapter 3 measures it, in the year the member whose depositors are collecting has itself failed, at a correlation of 0.30 with the loss and the advance dividend depending on the state of the common factor. Table 12 gives the layer that completes payment, and the shortfall.

Copula	Member	Design	Reserve alone	Plus special assessment	Plus tail	Short	Severity	Stack exhausted, any year
Gaussian	median	Reference	87.41%	1.99%	9.61%	0.99%	21.2%	0.0128%
Gaussian	median	Alternative	87.41%	2.79%	9.77%	0.0320%	12.8%	0.0002%
Gaussian	largest	Reference	75.19%	4.50%	18.76%	1.55%	20.5%	0.0128%
Student-t(5)	median	Reference	47.97%	3.03%	28.81%	20.19%	37.0%	0.15%
Student-t(5)	median	Alternative	47.97%	4.35%	42.73%	4.95%	22.6%	0.0239%
Student-t(5)	largest	Reference	38.13%	3.92%	35.05%	22.90%	36.6%	0.15%

Table 12. Reliability conditional on the named member failing in the year, correlation 0.30, state-dependent loss and advance dividend, mixed settlement rule, cash tested on the day it is due under the rule that applies. The first four columns sum to 100 percent; severity is the expected unpaid share of the cash due when the stack is short; the last column is the unconditional probability that the public layer is reached in any year. Source: the mutual engine’s results, Section 9.

In the year the median member fails, the reserve alone pays in nearly nine cases in ten under the Gaussian copula, the tail is needed in about one in ten, and the stack is short about one time in a hundred, leaving about a fifth of the cash due unpaid when it is. Under the Student-t(5) copula the reserve alone pays in fewer than half the cases, the tail is needed in nearly three in ten, and the same design is short one year in five, leaving nearly two fifths of the cash due unpaid. The public layer is reached in about one year in 8,000 under the first copula and one in 670 under the second; the alternative design brings the t shortfall to 4.95 percent and the public layer to about one year in 4,000, for 10.0 basis points more. A cross-guarantee among regional banks of the same size class in the same rate cycle is exactly the structure in which the members fail together if they fail at all, so the copula question weighs more here than it did for the vehicle, and the record is silent on it. Under moderate clustering the mutual saves about 17 basis points; under fat tails, where it is at its weakest, it saves about 10, since at the Student-t(5) 5 percent standard the stack costs 31.0 basis points under the mixed rule against 40.7 for the vehicle under gross par.

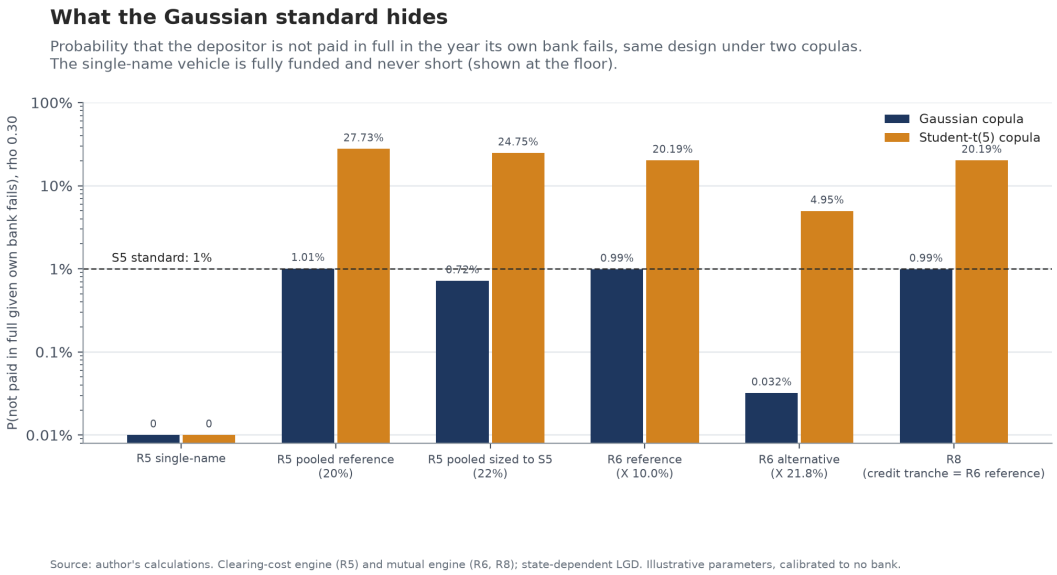


Figure 9. The probability that the depositor is not paid in full in the year its own bank fails, for the same design under the Gaussian and the Student-t(5) copulas, for every computed design; the single-name vehicle is fully funded and never short, shown at the floor. Source: author's calculations; the clearing-cost engine for R5 and the mutual engine for R6 and R8; state-dependent loss; illustrative parameters, calibrated to no bank.

The special-assessment cap is the smallest of the design's levers, because the collectable call is a few tenths of a percent of covered excess against a reserve of several percent, and what moves reliability is the reserve and, above it, the tail (Figure 10). The settlement rule sits in between, since at the same exhaustion point rule B alone falls short about one and a half times as often as the base mixture under either copula, and rule M alone well under half as often under the Gaussian and about three fifths as often under the t.

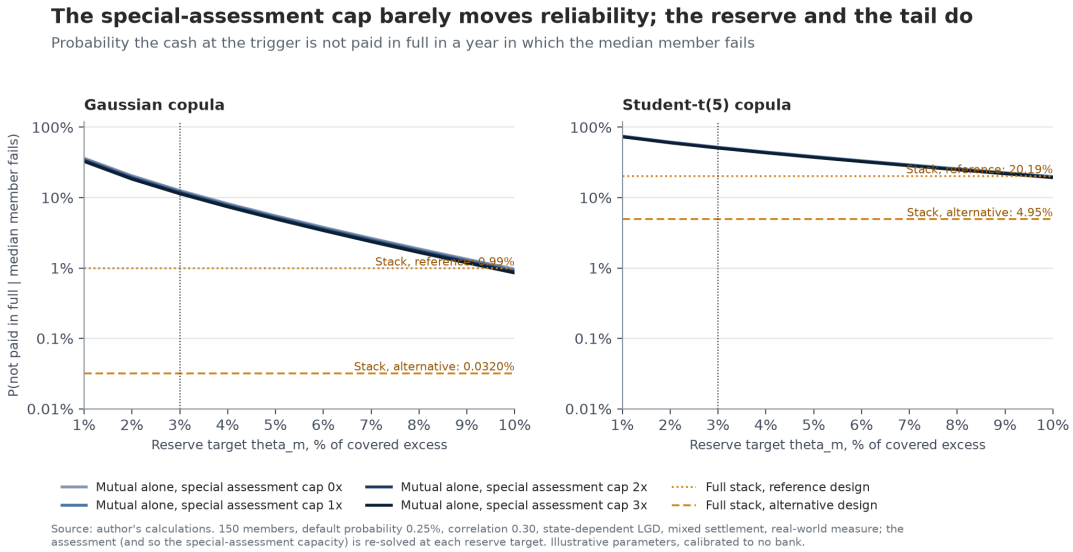


Figure 10. Probability that the cash at the trigger is not paid in full in the year the median member fails, against the reserve target, for the mutual alone at special-assessment caps of zero to three times the annual assessment, with the full stack's reference and alternative results as horizontal lines; Gaussian copula left, Student-t(5) right. Source: author's calculations from the mutual engine; illustrative parameters, calibrated to no bank.

The reserve is paid in at entry because a reserve built from 1 percent by assessments never reaches the target in the median ten-year path (1.71 percent at year ten). After a loss year the mutual restores a shortfall by special assessment up to the unused cap and then by a replenishment assessment in equal annual installments, and with a three-year replenishment the reserve is below target at a year end about 7 percent of the time under the Gaussian copula at a correlation of 0.20. After a 2023-like cluster drawing 0.84 percent of covered excess, the Gaussian shortfall in the second year is 1.23 percent unrestored against 0.99 in the first year, and 0.93 fully restored.

5.6 Where the challenger wins, and where neither does

The challenger wins outright if the capital-contribution question resolves against the mutual, and it wins for a depositor who wants par in every state of the world, for whom the single-name vehicle at 45.74 is the only design on the list that never falls short. Beyond those cases the answer changes with the depositor and the bank. A public-unit depositor passes S1 on the letter of credit, the cheapest route that keeps the dollar lendable, with the super-lien's cost on the fund. A bank whose covered excess fits under the reciprocal allowance, with a depositor whose policy will take the network's many bank names, has reciprocal placement at 12.5 plus the uplift, and loses it only when its capital category or its rating slips.

For the settlement-clock problem none of the priced routes qualifies under the strict reading, because the facility is open on S4, the joint account on S1 and the one-ledger structure on both; an unpriced route on FedNow or RTP rails might qualify (Section 6.14). With the open items resolved the facility wins at 10 to 34 basis points (base 22.0) with the joint account as challenger at 25.35, where the facility's Sunday settlement cash is Reserve Bank credit against loans already pledged at

the Reserve Bank, and the joint account wins at every share where that cash must be new reserves or bought securities (Section 4.9). Applying S2 as well leaves nothing under either reading, since both liquidity products lend 0.80 of the balance at a 20 percent share and pass only at a tenth or less, and the one-ledger structure lends 0.77. What an uninsured tokenized balance, the one on both clocks, should be charged, rule or no rule, is Section 7.3.

5.7 The base layer as an option

The base layer of Section 5.2 is outside the model, and the manual has so far treated it as fixed, whatever the depositor's policy and the bank's allowance admit; a bank can instead run it as an option across the cycle. The network fee is the bank's to pass or absorb, since the home bank sets the rate the swept balance earns (12 CFR 337.6(e)(2)(i); Ryfe and Saretto 2023), so in a calm period a bank can charge the fee, explicitly or as a lower rate on swept balances, and pay more on the unswept account. One bank was reported in April 2023 to charge no fee but to pay a lower rate on swept balances (Cross 2023), though that was in the stress and not before it. Only the depositors who care about insurance then stay swept, and the bank keeps a larger pool of uninsured balances at a lower all-in cost. In stress it absorbs the fee and sweeps hard. That is a timing option on the base layer, and because no study I have found tests cyclical fee pricing, what follows is my own reasoning rather than a finding.

The option is real only for a bank that set it up in calm. Kim, Kundu and Purnanandam (2024) report industry accounts that joining a network takes at least three to six months, and they find that only 3.3 percent of banks outside a network at the start of 2022 had joined by the end of March 2023, so in the runs it was in effect existing members that could sweep. Lu and Scida (2026), working from Fedwire data, find that banks already in a network had steadier total funding through those runs. Reciprocal deposits rose from about $157 billion at the end of 2022 to over $300 billion by the middle of 2023 (Ryfe and Saretto 2023). The largest network is reported to have added more than 125 banks after the March failures (Cross 2023), so the product a bank should want is pre-enrollment, papers signed and balances not yet swept.

Even for a bank that enrolled in time, the option is weakest where it is needed. A member below well capitalized or at CAMELS 4 or 5 may not grow past its special cap (Section 8.2), and the cap's four-quarter window is anchored at the first finding that the bank is no longer rated 1 or 2 or no longer well capitalized, so that a bank downgraded from 2 to 3 and years later to 4 has its cap set on the holdings before the first downgrade (FDIC 2026a, the example in footnote 15). Placement is end-of-day while a run moves in hours (Section 1.1), and the switch is visible, since a bank that charged for the sweep in calm and waives the fee in stress, or calls its depositors to ask them to sweep, has told them what it fears, and the campaign can signal the stress it answers.

Where the option does work, it sharpens the S4 reading of the base layer. The assessment base is average consolidated total assets less average tangible equity (12 CFR 327.5(a)), not deposits, so a one-for-one reciprocal swap leaves the base unchanged while the fund's insured exposure rises. The brokered-deposit adjustment, which reciprocal deposits within the 337.6(e) exception do not enter,

is taken from quarter-end Call Reports (12 CFR 327.16(e)(3)), and the invoice for a quarter is paid in the quarter after (12 CFR 327.3(b)). A balance swept within the exception in the stress quarter is therefore covered from the day it is placed and adds nothing to the bank's assessment, so the fund writes the timing option at no marginal charge. At a large bank the sweep can lower the rate, since the large-bank scorecard counts in-cap reciprocal deposits as core deposits and excludes uninsured non-brokered time deposits (12 CFR Part 327, Appendix A to Subpart A), so the effect on the assessment is zero or negative, never positive. Reciprocal deposits above the exception can enter the charge, but only through the quarter-end Call Report and a bill a quarter later. That changes no verdict in the price list, since the base layer is outside the model, but it turns what Section 4.3 called a priced public cover into a cover priced on average and not at the margin. Huberdeau-Reid and Pennacchi (2025) report, in a conference poster, that small and midsize banks that were riskier at the end of 2022 were more likely to gain insured deposits through reciprocal and sweep deposits in each quarter of 2023, which they read as adverse selection against the fund.

Against that, the bank's profit from the option is small. The calm-period saving is at most the fee on the balances that leave the network, about 12.5 basis points (Chain Bridge Bancorp 2026), the level the network's operator itself quoted in 2013 for banks meeting its minimum use thresholds (Battey 2013). At the manual's $5 billion bank, with covered excess of $0.90 billion (Section 7.1), that is at most about $1.1 million a year if all of it were taken out of the network. What the saving buys is a bet that the bank can re-sweep before the run, and losing the bet costs the retention value of Section 3.6 and the replacement spread of Section 3.7 on every dollar that leaves. The risk that the 21 basis points of Table 2 price does not go away either; it sits with the depositor in calm and moves to the fund in stress.

The same option, exercised by members, makes the champion's base pro-cyclical. Members that can still grow reciprocal deposits move excess out of the mutual in stress, members that cannot stay in, and because the special assessment is levied on the survivors' covered excess, survivors that swept shrink the call (the 0.42 percent of Section 5.2) in the year it is needed. The design answer is the one the vehicle already uses against stuffing during a run (Section 8.3), which is to measure the covered excess on which the annual and the special assessments are levied on a trailing average, with no refund for exits inside the year; I adopt it as a design rule and leave it unpriced, so the engine's numbers do not change. The same rule governs the capital contribution: each member's is measured on the same trailing average and returned only on exit, after a year's notice, from a reserve at target after that year's losses, because a year-end rebate on current covered excess would return capital to the members that swept out, in the year after a stress; that too is unpriced.

Who gains from the option depends on the caps, and the binding ones are not the bank's, since at a 20 percent covered share the allowance does not bind at either illustrative bank (42.2 and 32.4 percent of liabilities, Section 7.1). The limits that bind are per depositor, where the largest network now publishes no dollar figure (IntraFi n.d.); one bank gave $150 million in March 2023 as the network's maximum as then posted on its website, and another sets a limit of $250 million per client across products in 2026 (Axos Bank 2023; Dime Community Bank 2026). A middle-market

depositor below those limits can be covered whole but has little bargaining power, so in calm the fee falls mostly on it, whereas a very large corporate is covered only in part, its investment policy may refuse hundreds of bank names, and it has the government fund as its outside option. In stress I would expect scarce capacity to go to the largest relationships first, so that the largest depositors gain on priority and bargaining and the middle market on completeness of cover, an incidence I infer and have not seen measured.

What the models did here. The mutual was designed in stage 2 in a structured interview of four rounds, each a set of decisions I made, and built twice. The first build showed that the design as briefed did not work: the stated exhaustion point, a cluster of ten members, sat inside the mutual's own attachment at 150 members, so the tail layer was empty. A reserve built from 1 percent by assessment never reached its target in ten years, and mutual frictions of 10 basis points, carried over from the single-vehicle convention, were a third of the price and far above the two live excess insurers. The engine-building model flagged all three rather than building around them; I decided each, and the second build is the one of record. A draft written on the second build then inherited one sentence from the first ("saves nothing under fat tails"), which only the fact check against the results file caught. Stage 4 added the deciding question in its national-bank form, which stage 2 had asked only for a Texas state bank, and the sized vehicle variant, so that the two champions are compared at the same standard.

Part 3. The settlement clock

Chapter 6. When the deposit never sleeps

6.1 The mismatch, restated

The facility of route 4.7 is built for the window in Figure 11, Friday's Fedwire close to Sunday's opening, during which a bank receiving a transfer of tokenized deposits holds a claim on the sending bank that FedNow could settle only as a credit transfer of up to $10 million or a liquidity management transfer in support of one. A larger claim waits for the Fedwire Funds Service, and the sending bank's funding side gives no help until Monday, for the reasons Chapter 1 gives. The question this chapter takes up is what a private arrangement could do before the public rail is extended, for balances that are large, between banks that would rather not extend each other unsecured credit, and in a weekend during which one of them may be closed.

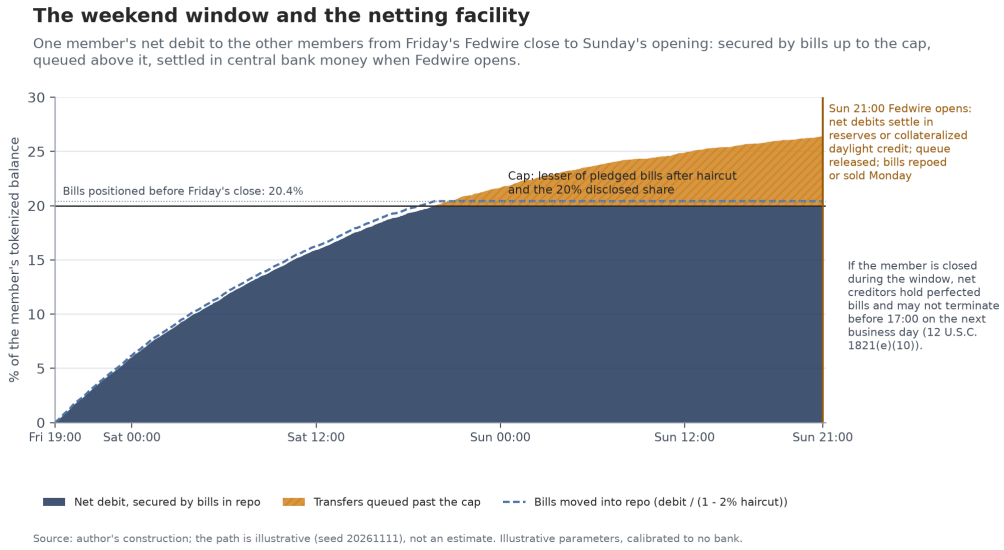


Figure 11. The weekend window and the netting facility: one member's net debit to the other members from Friday's Fedwire close to Sunday's opening, secured by bills up to the cap, queued above it, settled in central bank money when Fedwire opens. Source: author's construction; the path is illustrative (seed 20261111), not an estimate; illustrative parameters, calibrated to no bank.

6.2 The facility, and why it is a repo

The facility is run by a consortium of banks, because the problem is between banks and the asset that solves it sits on bank balance sheets. Members sign standard repurchase agreements with each other and a multilateral netting contract, appoint one calculation and collateral agent and one tri-party custodian, and no new entity holds a balance sheet; the bank-owned clearing-house model would be the form at scale. The other form is a prefunded pool, cash each member pays into an account before Friday's close and draws on through the weekend. A pool needs no entity of its own to hold it, since a Reserve Bank joint account is held by the eligible member banks themselves, an

operator that is not a bank may run it, and the live instant-payment network settles against one, so the prefunded form exists and is priced here as the ninth route. What separates the pool from the facility is the price, since cash in a joint account earns the rate on reserve balances where a positioned bill keeps the bill's yield, 23 basis points more on 3 November 2026, so at the same share the pool costs more above a share of 5.4 percent (Section 4.9); both commit their slice every weekend whether or not anyone draws. Nor do I claim that the collateral form's bills stay eligible as high-quality liquid assets until a debit arises; whether a positioned slice is encumbered before a draw turns on the facility's control, substitution and withdrawal terms, an open item of Section 8.4. Before Friday's close each member positions a slice of its Treasury bills, maturities under one year, at the custodian. During the window the agent records every transfer of tokenized deposits between members as a bilateral claim and keeps a running multilateral net position for each, and when a member's net debit arises the custodian moves bills of the member's into a repo in favor of the agent for the benefit of the net creditors, so that every net debit is secured at all times by bills worth, after a haircut, at least the debit. At 9:00 p.m. on Sunday the agent submits the net positions for settlement in central bank money, and the design has to say where a net debtor's cash comes from. The pledged bills are not reserves, and the Fedwire Securities Service through which they could be repoed or sold opens on Monday at 8:30 a.m., so every net debtor, not only a failed one, pays at the Sunday opening from its own reserve balance or from a collateralized Reserve Bank daylight overdraft, free of charge under the payment system risk policy (50 basis points a year if uncollateralized). It repays the overdraft on Monday from a delivery-versus-payment sale or a bilateral repo of the bills after the 8:30 a.m. securities opening (Board of Governors 2023d). Each member therefore needs, at the Sunday opening, reserves or collateralized intraday credit capacity at least equal to its cap, within its Reserve Bank net debit cap, which pledging collateral does not raise unless the Reserve Bank has granted collateralized capacity above it and which its other payment activity shares, from collateral pledged at the Reserve Bank and separate from the bills pledged to the facility (Board of Governors 2023d). The fee is negligible at either setting, 0.0000 basis points a year collateralized and 0.2919 uncollateralized on the expected Sunday debit, on the policy's 360-day year; the requirement is not, since it holds the lendable share at 0.80 only if the Reserve Bank collateral is loans the bank already holds and takes it to 0.60 if the bank holds reserves or buys securities to pledge. The repos unwind on Monday and the bills return.

The term that carries the design is the cap, under which each member's net debit is limited to the lesser of its pledged bills after the haircut and a disclosed share of its tokenized deposit balances, a share the member chooses within a system ceiling and publishes to its depositors; transfers that would take the member past its cap are queued and released at the Sunday opening, in the order received. The haircut is 2 percent on bills, for a horizon of three days from a Friday pledge to a Monday sale, four if the member is closed and delivery waits for Tuesday's securities opening (Section 6.9), and both the share and the haircut are illustrative inputs, calibrated to no bank, whose only job here is to make the arithmetic concrete. Cover is full, with no loss-sharing rule, so no member is exposed to another's credit beyond the chance that a bill portfolio loses more than 2

percent between Friday and Monday, which falls on the net creditors; the alternative of thinner cover with survivors paying would reintroduce the member failure risk that Chapter 2 prices at about 21 basis points, and the point of the collateral is to make that term zero.

The repo form is chosen for what happens if a member is closed over the weekend. A repurchase agreement is a qualified financial contract under the Federal Deposit Insurance Act, and the Act's safe harbor lets a counterparty net and liquidate its collateral notwithstanding the receivership (12 U.S.C. 1821(e)(8), (9) and (10)), subject to a stay until 5:00 p.m. Eastern time on the business day after the appointment, during which the FDIC may transfer the whole book to one institution, a bridge bank included, with the repo continuing. The stay has two parts: 1821(e)(10)(B) stays the counterparty's rights to terminate, liquidate or net the contract, and 1821(e)(8)(G)(ii) suspends the payment and delivery obligations due under the contract itself, so neither suspends payments in general. For a member closed on a Friday evening the net creditors therefore hold perfected collateral and wait. That is cover against loss, which is what the facility promises, and not cover against Monday-morning liquidity; a net creditor that needs the cash at Sunday's opening borrows against its own bills in the meantime, and Section 6.9 sizes the wait on the legal and the operational clock. The netting itself needs a contract, because bilateral repos between pairs of members do not discharge a chain in which A owes B, B owes C and C owes A, which nets to zero on the agent's screen while three contracts still owe three payments; the facility needs a multilateral netting contract, or clearing rules that provide for the netting, naming the agent's collateral beneficiaries and the default treatment. An all-bank consortium can fit the clearing-organization definition of 12 U.S.C. 4402, and 4404 then enforces the netting in accordance with that contract, still subject to the 1821(e) stay, which names those sections expressly; whether a given consortium qualifies is an opinion for counsel, and Chapter 10 asks for it. The reading is a practitioner's and the arrangement would need counsel's.

A net debtor pays each net creditor interest on the claim for the hours it is outstanding at an overnight secured rate, since the creditor has taken on a deposit liability and holds a secured claim in its place, and the charge nets to zero across the membership so that it belongs to the economics of a weekend rather than to the price of the facility. Operating balances that pay no interest during the week pay none at the weekend either; the charge is between banks.

6.3 The price in four terms, and the share that passes

The four terms of Chapter 2 fit the facility with the terms rearranged. Credit is 0 to 1 basis points, because every net debit is fully collateralized and what remains is the haircut-breach risk on the net creditors. Access, 2 to 5, is the cost of belonging, the agent's and custodian's fees and the legal and operational work. Standing capacity, 5 to 20, is the main term, the cost of holding a stock of bills sized to the cap rather than loans or less liquid assets, written as an encumbrance spread of 25 to 100 basis points between what the bank earns on a bill and what it would earn on the asset the bill displaces, times the 20 percent share the member stands ready to settle. Whether the slice counts as encumbered before a draw depends on the facility's terms (Section 6.2); either way, a stock the bank

must keep available every weekend is a stock it cannot plan to lend, and the price does not include the reserves or Reserve Bank collateral the Sunday settlement needs. The 22.00 base, and the facility's place ahead of the joint account, hold where that Sunday cash is Reserve Bank credit against loans the bank already holds there, within its net debit cap headroom. Where it must be new reserves or newly bought securities, the facility costs 39.10 or 34.50 at a 20 percent share, lends 0.60, and the joint account at 25.35 is cheaper at every share (Section 4.9; the routes engine's results, Section 7). Frictions, 3 to 8, are the haircut's funding, the cost of queueing to the depositors who meet the cap, and the documentation. The sum is 10 to 34 basis points a year on the whole tokenized balance, or 50 to 170 per always-on tier dollar. A bank writing a committed facility under the liquidity and capital rules as Chapter 3 reads them faces a cost floor of 55.4 to 117.0 on the undrawn amount, which is one reason the facility is a collateral arrangement among members and not a line from any one of them. The 10 to 34 is a price for a different thing from the 38 to 46 basis points of Chapter 3, since it makes a fifth of the balance movable outside Fedwire hours and does nothing for the balance if the bank fails except deliver the net creditors their collateral.

The standing-capacity term scales with the always-on share and the others do not (Table 13), which decides the facility's fate under S2. At a 5 or 10 percent share the balance stays nine tenths lendable and the facility passes, provided the Sunday settlement runs on Reserve Bank credit against loans the bank already holds (Section 6.2), at 20 percent it does not, and at 50 percent the member has turned half its deposit into a bill portfolio.

Always-on share	Price, whole balance	Per tier dollar	Lendable share	S2
5%	6.3 to 19.0	125.0 to 380.0	0.95	pass
7%	6.8 to 21.0	96.4 to 300.0	0.93	pass
10%	7.5 to 24.0	75.0 to 240.0	0.90	pass
20%	10.0 to 34.0	50.0 to 170.0	0.80	fail
35%	13.8 to 49.0	39.3 to 140.0	0.65	fail
50%	17.5 to 64.0	35.0 to 128.0	0.50	fail

Table 13. The facility by always-on share, basis points a year at a 2 percent haircut; the standing-capacity term scales with the share and the other three do not; bills positioned are 20.41 percent of the balance at a 20 percent share. Source: the routes engine's results, Section 7.

6.4 How often the cap binds, alone and together

A cap that is never reached is a disclosure, and a cap reached every other weekend is a product failure, so the frequency with which a member's weekend net debit reaches its cap is the number a depositor would want before accepting the terms. The record of weekend outflows of tokenized deposits is too short to say, so every figure in this section is illustrative in the strict sense. The quantity modeled is each member's peak net debit during the closed window, the largest net debit it runs at any point in the weekend to all counterparties, members and non-members alike, because a

cap limits a running position and not where the position ends: a bank that sent a quarter of its balance early and received a quarter late would finish the weekend flat and still have queued payments against a cap of a fifth of the balance in between. That peak, as a share of the always-on tier, is taken as lognormal with a median of 25 percent and a dispersion of 0.8, which at a 20 percent tier is a median of 5 percent of the tokenized balance under a cap of four times the median; payment paths within the weekend are not modeled. At those inputs a member's cap binds in 4.156 percent of weekends, 2.16 a year on 52 independent weekends; at a dispersion of 0.5 it binds in 0.278 percent of weekends and at 1.2 in 12.399 percent. Those per-member figures, and the backing and median boundary of Section 6.5 that follow from them, are the headline of this chapter, conditional on the assumed distribution.

Outflows at many banks on the same weekend are driven by the same news, so the question has the shape of the cluster question of Chapter 3, and the settlement engine answers it the same way. A one-factor copula runs over 20 members of equal size, under a Gaussian factor and a Student-t(5) factor at pairwise latent correlations of 0 to 0.50, all illustrative, and every member is drawn as a net debtor. In a closed membership the signed net positions would sum to zero, so what the engine describes is an open debtor cohort, twenty banks that are net senders on the same weekend to receivers outside the modeled set, and every multi-member statistic in this chapter, the joint counts, the queues, the system cap ladder and the cohort-average debit, is a statement about that cohort and not a result for the closed membership of Section 6.2. Each member keeps the lognormal marginal above, so the expected number binding on a weekend is 0.831 whatever the factor; what the factor moves is the joint count and the size of the queue (Table 14, Figure 12). With independent members the cap binds somewhere in the cohort on 57.2 percent of weekends and at a quarter of the members, five of the twenty, on 0.114 percent. At a correlation of 0.30 the Gaussian factor lowers the first figure to 39.6 percent and raises the second to 3.36 percent, and the Student-t factor puts a quarter of the cohort at its caps on 5.10 percent of weekends, about one in twenty. The queue is where the factor shows most. On the worst weekend of a year, at the 99th percentile across years, the amount queued above the caps is about a third of the cohort's always-on tier with independent members, three quarters of it under the Gaussian factor at 0.30, and nearly twice the tier under the Student-t factor.

Copula	Correlation	At least one of 20 binds	At least five (a quarter) bind	Queued on the worst weekend of a year, 99th percentile
Gaussian	0.00	57.2%	0.114%	36.37%
Gaussian	0.15	47.7%	1.51%	47.90%
Gaussian	0.30	39.6%	3.36%	74.78%
Gaussian	0.50	30.0%	5.16%	128.97%
Student-t(5)	0.00	41.5%	2.76%	106.80%
Student-t(5)	0.15	36.0%	4.12%	142.70%
Student-t(5)	0.30	30.8%	5.10%	179.29%
Student-t(5)	0.50	24.1%	5.89%	225.75%

Table 14. Same-weekend binding in an open debtor cohort of 20 equal members at a dispersion of 0.8, each drawn as a net debtor with its receivers outside the modeled set: the probability that at least one, and that at least a quarter, reach their caps on one weekend, and the peak net debit queued above the caps on the worst weekend of a year at the 99th percentile across years, as a share of the cohort's always-on tier; the expected number binding is 0.831 in every row. Source: the settlement engine's results, Section 3.

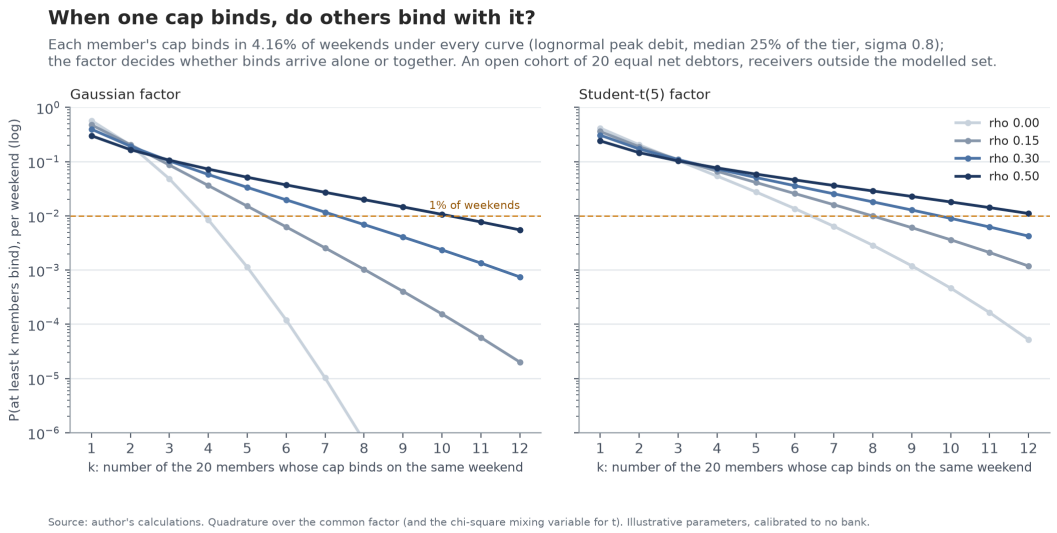


Figure 12. Probability that at least k of 20 members of an open debtor cohort reach their caps on the same weekend, by correlation, under a one-factor Gaussian and a one-factor Student-t(5) factor; each member's own rate is 4.156 percent of weekends under either. Source: author's calculations from the settlement engine; illustrative parameters, calibrated to no bank.

6.5 The cap ladder

The question a depositor would put to Table 14 is what backing it would take for the cap to bind rarely. The engine holds the weekend peak net debit fixed in balance terms, a median of 5 percent at a dispersion of 0.8, and moves the backing, the always-on share with the cap equal to it, until a bind-

frequency target is met. Each backing is priced with the routes engine's terms, and the lendable share is one minus the share. For a member's own cap to bind in no more than 1 percent of weekends the cap has to be 6.43 times the median peak net debit rather than four: a backing of 32.2 percent of the balance, priced at 13.0 to 46.2 basis points, with a lendable share of 0.678. For once in ten years, one weekend in 520, the multiple is 10.10, the backing 50.5 percent and the lendable share 0.495. Holding the whole cohort to the same targets costs more: for the chance that any of the 20 members binds to be at most 1 percent of weekends, the backing is 68.2 percent of the balance under the Gaussian factor at 0.30, and up to 98.7 percent for the ten-year target, both statements about the open debtor cohort. Of the 34 rows the engine prices across the two targets, three criteria, two factors and four correlations, none passes S2.

The screen allows a backing of at most 10 percent of the balance, a cap of twice the median at this peak net debit, and at that cap a member's own cap binds in 19.3 percent of weekends, one in five. Turning the ladder around gives the condition a bank would need to observe before the facility could pass the lendable screen and bind rarely at once. At a dispersion of 0.8 the two hold together only if the median weekend peak net debit is at most 1.56 percent of the tokenized balance for the 1 percent target, and 0.99 percent for the ten-year target, under a third of the 5 percent the routes engine assumes. At a dispersion of 0.5 the ceilings are 3.12 and 2.36 percent, and at 1.2 they are 0.61 and 0.31; at those two dispersions a member's cap at a 10 percent backing binds in 8.28 and 28.2 percent of weekends (the settlement engine's results, Section 3, Table A4). The lendable screen and a rare cap cannot both hold at the peak net debit this manual assumes. Whether they can hold at a real bank's peak net debit turns on that bank's own weekend record, its median read with its dispersion, which is why Chapter 10 puts it to any bank that can observe one.

Table 15 shows how narrow the margin is. Holding the cap at the 10 percent of the balance that S2 allows and moving the median peak net debit from 1 to 5 percent of the balance at each dispersion, the verdict at a dispersion of 0.8 turns between a median of 1.5 percent, where a 1 percent bind needs a cap of 9.65 percent and the facility passes, and 2 percent, where it needs 12.86 and fails; in the full grid of the results file, at 1.2 every median fails and at 0.5 every median up to 3 percent passes. The conclusion that the facility cannot be both nine tenths lendable and rarely capped therefore rests on where a number sits that no bank reports, since no public source carries participant-level peak debits on tokenized balances and the aggregate instant-payment statistics are gross volumes without signed paths or a tokenized-balance denominator, which is why the median is the first thing a pilot would have to observe.

Median peak net debit, share of the balance	Bind probability at a 10% cap	Cap for a 1% per-member bind	Lendable share at that cap	S2
1.0%	0.200%	6.43%	0.936	pass
1.5%	0.886%	9.65%	0.904	pass
2.0%	2.21%	12.86%	0.871	fail
3.0%	6.62%	19.29%	0.807	fail
5.0%	19.3%	32.15%	0.678	fail

Table 15. Median sensitivity, per member, at a dispersion of 0.8: the bind probability per weekend at a cap of 10 percent of the balance, the largest backing S2 allows, and the cap a 1 percent per-member bind needs, with the lendable share and the S2 verdict at that cap; the full grid of three dispersions by five medians is in the results file. Source: the settlement engine's results, Section 3, Table A5.

6.6 The closed-window calendar

The weekend is not the only window, and the facility has to be sized for the longest one. The Fedwire Funds Service opens at 9:00 p.m. Eastern time on the calendar day before each business day, holidays included, and closes at 7:00 p.m., so the closed window between two consecutive business days is 24 hours for each calendar day from one to the next, less the 22 the service is open; the Reserve Banks' weekday closings are the holidays (Federal Reserve Financial Services 2026a, 2026f). In 2026 that gives 251 business days and 3,238 closed hours: 196 weekday evening gaps of two hours, 45 normal weekends of 50 hours, seven three-day weekends of 74 hours and three midweek holidays of 26 hours, with no window longer than 74 hours (Figure 13). Those are nominal schedule hours; the two weekends a year that span a daylight-saving change run 49 and 51 elapsed hours (6 to 8 March and 30 October to 1 November in 2026), the annual total is unchanged, and of the 36 printed calendar figures only one moves when recomputed on elapsed hours, the 2026 expected binds a year under square-root scaling, 2.41 nominal against 2.42 elapsed. How the peak net debit scales with the hours is not known, so the engine carries two illustrative scalings, the median peak net debit proportional to the hours of the window and proportional to their square root.

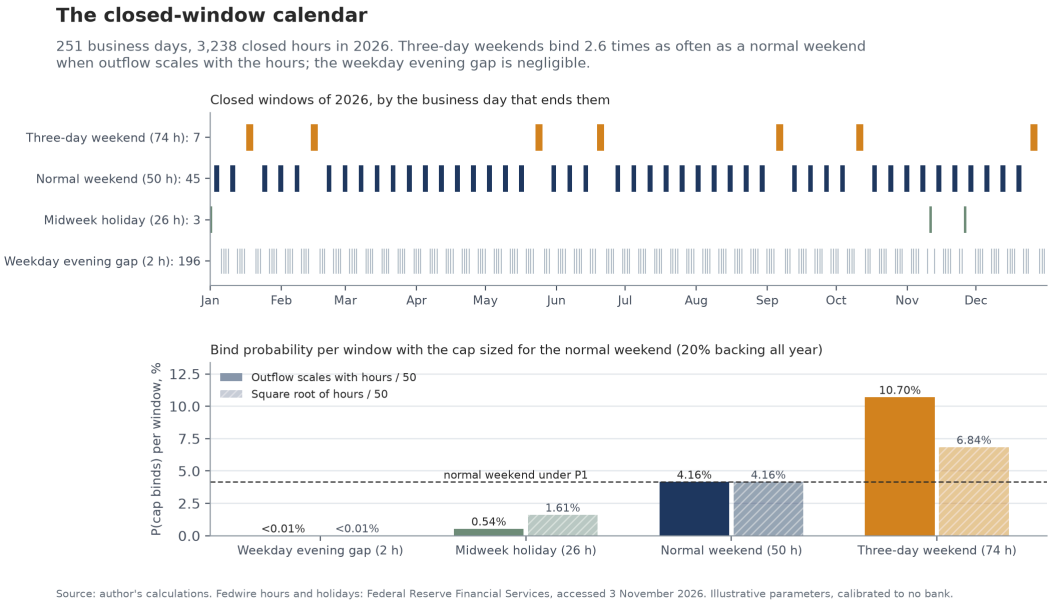


Figure 13. The closed windows of 2026 on a strip, with the bind probability for each window type at a 20 percent backing under linear and square-root scaling of the peak net debit with the hours. Source: author's calculations from the settlement engine; Fedwire hours and Reserve Bank holidays from Federal Reserve Financial Services (2026a, 2026f); illustrative parameters, calibrated to no bank.

If the cap is sized for the normal weekend and held all year at a 20 percent backing, a three-day weekend binds 2.57 times as often as a normal one under linear scaling, 10.7 percent against 4.156, and 1.65 times as often under square-root scaling; a midweek holiday binds 0.13 times as often, and the weekday evening gap almost never. The calendar adds 0.47 expected binds a year to the routes engine's 2.16, nearly all of it from the seven three-day weekends.

The Board's notice of 17 November 2025 expands Fedwire Funds and the National Settlement Service to Sunday through Friday, weekday holidays included, with Saturday closed, to be implemented in 2028 or 2029 and voluntary for participants; it is announced and not yet in effect, and the facility gains the shorter window only if every member takes part (Board of Governors 2025a, 2025b). Under it the 2026 calendar has 261 two-hour gaps and 52 Saturday closures of 26 hours, and the backing that holds that normal-weekend bind rate on every window falls from 29.6 percent under today's calendar to 10.4 under linear scaling (14.4 under square-root), priced at 7.6 to 24.4 basis points. The standing-capacity term falls by 48 percent against the 20 percent backing on the hours alone, yet the lendable share at the linear backing is 0.896, which misses S2 by 0.004, and by 0.044 under square-root scaling. Seven-day operation, a hypothetical the Board has said it stands ready to propose no sooner than two years after the six-day schedule, leaves only the 365 two-hour gaps. Held to 2.16 binds a year, the backing is then 1.5 percent of the balance (7.5 under square-root), priced at 5.4 to 15.5 basis points, of which access and frictions are nearly all, and the lendable share of 0.985 passes S2 (0.925 under square-root). That pass is conditional on the Sunday cash source of Section 6.2, because seven-day Fedwire Funds alone does not reset bill-backed capacity while Fedwire Securities and the repo market keep weekday hours: it holds if Reserve Bank

credit against loans the bank already holds carries each weekend debit from the evening it settles to Monday's securities opening, which is overnight credit over Saturday and Sunday and not the intraday credit of Section 6.2, and fails in every row if the bank holds reserves against the outflow instead, at 0.785 under linear scaling and 0.725 under square-root with the reserves sized over the 50-hour weekend (0.724 and 0.697 sized over the 65.25 hours from Friday's securities cutoff to Monday's securities opening), because the reserves cost the lendable share the thinner bill backing had saved.

6.7 The liquidity buffer, as a sensitivity

Everything above counts a bill positioned for the facility as a dollar the bank cannot lend, which is the routes engine's convention and the base case of this manual. A bank that holds a liquid-asset buffer anyway could draw the facility's backing from it, and the engine tests what that does. It does not change the lendable share, because a deposit dollar that funds a bill does so whether the bill was bought for the facility or sat in the buffer, so on the absolute convention S2 reads 0.800 at a 20 percent share whatever the buffer and fails; a reading of 0.900 with a 10 percent buffer and 1.000 with 20 percent measures instead one minus the lending the facility displaces on top of a buffer the bank holds anyway. That incremental displacement is reported under its own name: relative to a bank that already holds the buffer, the facility displaces 10 percent of the balance from lending with a 10 percent buffer and nothing with 20 percent, and the standing-capacity term on the covered part falls to a custody-and-positioning spread of 0 to 5 basis points in place of the encumbrance spread of 25 to 100, both author assumptions, so the price is 7.5 to 24.5 basis points with a 10 percent buffer and 5.0 to 15.0 with 20 percent, and at a buffer of zero the price and the lendable share are the routes engine's (Figure 14).

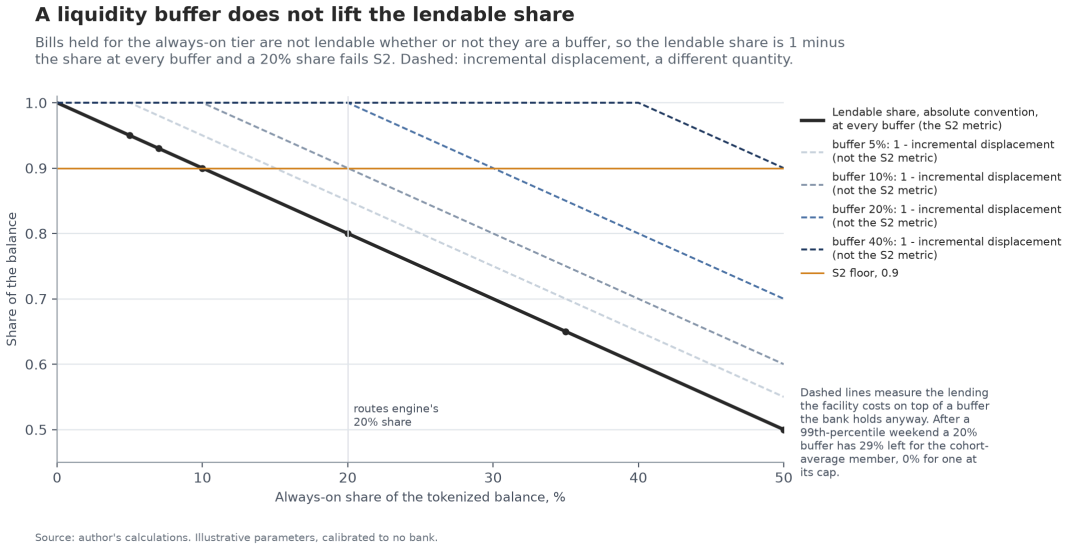


Figure 14. Lendable share against the always-on share on the absolute convention, one minus the share at every buffer, with the S2 line at 0.9, and, dashed, one minus the incremental lending displacement relative to a bank that already holds the buffer, which is not the S2 metric; a sensitivity, not the base case. Source: author's calculations from the settlement engine; illustrative parameters, calibrated to no bank.

A buffer is held for the bank's own outflows, and a weekend on which the facility has drawn on it is a weekend after which the bank meets Monday's outflows from what is left. On the 99th-percentile weekend of the open debtor cohort, the worst in a hundred rather than the worst of a year of Section 6.4 (Gaussian factor, correlation 0.30, 20 members), the cohort-average member's net debit is 70.8 percent of its tier, 14.17 percent of its balance, and a member at its cap has settled the whole tier. With a buffer of 5 or 10 percent nothing is left for Monday in either case; with a buffer of 20 percent the average member has 29.2 percent of its buffer left and the member at its cap has none; with 40 percent they have 64.6 and 50.0 percent. For a bank subject to the liquidity coverage ratio the rule adds its own bar, since an encumbered asset is not eligible as a high-quality liquid asset (12 CFR 249.22(b)(1)) and Section 8.4 lists the question whether a positioned slice is already encumbered before a draw. Counting the grossed-up bills, the 20.41 percent of Table 13, as non-lendable instead of the 20 percent share, the buffer at which a 20 percent share's incremental displacement falls to a tenth is 10.41 percent of the balance rather than 10.00, and the S2 boundary of Section 6.3 is a 9.8 percent share rather than 10. The manual keeps the routes engine's convention.

6.8 Non-members through a sponsoring member

A bank too small to carry the membership can reach the facility through a member that carries the non-member's net debit on its own cap, backed by bills the non-member pledges to the sponsor in a back-to-back repo. The non-member pays the facility's terms plus a sponsor margin of 2 to 6 basis points per always-on tier dollar and the sponsor's extra access cost of 1 to 3 on its whole balance, author assumptions. At a 20 percent share sponsorship adds 1.4 to 4.2 basis points on the whole balance, taking the non-member to 11.4 to 38.2; at a 5 percent share it adds 1.1 to 3.3, taking the non-member to 7.4 to 22.3, or 147 to 446 per tier dollar, because the access cost is fixed and the tier is small. What the sponsor bears, on my reading, is cap room rather than credit while the back-to-back repo is full. Every dollar of the non-member's net debit is a dollar the sponsor's own depositors cannot use that weekend, so a sponsor would want a sub-limit; its credit exposure is the haircut-breach risk on the pledged bills and the Monday gap of Section 6.9 if the non-member is closed. The precedent is the sponsored membership of the Government Securities Division of the Fixed Income Clearing Corporation, in which a sponsoring member brings firms into a limited form of membership and unconditionally guarantees their performance to the clearing house. An SEC order of 12 December 2025 approved a rule change under which a lien on collateral generally stands in for that guaranty for one class of trades, collateral-in-lieu sponsored general-collateral repo, and the service launched on 23 December 2025 with the first trade executed on the custodian's collateral platform; neither document says anything about weekend operation (U.S. Securities and Exchange Commission 2025; BNY 2025).

6.9 Monday cash when a member is closed

Section 6.2 says that the net creditors of a member closed over the weekend hold perfected collateral and wait, and the engine sizes the wait on two clocks. The counterparty to a repurchase agreement, a qualified financial contract, may not terminate, liquidate or net it solely because of the receiver's appointment until 5:00 p.m. Eastern time on the business day after it, a business day being any day other than a Saturday, a Sunday or a day on which the New York Stock Exchange or the Federal Reserve Bank of New York is closed (12 U.S.C. 1821(e)(10)(B)(i) and (e)(10)(D)). For a member closed on the last business day before a weekend the legal stay interval therefore runs from the scheduled Fedwire opening at which the net positions would have settled to the end of the stay, Sunday 9:00 p.m. to Monday 5:00 p.m., 20 hours. The end of the stay is not cash in hand, though, because 5:00 p.m. falls after the Fedwire Securities Service's 3:15 p.m. cutoff for secondary-market transfers, so a creditor that is free to liquidate at 5:00 p.m. cannot deliver the bills against payment through Fedwire Securities until the service opens on Tuesday at 8:30 a.m., unless the sale settles on the tri-party custodian's own books, which no source I have read describes for a weekend default; the operational collateral-to-cash interval is 35.5 hours (Federal Reserve Financial Services 2026g). A Monday holiday lengthens neither, because the opening moves with the holiday, the stay ends on the Tuesday and delivery waits for the Wednesday. A net creditor bridges the gap by borrowing against its own bills at the overnight secured rate of 4.3 percent plus 10 to 50 basis points, author assumptions, and a member is closed in a year with the 0.25 percent probability of Chapter 2, its net debit spread equally over 19 creditor banks of its own size, which in the open-cohort reading of Section 6.4 sit outside the modeled set.

With the closed member at its cap its net debit is 20 percent of its balance, 1.053 percent of each creditor's balance if spread evenly and the whole 20 percent at one creditor that received it all. The expected bridging cost is 0.00050 to 0.00055 basis points a year on the whole balance over the legal interval and 0.00089 to 0.00097 over the operational one, before the usage charge the claim earns while it waits; per event each creditor, with the closed member at its cap, bears 0.0106 to 0.0115 basis points of its balance over 20 hours and 0.0188 to 0.0205 over 35.5. The arithmetic assumes a bridge is there to borrow from, which is a Sunday-evening question of the same kind as the net debtor's in Section 6.2. The facility's collateral sits with a tri-party custodian, and no primary source I have read describes the mechanics of re-use by a cash lender at such a custodian beyond the Office of Financial Research's description of re-use as legally feasible within the tri-party market and restricted to certain participants (Office of Financial Research 2025, 2016). The stay bars termination without saying what a holder may do with collateral it already holds, so the manual counts the gap in full and leaves re-use to counsel.

6.10 The deposit tranched by settlement time

The cap of the facility is a term the facility imposes, and it can instead be a term the deposit carries. A tokenized deposit contract could offer the holder two tiers of the same balance, an always-on tier redeemable to another bank at any hour up to a stated amount, backed one for one by bills the issuing bank has positioned at the facility, and a business-hours tier redeemable at the next Fedwire

opening like any deposit today. The holder chooses the split when the account is opened and can change it on notice, and pays for the always-on tier through the rate, the earnings credit or a fee set from the facility price, 50 to 170 basis points on the tier dollar at the illustrative inputs, so that Table 13 becomes the fee schedule and the business-hours tier pays what an operating balance pays now. Writing the limit into the deposit in advance tells the holder what it bought and lets the bank price it. The depositor bears the queue it chose and pays for the speed it uses, the bank bears the encumbrance on the always-on share and nothing on the rest, and the standing-capacity term that Section 6.3 kept inside the bank's cost of funds becomes a line on the depositor's statement. In this form the mismatch is a priced feature, the cap is a term the depositor signed, and software acting for the depositor can set the split from the depositor's own payment calendar rather than from a bank's guess.

Against that, a bank which sells too much always-on tier has built a narrow bank inside itself, since every dollar in that tier is matched by a bill it cannot lend, which is why Table 13 fails S2 at a fifth of the balance and passes at a tenth. A depositor who has paid for the right to leave at 2 a.m. is likely, I think, to use it. And under stress the split itself gives way, because a depositor who moved its whole balance into the always-on tier on the Thursday has told the bank what it intends, and the notice period on changes to the split is the only brake, which makes that period a supervisory variable as much as a commercial one.

6.11 One ledger

The one-ledger structure of route 4.8 is the always-on module and the champion on one membership. The discipline that holds it together is that the two tranches share a ledger and never a reserve, since a stack that counted the same capital behind both would be thinner than it looked in exactly the state of the world it exists for. It saves the second agent, the second custodian and the second set of legal work. The routes engine reads the smaller structure's access term as the mutual's access, 0.37 basis points, and saves half of it plus 1.0 basis point of frictions, 1.19 in all, or 2.0 to 3.5 if the facility's own membership access is the term. None of it changes the ranking, because the one-ledger structure contains the facility and is dearer than it at every point of the range. Nor does it rescue the lendable share, which reaches 0.9 only at an always-on share of 7 percent or less. The governance cost is the one to watch, since a bank-owned clearing house that holds its members' collateral and writes protection on them has the incentive conflict Chapter 3 describes for a bank standby letter of credit at the scale of the system, and Chapter 8 says what it would need.

6.12 A weekend secured rate

The usage charge is a rate the net debtor pays the net creditor for hours in which no market is quoting one, and the facility can publish it. From its own repos the agent observes, every weekend, the rate at which members financed net debits against bills for 26 or 50 hours, and a volume-weighted figure, published on Monday, is an observable price of secured liquidity outside Fedwire hours, a weekend basis over the overnight secured rate that today is hidden in earnings credits, relationship pricing and the lines banks extend each other. It gives a funds transfer pricing desk a

curve point for always-on balances. Chapter 3's argument that the clearing cost belongs in transfer pricing whether or not anyone charges it applies here with more force, since this rate is paid, and it lets a depositor compare the fee for the always-on tier with what the window actually cost. A thin market makes a noisy rate, however. A published rate from a small membership can be moved by one member's weekend, so it needs a fallback, the overnight secured rate plus a fixed spread for any weekend on which volume falls below a floor, and it should stay an information product until the volume says otherwise, since if anything is indexed to it beyond the facility's own charges the thinness becomes someone else's problem.

6.13 The extensions, caveats first

Settling in the bills themselves is the step beyond netting. The sending bank delivers tokenized Treasury bills of equal value to the receiving bank outright, delivery against payment on the same ledger, so that no claim between banks exists at any point and the bill itself is the weekend money. Tokenized Treasury products that transfer between approved holders at any hour already exist (BlackRock 2024), with roughly $15.9 billion outstanding in July 2026 (rwa.xyz 2026), so the mechanics are not speculative, though a market of that size is not depth to rely on in a cluster year. The form pays for its finality, since the run is then settled in real time and the queue that netting makes visible never forms, the sending bank's liquidity ratio falls during the window with no opening at which to rebuild it, and a bill delivered at 2 a.m. is valued at a price nobody is quoting. I would offer outright delivery as the layer above netting, for members and balances that will not accept a capped claim at any price, and keep netting as the cheaper default.

A resolution protocol under which the agent notifies a member's primary regulator when its cap binds, and the FDIC can step a bridge bank into the member's position at Sunday's opening with the repos continuing, would turn a weekend run into one the resolution authority watches from its start, and only the FDIC and the supervisors can agree to it.

6.14 Routes not priced

In a private settlement bank, against the facility's rule that no new entity holds a balance sheet, a correspondent could settle the members' weekend transfers on its own books and commit its own cash until the public rail and the securities market reopen, supplying the Sunday-evening counterparty that Section 6.2 has the Reserve Bank's daylight credit supply. It adds the correspondent's credit, a concentration of the membership's weekend liquidity in one name and the capital and liquidity cost of the commitment, and an executable quotation and a default waterfall would rank it against 10 to 34 basis points, neither of which I have.

A route on FedNow, through which a transfer of up to $10 million already goes as a customer credit transfer at any hour (Chapter 1), or on the existing RTP joint account, with payments above $10 million split into messages under each network's $10 million limit, needs no new approval for a bank already on either network. It would therefore pass S1 as scored, the screen on which the new joint account of Section 4.9 is open; its S3 and S4 are unscored. It still needs reserves in the sending

bank's account, as the joint account does, since the message limit creates none, so it would carry a capacity cost of the same kind as the joint account's, but it is unpriced, no observed price exists for it, and whether either network's rules permit one payment to be split into several is unverified (Federal Reserve Financial Services 2026b; The Clearing House 2026). A restricted service that routes transfers under the limit live and defers larger ones to business hours within a weekend budget the customer selects is the same route without the split.

The seven-day pass of Section 6.6 rests on Reserve Bank credit that carries weekend debits to Monday's securities opening, which is overnight credit and not intraday, and a standing Reserve Bank weekend credit window would be the public form of it. The discount window's published access hours are Monday to Friday, 8:00 a.m. to the close of Fedwire Funds, and the Board's 2025 notice says only that it is exploring whether to expand discount window operating days with the Fedwire expansion, so the window is a scenario until adopted terms exist (Federal Reserve Bank of New York 2026; Board of Governors 2025b).

On 13 October 2026 a custodian bank completed a tri-party repo in which a two-year Treasury note, settled within minutes as a tokenized security on the bank's private ledger, was used as collateral in an overnight repo with the bank providing the cash; the cash leg is not described as tokenized, the operating-hours extension the release anticipates is aimed at European and Asian trading hours, and nothing in it speaks to weekend operation or default liquidation, so it shows that collateral can move faster, not that the Sunday settlement is solved (BNY 2026). Tokenized reserves, if a Reserve Bank ever offered them, would be the piece that mattered.

6.15 The objection, faced

The facility lets uninsured money leave a weak bank on a Saturday and lets the bank's Treasury bills leave with it, since every dollar of net debit is secured by bills the net creditors will keep if the bank is closed. A receivership opened on Monday therefore finds a bank with fewer liquid assets and the same insured deposits, and the deposit insurance fund bears more of the loss than it would have had the run waited for the opening. The facility also adds about 50 hours to the time a run has to work in, and in March 2023 the largest single-day outflows at the failing banks were about a fifth of deposits, so a weekend is not a small addition. The routes engine bounds the transfer to the fund at 1.38 basis points a year under the real-world measure and 4.15 under the pricing measure on the whole balance, assuming every failure follows a weekend in which the capped outflow left. That bound is why the facility's S4 is open rather than a pass.

As a description the objection is accurate, but a weekday run through Fedwire does the same thing with no cap at all. On a Thursday the uninsured depositors wire out, the bank funds the wires with Home Loan Bank advances secured by its best assets under a lien that outranks the receiver, and Friday's receivership finds the same thinner bank with nothing having limited the pace. The facility's cap is a disclosed speed limit, set by the member within a system ceiling and backed by collateral the member chose to position, and it binds first at the bank with the least pledged collateral relative to its tokenized balances, which is the bank that should be slowed; a supervisory

freeze on a member's net debit would likely be worse, since a freeze announced on a Saturday reads as a run signal. What the cap cannot do is prevent the queue from forming, and a queue is visible, which is what the resolution protocol above would use.

The public schedule changes the arithmetic of Section 6.6 and not the argument. Under the six-day schedule the window the cap guards is the 26-hour Saturday closure; under seven-day operation the Funds gap is two hours a day, but Fedwire Securities and the repo market still close for the weekend, so netting is a formality only if weekend Reserve Bank credit carries each debit to Monday (Section 6.6), and the facility's remaining use is as the ledger and the collateral rules that the tranched deposit and the one-ledger structure are built on. I expect the facility to be worth building for a window of 26 hours and to be worth keeping, as a settlement-time tranching and clearing arrangement, after the window closes.

6.16 The settlement-clock answer

Read strictly, none of the priced routes qualifies on the settlement clock, because the facility is open on the subsidy screen, the joint account is open on its Reserve Bank approval and the one-ledger structure is open on two screens. That describes the nine routes priced here with open legal scores counted as failures, and it is not a proof that no product can serve the clock, since the arrangements of Section 6.14 were not priced at all. One of them, a route on FedNow or the existing RTP joint account with payments split under the networks' limits, would pass S1 as scored and might qualify. With the open items resolved the facility wins at 10 to 34 basis points a year on the whole tokenized balance, base 22, for a fifth of the balance movable at any hour, with the prefunded joint account as challenger at 25.35 and the one-ledger structure third at 41.8, where the facility's Sunday settlement cash is Reserve Bank credit against loans the bank already holds there, within its net debit cap headroom. Where that cash must be new reserves or newly bought securities the facility costs 39.10 or 34.50 and the joint account wins at 25.35 at every share (Section 4.9). That price buys movement, and the facility keeps the balance nine tenths lendable only at an always-on share of 10 percent or less (9.8 counting the grossed-up bills), and the cap ladder shows how often the cap binds at that boundary. At the weekend peak net debit this manual assumes, a cap that binds in 1 percent of weekends needs backing of about a third of the balance, and at the assumed dispersion the two hold together only at a median weekend peak net debit of at most 1.56 percent of the balance, a number no bank has reported, and one on which the verdict turns between 1.5 and 2 percent (Table 15). A buffer held anyway makes the facility cheaper, at 7.5 to 24.5 with a tenth of the balance, not more lendable, and the price leaves out the reserves or Reserve Bank collateral each member needs to settle its net debit at the Sunday opening, which hold the lendable share at 0.80 only if the collateral is loans the bank already holds. Under the announced six-day Fedwire schedule, if the peak net debit scales with the hours, the backing that holds today's bind rate on every window falls to about a tenth of the balance, at 7.6 to 24.4, and the facility misses the lendable screen by 0.004; under seven-day operation it passes at about 1.5 percent of the balance and 5.4 to 15.5 if weekend Reserve Bank credit carries each debit to Monday's securities opening, and fails at 0.785 if the bank holds reserves against the outflow instead.

So the settlement-clock answer turns on the median and dispersion of the weekend peak net debit on tokenized deposits, which a bank can observe and this manual cannot; on the Reserve Banks' schedule, announced for six days and open for seven, and on whether their credit will be there at a Sunday opening and, under seven-day operation, through the weekend to Monday's securities opening; and on whether a depositor will pay 75 to 240 basis points on the tier dollar for a tenth of its balance to move at any hour. None of the three is in the design's control. The decision is the bank's and its depositor's once the first of the three is known, and the Reserve Banks' before then.

What the models did here. The settlement clock entered the work late, and from my side of it. Through twelve drafts of stage 1, two of them from outside model families, four gates and an audit, every model had treated live-bank access only as the queued-wire problem during a run; the gap a treasury that pays around the clock notices first, cash access at a bank that is closed but has not failed, came from my asking whether the design covered payment speed and always-on withdrawals. Stage 3 built the facility around it. Its fact check was the one that found, after thirty stage-1 versions, that the reciprocal-deposit allowance had been quoted at a limit the FDIC replaced on 1 September 2026; its cap-exceedance figure was never built, and the placeholder's caption had to be rewritten to the weekend window the routes engine later drew. In stage 4 the engine showed the facility failing the lendable screen at a fifth of the balance, which is the finding this chapter now turns on. The stage-5 correction is the one in Chapter 1: every stage before it described the window as the hours in which the central bank was closed, and FedNow runs through it. The settlement engine of stage 6 was built after I reread this chapter and judged that its answer read as a fix for the uninsured deposit rather than for the always-on mismatch. A Claude Opus instance read the primary pages for it, and found the FedNow liquidity management transfer limit raised to $500 million in the October 2026 operating procedures and the Sunday-to-Friday Fedwire schedule announced for 2028 or 2029; a second instance built the engine. Its first run measured the Monday-holiday cash gap from the wrong opening, 44 hours where the answer is 20, and counted the lendable share two ways inside one engine, as one minus the share and as one minus the grossed-up bills; I chose the routes engine's convention, and the 9.8 percent boundary is disclosed in Section 6.7. The stage-5 text put the fall in the capacity term under the six-day schedule at about half; the engine gives 48 percent and shows the lendable share still short. Claude Fable drafted the new sections. Stage 7, an outside review by GPT Astra, reproduced every figure in this chapter that it recomputed, the extreme Student-t queue percentile only within simulation error, and found four of the definitions behind the chapter's numbers wrong, the Sunday cash source, the debtor cohort, the peak debit and the buffer metric; Chapter 9 lists the findings.

Part 4. Both clocks together

Chapter 7. What it costs two banks

7.1 The dollars

The prices become dollars on a bank’s covered excess, which the routes engine takes as 20 percent of liabilities at two illustrative banks of $50 billion and $5 billion in assets, each with liabilities at 90 percent of assets, so covered excess of $9.00 billion and $0.90 billion. At both the reciprocal allowance exceeds the covered excess ($14.60 billion and $1.90 billion), so at a 20 percent covered share the cap does not bind. The tier changes the arithmetic between them, since the allowance is 32.4 percent of liabilities at the larger bank, whose marginal tier is 30 percent, and 42.2 percent at the smaller, whose marginal tier is 40 percent, so the smaller bank has proportionally more room before its placements become brokered. Existing reciprocal holdings use the same allowance, and the mutual’s 3 percent capital contribution is $270 million at the larger bank and $27 million at the smaller.

Table 16 puts every route on the covered excess in dollars a year. Fees in basis points are the same at both banks, so the small bank’s column is a tenth of the large bank’s throughout, and the fixed-cost terms (the vehicle’s frictions, the facility’s access, the mutual’s frictions) would be dearer per dollar at the small bank, which the table does not model. Against every route stands the cost of letting the balance go, $45 million to $270 million a year at the larger bank and $4.5 million to $27.0 million at the smaller if the whole covered excess left and were replaced at 50 to 300 basis points; the champion under the settled reading costs $18.9 million and $1.9 million a year, the strict champion $34.1 million and $3.4 million.

Route	Variant	$50bn bank	$5bn community bank
R1	Pledged from securities already held	9.0 to 27.0 (18.0)	0.9 to 2.7 (1.8)
R1	Securities bought for the purpose	148.5 to 297.0 (222.8)	14.9 to 29.7 (22.3)
R2	Public-unit letter, standard structures	9.5 to 16.2 (12.8)	0.9 to 1.6 (1.3)
R3	Placement fee plus rate uplift (base 10.0)	11.2 to 33.8 (20.3)	1.1 to 3.4 (2.0)
R4	Replacement spread on each dollar replaced	45.0 to 270.0 (157.5)	4.5 to 27.0 (15.8)
R5	Single-name funded vehicle	41.2	4.1
R5	Pooled all-funded reference design (theta 20%)	34.0	3.4
R5	Pooled, sized to the S5 standard (theta 22%)	34.1	3.4
R6	Reference design (X 10.0%)	18.9	1.9
R6	Alternative design (X 21.8%)	27.9	2.8
R7	Facility at a 20% always-on share	9.0 to 30.6 (19.8)	0.9 to 3.1 (2.0)
R8	R6 reference design + R7 less the saving	26.7 to 49.5 (37.7)	2.7 to 5.0 (3.8)
R9	Joint-account position at a 20% prefunded share	12.2 to 33.4 (22.8)	1.2 to 3.3 (2.3)

Table 16. Annual cost of each route on the covered excess, $ million a year, low to high with the base in parentheses. Source: the routes engine’s results, Section 10.

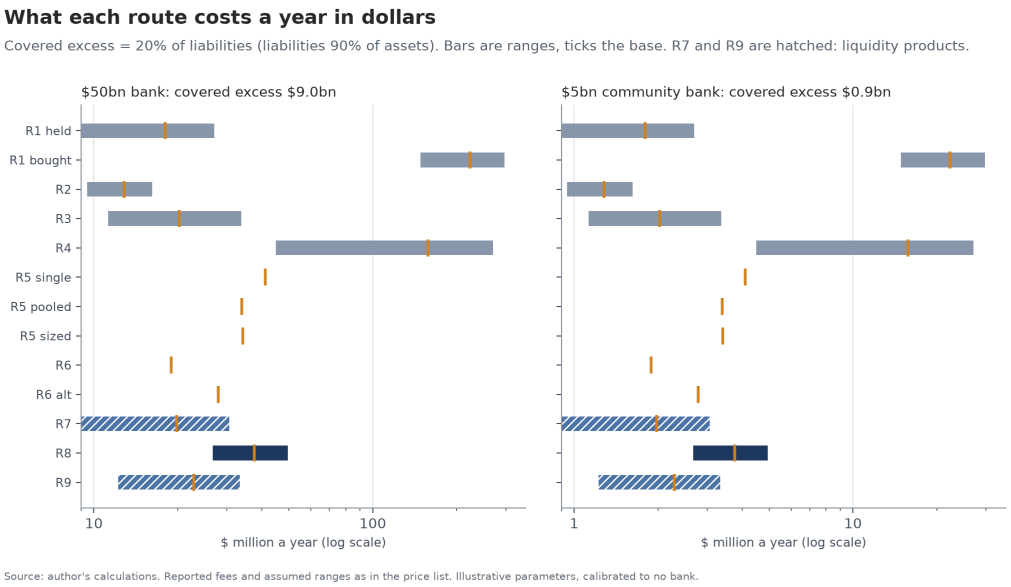


Figure 15. What each route costs a year in dollars, on a covered excess of 20 percent of liabilities at a $50 billion and a $5 billion bank; bars are ranges and ticks the base; R7 and R9 are hatched as liquidity products. Source: author's calculations; reported fees and assumed ranges as in Table 6; illustrative parameters, calibrated to no bank.

7.2 The bank's ceiling and the depositor's advantage

The three-party condition of Chapter 3 extends with the bank's replacement cost, so that protection clears when the depositor's advantage, the bank's retention value and its replacement value together are at least the price. Counted in full, the replacement spread is paid every year the balance is gone, the right reading when the counterfactual is that the balance leaves and is replaced, and it is the ceiling on what the bank should pay. That ceiling runs from about $18 million a year at the larger bank at a 20 basis point spread with no retention value to four times that at 80, and a tenth as much at the smaller bank. Counted probability-weighted, it is the cut in run probability times the spread times the five-year annuity of Chapter 3 (4.067), the right reading when the balance would mostly have stayed. Under the full count a replacement spread of 20 basis points leaves 1.0 basis point of depositor advantage to find at the mutual's price and 17.9 at the sized vehicle's, and at 40 nothing is left to find for either. Under the weighted reading the spread barely matters, and with no retention value the depositor needs 21.0 basis points at the mutual's price and 37.9 at the vehicle's whatever the spread, the gap closing only in the larger retention cell of Chapter 3. Which reading applies is the behavioral question the pilot in Chapter 10 would test, and each basis point of depositor advantage is $0.90 million a year at the larger bank and $0.09 million at the smaller.

There is a quiet-year value the pilot would also test. If verified stability let a bank release 5 to 25 percent of a protected balance into its structural ladder at a term premium of 50 to 80 basis points, anchored on the Kim and Wright estimates of 23 October 2026, the value is 2.5 to 20 basis points a year, and zero for a bank bound by the standardized LCR (Kim and Wright 2005; Federal Reserve Bank of St. Louis 2026b, 2026c). With a one-point cut in run probability that covers a bank's half-

share of the vehicle's premium only at a 25 percent release and a term premium of 65 basis points or more, and the full price in no cell. Sponsorship by the bank alone therefore pays only if protection also keeps balances that would otherwise drift to a large bank in quiet years, which is untested.

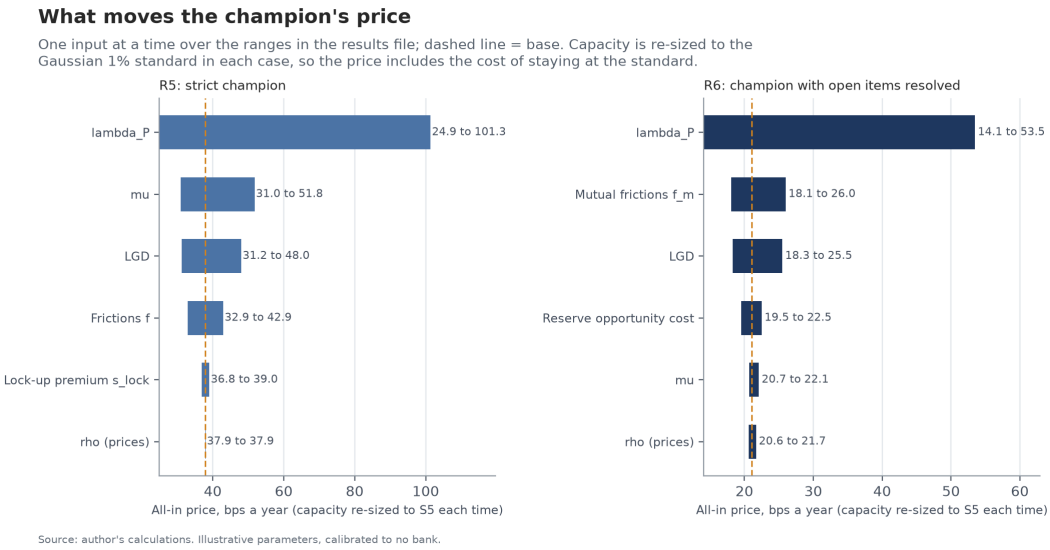


Figure 16. What moves the champion's price: one input at a time over the ranges of Table 11, with the dashed line at the base; capacity is re-sized to the Gaussian 1 percent standard in each case, so the price includes the cost of staying at the standard. Source: author's calculations; illustrative parameters, calibrated to no bank.

7.3 What each kind of deposit should be charged

The prices in this manual are the charge for each risk whether or not any route exists to move it. The self-insured cost of Section 2.2, 20.74 basis points a year under the pricing measure and 6.91 under the real-world one, is what a plain uninsured balance costs the bank that keeps it with no route bought, and each route's price is what moving that risk would cost if the route existed. The self-insured cost is a usable internal charge for the run clock, and a route's price for the clock that route serves, and on my reading the deposit should be charged for each clock it runs on and credited as funding only for its lendable share; whether a funds transfer pricing desk should stack the charge that way is the question Chapter 10, item 7 puts to practitioners. Table 17 sets the charges out by kind of deposit, with the routes stacked where the risks stack. The one-ledger structure is the mutual plus the facility less the shared ledger, and the mutual with the joint account is the pairing with no shared ledger, for which no saving is priced.

Deposit	Clock(s)	Route on the list	Price, bps a year on the whole balance (base; range)	Lendable share
Plain insured balance	neither	none needed	0	1.00
Plain uninsured balance	run	R6 mutual with tail, reference design (open items resolved)	21.03	0.97
Plain uninsured balance	run	R5 pooled vehicle sized to the S5 standard (strict champion)	37.94	1.00
Insured tokenized balance	settlement	R7 facility, Sunday cash as Reserve Bank credit against loans already pledged	22.00 (10.0 to 34.0)	0.80
Insured tokenized balance	settlement	R9 joint account	25.35 (13.6 to 37.1)	0.80
Insured tokenized balance	settlement	R7 facility, Sunday cash from bought securities	34.50 (15.0 to 54.0)	0.60
Insured tokenized balance	settlement	R7 facility, Sunday cash from new reserves	39.10 (19.6 to 58.6)	0.60
Uninsured tokenized balance	both	R8 one ledger (R6 + R7 less the shared ledger), Sunday cash as Reserve Bank credit	41.84 (29.7 to 55.0)	0.77
Uninsured tokenized balance	both	R6 + R9, no shared ledger	46.38 (34.6 to 58.1)	0.77
Uninsured tokenized balance	both	R8 one ledger, Sunday cash from bought securities	54.34 (34.7 to 75.0)	0.57
Uninsured tokenized balance	both	R8 one ledger, Sunday cash from new reserves	58.94 (39.3 to 79.6)	0.57

Table 17. What each kind of deposit should be charged: the clocks it runs on, the routes on the list for those clocks, the price in basis points a year on the whole balance (base, with the range where the price is a range) and the lendable share on the absolute convention. Open items resolved and S2 not applied, the strict R5 row excepted; the rows price

the risk each kind of deposit carries and do not choose a route; the Sunday cash cases are those of Section 4.9; the settlement rows take the same weekend peak net debit for insured and uninsured balances. Illustrative parameters, calibrated to no bank. Source: the routes engine's results, Section 10b.

The settlement clock adds 20.81 to 25.35 basis points a year to the mutual's 21.03 and takes 0.20 off its lendable share of 0.97, which roughly doubles the cost of keeping the dollar. The cheapest route for both clocks is the one-ledger structure at 41.84 where the facility's Sunday cash is Reserve Bank credit against loans already pledged, and the mutual with the joint account at 46.38 where that cash must be bought securities or new reserves, since the one-ledger structure then costs 54.34 or 58.94 and lends 0.57. On these parameters, which are illustrative and calibrated to no bank, and on that reading, a bank that charges an uninsured tokenized balance the mutual's price for a plain uninsured balance undercharges it by about 21 to 25 basis points a year and overstates its lendable share by 0.20 of the balance, 0.97 against 0.77. The stacking is the arithmetic of two routes with separate encumbrances on the absolute convention, and the engines price no event in which the two clocks strike together. The two tranches of the one-ledger structure share a ledger and never a reserve, because the weekend on which many members are net debtors is the weekend before the Monday on which one of them is closed (Section 4.8), and the case in which the two clocks strike together, a run that starts on a weekend in an uninsured tokenized balance, is priced nowhere in this manual (Section 8.4, item 19).

> **What the models did here.** The reading of the depositor's side in this chapter is the second of the three things only I caught. Through sixteen stage-1 versions, their gates and an audit, the depositor's advantage for an operating balance was read as a rate spread over a large bank's account, so a reader could take the 37.7 basis point price as a yield the depositor would earn. An operating account is paid mostly in services, the earnings credit offsets fees and is not interest, and cash in a fund neither makes payments nor reduces a loan's cost. The bank's side went wrong in the other direction. A round drafted outside the main workflow in stage 1 put a core spread of 100 to 150 basis points forward as deposit-stability value when the Treasury slope on 28 October 2026 was 73 to 78 and the term premium sits below it, and claimed a table covered "the full 37.7 basis point clearing cost" when its own cells read 31.2, 35.3 and 37.5. A fresh review rebuilt it at a 50 to 80 basis point term premium as a hypothesis. In stage 2 a draft priced reciprocal placement as replacement funding, which it cannot be. The third of the three, the charge by clock of Section 7.3, came on my own rereading of the thirteenth draft after the outside reviews.

Chapter 8. How it would be built, and what the regulator will say

The readings below are a practitioner's readings of public statutes and regulations, not legal advice; each structure would need counsel in every state its members and depositors sit in.

8.1 Wrappers and licensing

The vehicle has four possible wrappers. A security-based swap facing an eligible contract participant brings the SEC regime and dealer registration if the vehicle's gross notional crosses the de minimis threshold ($8 billion for credit default swaps, a phase-in level kept until 8 May 2028; U.S. Securities and Exchange Commission 2026). A bank standby letter of credit under 12 CFR 7.1016 carries lending and counterparty limits, a 100 percent credit conversion factor and the conflict of a large bank guaranteeing against the flight to quality it benefits from. A policy from a licensed or special-purpose insurer is the third form, and a funded vehicle guarantee the fourth, which is financial guaranty insurance under state law if sold to the party holding the exposure (New York Insurance Law Article 69 is the usual reference). The mutual's form is the first question for its organizers, because the insurance form comes first for a guarantee sold to the party holding the exposure and the swap does not fit a mutual whose members are the banks rather than the depositors. The Ohio form answers it directly. American Mutual Share Insurance Corporation is a licensed share guaranty corporation under Chapter 1761, owned by its member credit unions with one vote each and funded by a capital contribution of at least 1 percent of insured shares with special assessments permitted. Its subsidiary is a property and casualty insurer licensed in 35 states and the District of Columbia writing excess cover up to $10 million per account relationship (American Mutual Share Insurance Corporation and Excess Share Insurance Corporation 2026). Whether a bank mutual takes that route as an admitted insurer, forms a risk retention group, or needs enabling legislation in its chartering state, and how each treats the capital contribution and the special assessment for solvency purposes, is a question for counsel that this manual states and does not answer, as is the national-bank question that decides the champion.

The facility is documented in standard repurchase agreements with a calculation and collateral agent and a tri-party custodian, and in a multilateral netting contract or clearing rules that provide for the netting and name the agent's collateral beneficiaries, since the repos alone discharge nothing through the net (Section 6.2); an all-bank consortium can fit the clearing-organization definition of 12 U.S.C. 4402 and rely on 4404 for the netting, subject to the receivership stay, on an opinion counsel would have to give. The joint-account route needs the account-holding Reserve Bank's approval of the arrangement and of each holder under the Board's 2017 guidelines, which expect the arrangement to manage risk to the payment system risk policy's standard and may limit the level or volatility of the balance, and an interest determination of its own if the balance is to earn anything (Board of Governors 2017, 2023c). A structure that both holds members' collateral and writes protection on them needs the vehicle's capital outside the membership and the governance a designated clearing arrangement is held to. Section 18(a)(4) of the Federal Deposit Insurance Act and 12 CFR Part 328 Subpart B prohibit misrepresenting insured status, and the National Credit Union Administration's rule for private share insurance, 12 CFR 740.3, gives the pattern (National Credit Union Administration 2002). The mutual pays no third party a fee tied to deposits placed or retained, so on my reading the covered deposits are not brokered under 12 CFR 337.6, while for a vehicle in which the bank shares the premium the facilitation question remains open (86 FR 6742; the 2024 proposal at 89 FR 68244 was withdrawn at 90 FR 12115).

8.2 The member below well capitalized

The base layer is the one that stops growing when a member weakens, and it is why the mutual sits on top of the reciprocal base rather than beside it. A member that falls below well capitalized, or to a CAMELS rating of 4 or 5, may keep the reciprocal deposits it holds but may not receive new ones that take its holdings above its special cap, the average of four earlier quarter-ends anchored at the first finding that it is no longer rated 1 or 2 or no longer well capitalized (Section 5.7). The cap limits what the bank may receive, not what it holds. If it does, every reciprocal deposit it holds is counted as brokered, which an adequately capitalized bank may not renew without an FDIC waiver and an undercapitalized one may not renew at all (12 U.S.C. 1831f(i); 12 CFR 337.6). A member below well capitalized also may not pay above the applicable rate cap (12 CFR 337.7). The mutual's cover turns on its own covenants rather than on the member's capital category. The remedies of the vehicle's term sheet, which freeze new cover at a bank on a covenant breach while existing cover runs to term, would make its cover stop growing for a weakening member in the same way, though, an open point for the mutual's terms. The facility's cap rests on the member's own pledged bills with no capital condition; the design expects a weak bank to use it and slows it through the cap.

8.3 The objections register

Market discipline and moral hazard come first. My reading of the 2023 record is that the monitoring large depositors actually did was binary, a decision to run taken in an afternoon, and a priced protection could turn that exit into a continuous signal. The design's answer is a covenant set (liquidity, concentration, capital adjusted for unrealized losses, interest-rate risk, growth and commercial real estate), tested quarterly on public filings with a two-stage remedy and no cliff, and the answer is partial, since in the depositor-paid vehicle the bank is not a party at all.

Adverse selection is answered in price by the mutual's risk classes (with three classes the class rates run 18.22, 23.91 and 35.30 basis points against a pool price of 22.24), in membership by admission on public metrics, and, for stuffing during a run, by the vehicle's trailing-average covered balance.

The structure limits implicit support, since the depositor-paid vehicle keeps the banks out of it, and for the mutual each member's obligation ends at its capital contribution, which the Ohio statute requires, and its special assessment, which the design caps.

The hidden subsidy is conceded where it exists. The letter of credit fails the screen and the manual says so, the facility is open on it for the bounded weekend transfer, the reciprocal base grows the fund's exposure against the assessments the banks pay, and the mutual's rule M asks the receivership to carry the excess recovery at closure, which I read as ordinary resolution practice.

The reciprocal base is also a timing option the fund writes at no marginal charge (Section 5.7), because the assessment base is average assets less average tangible equity, which a one-for-one swap leaves unchanged, and the invoice is paid a quarter later, so a bank that charges for the sweep

in calm and absorbs the fee in stress moves the risk to the fund when it is largest, and the switch can itself signal the stress it answers. The mutual's answer is that section's trailing average of covered excess as the base for its own assessments, adopted and unpriced.

Liquidity rules give no credit for any of it. The LCR's run-off rates are set by counterparty type, operational status, FDIC coverage and brokered status (12 CFR 249.32(h)), none of which private protection changes, so neither the vehicle nor the mutual earns LCR credit.

Model governance reaches the design through its members. SR 26-2 is supervisory guidance for banking organizations, so a non-bank mutual or vehicle sits outside it unless supervised as one, but the member banks relying on its prices would be running a model in the sense of that guidance (Board of Governors, OCC and FDIC 2026), and the mutual should meet the same standard, with independent validation, documented limitations and a named person responsible for the copula choice as the minimum.

Above the mutual's exhaustion point and above the vehicle's stated cluster the loss is public on no public commitment, since no private balance sheet has reason to hold the cash a systemic cluster would call for, and the manual prices the layer below it and leaves the tail where it is.

8.4 What is unverified

The register below merges the open items of every stage with the conventions the routes engine adopted and I have not confirmed; each entry names what would resolve it.

1. The national-bank capital contribution (Chapters 4 and 5): the question that decides the champion turns on the incidental powers of 12 U.S.C. 24(Seventh), the non-controlling investment rules of 12 CFR 5.36 and the Comptroller's interpretive letters on bank investments in insurers, which this manual names and does not read; a letter from counsel, or an interpretive letter from the Comptroller, would resolve it (Chapter 10, item 1).
2. The rule and the screens (Chapter 2): that an open verdict is not a pass, that the settlement-clock problem is scored on S1, S3 and S4 alone, that the base of a range is its midpoint (R3 and R8 excepted) and that the reciprocal uplift is 10 basis points are conventions chosen before the run; a reader who prefers others can rerun the routes engine.
3. The mutual's subsidy screen (Chapter 5): the mutual passes S4 on the reading that rule M's receivership carry (0.99 basis points a year under the pricing measure) is a timing cost the FDIC carries in any all-deposit purchase and assumption; a resolution specialist's view would settle it (Chapter 10, item 4).
4. My other readings (Chapters 4 and 6): the facility's bounded transfer to the fund, the pledge's open S4, the letter's open S3, the lendable shares and the shared-infrastructure saving are readings, not findings, and whether a bill slice positioned for the facility is encumbered before a draw under 12 CFR 249.3 and 249.22 is open, since the exception in 249.3 for assets pledged to a central bank does not reach a private facility; a bank's own liquidity policy, receivership-level data on pre-failure weekend outflows, a quotation for a combined agent and custodian mandate

and the facility's control, substitution and withdrawal terms, read with the bank's supervisor, would resolve them, and the ranking does not depend on the saving.

5. The copula and the correlation (Chapters 3 and 5): nothing read for this manual decides between the Gaussian and the Student-t(5) copula, and the correlations of 0.20 for prices and 0.30 for reliability are illustrative; a receivership-level dataset of failure clusters would narrow both, and a rating agency's tail model would decide the capacity in practice (Chapter 10, item 5).
6. The default intensity, the loss and the multiplier (Chapters 3 and 5): the 0.25 percent intensity, the 25 percent loss given default and the multiplier of 3 are illustrative, and the intensity dominates every sensitivity; a dataset of deposit recoveries net of secured borrowing would resolve the loss, and annual repricing on public metrics is the design answer for all three meanwhile.
7. Frictions and market prices (Chapters 3 and 5): the frictions, the lock-up premium and any facility fee have no observed market price, and most of the champion's price is running cost; a licensed carrier, a rating review and a tail investor would supply the numbers.
8. Rule M and the plumbing (Chapters 3 and 5): the purchase and assumption with a private fund, the assignment of the claim and the advance dividend under the cutoff of 12 CFR 360.8, and the acquirer probability of one half are resolution questions that only a protocol agreed with the FDIC before the first member fails would settle (Chapter 10, item 4).
9. The weekend repo, the netting and the haircut (Chapter 6): the safe-harbor reading for a repo moved to a bridge bank, the reach of the payment suspension under 1821(e)(8)(G)(ii), whether the consortium's netting contract qualifies under 12 U.S.C. 4402 to 4404, and the 2 percent haircut over three days are open, and counsel's letter and a price history of bills over such weekends would resolve them (Chapter 10, item 6). The Sunday cash source is a requirement the price leaves out: reserves or collateralized Reserve Bank intraday credit of at least the cap, separate from the facility's bills and within its Reserve Bank net debit cap, which pledging collateral does not raise unless the Reserve Bank has granted collateralized capacity above it; only a bank's own reserve position and its Reserve Bank's collateral terms would settle it.
10. The always-on inputs and fixed costs (Chapters 6 and 7): the encumbrance spread, the always-on share, the haircut and the outflow distribution are round inputs that a bank's own numbers would resolve for that bank; the engines do not model scale effects in the fixed-cost terms or the activity-stock cost of a letter of credit; and whether any bank's tokenized deposits are large enough to need a secured arrangement is unproven.
11. The 2023 record (Chapters 1 and 2): the data carry no depositor type, and the link from the regionals' replacement funding to their originations is unmeasured; a bank-level study against banks that kept their uninsured deposits would turn the credit point into a number or retire it, and would leave the depositor type open.
12. Reported figures and legal readings (Chapters 4 and 8): the reciprocal placement fee is one bank's disclosure, the California collateral percentages are quoted from a secondary reproduction of the statute, and the insurance-law, swap, financial guaranty, facilitation and

clearing-house readings are a practitioner's; a second disclosure would resolve the fee, the statute's official text the California percentages, and counsel's opinion the readings.

13. Behavior: whether protection arranged in advance changes what a depositor does in a run, and whether a depositor who has paid for an always-on tier uses it, are untested (Chapter 10, item 8).
14. The settlement engine (Chapter 6): the assumed peak net debit drives the cap ladder, and the verdict turns between a median of 1.5 and 2 percent (Table 15). The variable is each member's peak net debit to all counterparties, with payment paths and queue release not modeled; the members are equal in size and form an open debtor cohort with receivers outside the modeled set, so every multi-member statistic describes that cohort and not a closed membership; the correlations and the two scalings with the hours are illustrative and neither is estimated; weekends are independent, and serial dependence would change annual incidence, maxima and run lengths in ways the engine does not model while leaving the expected count unchanged. The buffer sensitivity reports incremental lending displacement beside the absolute lendable share, which the buffer does not move, and double-counts the buffer against Monday's outflows. The lendable share is one minus the always-on share, with the grossed-up convention disclosed (an S2 boundary of 9.8 percent; a displacement threshold of 10.00 against 10.41 percent). Window lengths are nominal schedule hours, with the daylight-saving check disclosed. Tri-party re-use by a cash lender rests on the Office of Financial Research's description alone. A bank's record of weekend peak net debits and counsel's letter would resolve them (Chapter 10, item 9).
15. The mutual's rule-M advance and the ninth route (Chapters 4, 5 and 6): the tail's advance of the special-assessment slice at closure carries no charge for a survivor that fails to pay within the week, and the seven-day collection is an assumption; the joint account's price rests on an interest treatment that a Board order gives the live network's account and no other, on rates as of 3 November 2026, and on access, loan-spread and frictions ranges that are author assumptions; and whether a participant's share of a joint-account balance counts as a high-quality liquid asset has no public answer. A Reserve Bank's approval decision, an interest determination and the mutual's assessment covenants would resolve them (Chapter 10, items 4 and 9).
16. The base layer as an option (Chapter 5): no study tests cyclical pricing of the reciprocal fee, so Section 5.7 is reasoning, not evidence; the per-depositor limits come from two banks' disclosures; the trailing-average base for the mutual's assessments is a design rule the engine does not price; and the onboarding period and the network's enrollment after March 2023 are reported, not primary. A panel of sweep fees and rates across 2022 to 2024, the network's own terms, a run of the mutual engine with the trailing base, and the network's own enrollment records would resolve them in turn.
17. The tail's cap (Chapter 5): as built, the tail pays up to its layer and not up to the stack's exhaustion point, so when members fail a slice of its funded commitment goes unused and the stack exhausts below the stated 10.0 percent (Section 5.2). A contract that lets the tail pay up to the exhaustion point would use that slice. The engines do not price it, and a rerun of the mutual

engine with that contract would settle its effect on the exhaustion point, the price and the ranking.

18. The FedNow or RTP route (Chapter 6): a route on FedNow or the existing RTP joint account, with payments split under the networks' $10 million limits, would pass S1 as scored and might qualify strictly where none of the priced routes does (Section 6.14); the networks' rules on splitting one payment into several, the Reserve Banks' treatment of a weak bank's FedNow access (S3), and a price would settle it.
19. The correlation of the two clocks (Chapters 6 and 7): the stacked prices of Table 17 add two routes whose encumbrances are separate and whose events the engines price separately, and a run that starts on a weekend in an uninsured tokenized balance, the case in which both clocks strike at once, is priced nowhere in this manual; the one-ledger structure's shared ledger and separate reserves (Section 4.8) are a design reading of that case and not a price. A joint model of weekend net debits and failure, run on a bank's own weekend record, would resolve it; Chapter 10, item 7 asks how the charge should stack until then.

What the models did here. The legal readings in this chapter were the first review's strongest work and the place where citation drift was worst. The stage-1 red-team legal pass found that the original outside draft rested on two structures not permitted as drafted, a deposit secured by the bank's own collateral and a custodial workaround under the FDIC's recordkeeping rule, and its findings held through every later round. The drift came from both directions. The outside versions cited a Federal Housing Finance Agency rule for bank supervisory confidentiality and a Dodd-Frank swaps provision for security-based swaps; the Claude pipeline cited the 2011 model-risk guidance as current for two versions after SR 26-2 had replaced it, and its gate suggested adding the outdated entry to the references. In stage 2 the Texas and Illinois pledging statutes were hedged as unread after the official texts had been pulled, and a later style edit placed the Texas carve-out inside a "because" clause so that it read as a reason the bar holds. In stage 4 the first draft credited the cap on special assessments to the Ohio statute, which requires the contribution and permits assessments; the cap is the design's.

Part 5. The experiment

Chapter 9. How this manual was made

9.1 One experiment in seven stages

In October 2026 I had an AI-drafted white paper on private protection for uninsured bank deposits, written by one outside model family and extended by another, and I asked for a red-team review and a version I could post. The review said the idea held up and the draft could not go out. Over the next day the work was rebuilt from a verified-facts base, and partway through I changed the question. From the third round of decisions the work was also a test: can frontier AI models financially engineer a credible answer to a real problem for which no verified product exists, with empirical facts that check out and a design explicit, ranged and stress-tested enough to survive a regulator or a rating analyst? And where do they still need a human? This manual is the fifth stage of that one experiment, its settlement engine the sixth, the outside review that corrected both the seventh, and the stages before it are its working drafts, logged in the process record in the repository.

Stage 1, with corrections that ran into stage 2, produced the clearing-cost engine and the pooled vehicle of Chapter 3. Three model families took part. Two outside families wrote and extended the original draft and came back twice more, at the fourth and ninth drafts, each outside version built on the pipeline's latest draft so that the models were in effect reviewing and extending one another. Claude models wrote the briefs from my decisions, built the engine, drafted the prose, ran the gates and the fact checks, and carried out a clean-room audit that reran the engine in an empty folder and got byte-identical results. The draft went through 31 versions. From the first model run to an audited twelfth version took about 18 hours, and the process record keeps an error and challenge log of 66 entries.

Stage 2 produced the mutual with a tail of Chapter 5 and the surviving-bank panel of Chapter 1. The design came out of a structured interview of four rounds, each a set of decisions I made; the mutual engine extended the clearing-cost engine to a member pool with a mutual layer and a re-attached tail, with a parity check against the stage-1 figures before any new result was read, and was built twice. The draft went through eight versions in one day, a fact check of 501 items, three style rounds and a second fact check aimed at its premise, which it failed.

Stage 3 produced the always-on facility and the settlement-time tranching of Chapter 6, in six drafts. Stage 4 produced the routes engine and the eight-route comparison of Chapters 4, 5 and 7. It was a second test: whether a rule set in advance, run by a model, produces a result the author did not choose, and whether that result survives checks by models that did not produce it. No outside family took part in it; the engine was built in about 47 minutes, the draft went through four versions, a fact check of about 1,450 items found 31 fails and no model number wrong, and three style rounds ran to the cap. Stage 5 drafted this manual from the four stage drafts and the process record with the result frozen: no new modeling, no new route, no change to a screen, a rule or a parameter, and

every number as the engines print it. The draft went through a fact check of about 842 items that failed 24, a scoped recheck of about 340 items that failed nine, and three style rounds that failed 35, 17 and 7, the third scoped, with no earlier fix reverted and the long-sentence count left at the cap, and a confidentiality check that passed on all five points, and the code was reorganized for release with every number in the results files, about 11,000, compared before and after. Stage 6 deepened Chapter 6, for the reason that chapter's box gives. A sourcing pass read the primary pages for the Fedwire hours and holidays, the announced schedule, the FedNow limits, the liquidity coverage ratio, the FICC sponsored service and the qualified-financial-contract stay, marking each fact verified, partly supported or not found; a settlement engine was built on the routes engine, changing no route, screen, rule or parameter of stage 4; and the chapter was redrafted on its results. The changed passages then went through a scoped fact check of 204 items, of which 179 passed, 10 failed, 10 failed on inference and 5 could not be verified, a style round that failed 27, and a confidentiality check that soft-failed on internal folder names in two engines' figure-copy code, fixed by removing that code with the results byte-identical.

Stage 7 was an outside review of the whole manual and the release by GPT Astra, the first check of the settlement work by a model family that had built none of it, since stages 3 to 6 had run on Claude models alone. The review reproduced the arithmetic without importing the engines, the bind probabilities, the cap ladder to rounding, the calendar, the schedule backings, the Monday bridge cost, the vehicle's 37.94 and the mutual's 20.68, ran the release in a temporary folder and found every output byte-identical, and then returned ten findings, four blockers, five major and one minor, every one of which I confirmed against the code and the text. The blockers were that the Sunday settlement had no cash source, since opening Fedwire Funds does not turn pledged bills into reserves and Fedwire Securities stays closed; that every simulated member was a net debtor, so the system statistics described a debtor cohort with its receivers left out and not the closed membership the text described; that the lognormal described a weekend's terminal position where the cap limits a running peak; and that the mutual's reliability counted a special assessment as cash at closure when it is collected in the week after. The review got one detail wrong, corrected by the primary sources: the Fedwire Securities cutoff is 3:15 p.m. for secondary-market transfers, not 3:30, which is the cutoff for reversals and repo adjustments, and it does not change the 35.5-hour interval. The sources also sharpened its reading of the stay: 1821(e)(10)(B) stays termination, liquidation and netting rights, and (e)(8)(G)(ii) suspends payment and delivery only under the qualified financial contract itself. The review also named the alternatives of Section 6.14 and asked why a prefunded pool had been dismissed when a joint account exists. What changed is in this manual: the ninth route, the Sunday cash requirement, the peak-debit definition and the cohort labels, the tail's advance and the mutual's price from 20.68 to 21.03, the two stay intervals, the buffer section, the median-sensitivity table, the netting contract among the counsel questions, and the clustering sentence removed from the engine results. After the review I raised the reciprocal base as a timing option across the cycle, added as Section 5.7 with its sources checked by a fresh Claude model. Gemini Argon then reviewed the twelfth draft and the release, reran every engine and reproduced every result byte for byte. Its findings moved one number, the uncollateralized daylight overdraft fee

from 0.2898 to 0.2919 basis points on the payment system risk policy's 360-day year (Section 6.2). They also scoped three claims: the mutual's exhaustion point, which the stack as built stops short of (Section 5.2); the facility's Sunday-cash condition, without which the joint account is the cheaper settlement route (Section 4.9); and the strict settlement answer, which an unpriced route on FedNow or RTP rails might undo (Section 6.14). Smaller findings sharpened Sections 4.9, 5.7 and 8.2, and I added a sensitivity of the lendable bar (Section 4.11 and Table 19). Fresh Claude models checked each finding against the code and the primary sources before any change, and the findings, or parts of findings, that failed it were not applied, among them a claimed double count in the seven-day lendable share, a mislabeled Monday cost and a claim that a weak bank's reciprocal deposits must run off. After both reviews I added a table by kind of deposit to the routes engine's results, a recombination of priced variants that changes no route, screen, rule or parameter, and it became Section 7.3.

9.2 The model families and their roles

I set the problems, chose the routes, wrote the screens and the rule before any result was read, and made every design decision. Those decisions were that the client is regional banks together and not one bank, that reciprocal placement may be a layer although it leans on the deposit insurance fund, and that the mutual is a private licensed insurer and not a fund by statute. They were also that the tail is sized to the Gaussian standard with the fat-tailed result disclosed, that the Massachusetts settlement rule is primary, that the facility insists on full collateral rather than a mutual promise, and that the cap is a term the depositor signs. The design conventions behind the pricing were proposed by the models and reviewed and approved by me: the four-term decomposition, one pricing measure with no stacked premiums, wrong-way risk as a haircut to the protection's value, and the simultaneous timing standard.

Frontier AI models did the financial engineering under my direction. In stage 1 GPT Astra wrote the original draft and Gemini Argon extended it; Gemini produced the fourth outside version and Gemini Argon, with GPT Astra's feedback, the ninth; in stage 7 Gemini Argon reviewed the twelfth draft and the release after GPT Astra had reviewed the eighth. Claude models did the rest in every stage, in separate roles with separate instances: a coordinating model that wrote the briefs from my decisions and read every memo, a model that built each engine, a drafting model that wrote the prose, gate and fact-check instances that had not seen the draft before, and fresh grading instances for each style round. For this manual Claude Fable drafted the text from the stage drafts and the process record, fresh Claude Opus instances fact-checked it twice, graded its style three times and ran a confidentiality check, and what they found is in the box at the end of this chapter. In stage 6 one Claude Opus instance read the primary sources and another built the settlement engine, Claude Fable drafted the changes, and fresh Claude Opus instances fact-checked the changed passages once, graded their style once and ran a confidentiality check. In stage 7 GPT Astra reviewed the manual and the release with the settlement work as its focus and the run-clock work at a lighter depth, a Claude Opus instance read the primary sources its findings rested on, corrected one of them and sharpened another, another changed the three engines and recorded every number that moved, and

Claude Fable drafted that version. After Gemini Argon's review two fresh Claude Opus instances tested each of its findings against the code and the primary sources, another changed the two engines for the Sunday-cash rows, the fee and the lendable-bar sensitivity, and Claude Fable drafted this version. Reviews reached me throughout as short visual summaries with each decision put as a multiple-choice question with a recommended option, which made the process fast and means the models framed most of the choices I made, which a reader should know. The prompts and conversations I had with the outside models are not published; the process record rests on their outputs, the project's own files and the recovered outside workpaper.

9.3 The errors, by direction of failure

The stage-1 log of 66 entries is split by who erred and who caught it. Twenty-one are errors by the outside models that Claude caught, among them, found last, the reciprocal-deposit allowance stated at a limit the FDIC had replaced before the draft was written. Seven are errors or gaps in Claude's work that the outside models caught or exposed, such as out-of-date model-risk guidance cited as current. Twenty-nine are Claude errors caught by Claude's own reviews and checks, among them a stale risk-free rate caught only by a direct check against the Federal Reserve's data. Two were caught only by me, and Section 9.4 has them with a third that came after stage 7. Seven more came from a round drafted outside the main workflow, among them a citation that put two authors' finding under a Federal Reserve working-paper number belonging to a different paper, and fresh Claude checks caught all seven, each fixed in the next draft.

Stage 2 added 23 entries: three design errors the engine-building model flagged at the first build rather than building around, seven from the fact check of the first draft, four from the style rounds, five from the second fact check that found the premise wrong, and four from the scoped checks of the rewritten passages. Stage 4's log holds three findings of the routes engine that changed the work and were not errors, the reference design's 0.0084-point miss of the standard first among them; 31 fact-check fails in a draft in which no model number was wrong; 14 style fails of 49 checks on the first draft; five slips the style edits introduced; and three more slips from the scoped grade before the loop stopped at its cap. Stage 6 added two entries from the settlement engine's first run, fixed before any number reached the chapter: the Monday-holiday cash gap measured from the wrong opening, 44 hours where the answer is 20, and a lendable share counted two ways inside one engine. The scoped fact check of the stage-6 passages then failed 20 of 204 items, half of them on inference, and the style round 27. Stage 7 added ten entries, the outside review's findings by severity, four blockers, five major and one minor, all confirmed, with one detail of the review itself corrected by primary sources; they are the first entries since stage 1 in which an outside family caught Claude, and the first in which numbers that same-family checks had verified turned out to measure the wrong quantity.

The two failure modes ran in opposite directions, and that is the finding of the experiment. The outside models erred toward the confident number: an invented parameter stated as fact, a target set so the design passed, a result dropped because it was inconvenient, a reliability headline true

under one copula. Claude erred toward caution and stale rules: figures barred without opening the primary source, guidance six months out of date, a rate four tenths of a point off, a review that called a result not reproducible without asking for the script. Both have one check, which is to date-stamp what moves and open the primary source, and the primary source is the arbiter, never another model. Same-family review is a consistency check and not an independent one, because the gates and audits, which found real errors at every round (29, 31, 11 and 10 required fixes in stage 1, then four defects in the audit), shared their drafter's blind spots on facts and framing; the cross-family review caught what they could not, in stage 1 on facts and in stage 7 on what the engines measured, where every number had been verified and the definitions behind five of them were wrong: the Sunday cash source, the debtor cohort, the peak debit, the buffer metric and the mutual's cash at closure. Stage 2 added a lesson the counts above understate. A draft can pass a fact check of 501 items with every table cell right and its premise still wrong, because the checks verified the cells and nobody had asked whether the cells carried the claim; the inference question has to be asked separately, and asked of the sentence a reader will quote.

9.4 What only I caught, and what only I decided

Only three entries in the whole log were caught by me alone, and all three were questions the models had not asked rather than errors in what they had done: the cash-access gap that became the settlement clock, told in the Chapter 6 box; how an operating account is actually paid, told in the Chapter 7 box; and, on rereading the thirteenth draft after the outside reviews, that the two clocks are two different risks which not every deposit carries, so that an uninsured tokenized balance has to be charged for both and credited as funding for less than a plain uninsured balance, which became Section 7.3.

That understates the human part, because most of what I contributed was decisions. The models surfaced most of the problems; somebody still had to decide what the work was, what it would claim, and what was verified enough to stand behind. I turned the paper into a test, refused to have a model's 20 percent capacity presented as my judgment, set the frictions floor at 10 basis points in stage 1 and at 5 with benchmarks in stage 2, and accepted a 45 basis point buffer premium only as a sweep point. I held the posture rules (generic naming, protection and not insurance, the views-my-own and disclosure lines) each time an outside version broke them, and rejected most of the fourth outside version and much of the ninth while keeping their good ideas and requiring each to be rebuilt in the project's own engine. I made the all-funded buffer the reference design, put reliability in the year the depositor's own bank fails under both copulas in the abstract, decided that an open verdict is not a pass, and asked for the second fact check of stage 2 that found the premise wrong. Without the two outside versions the stage-1 work would still cite out-of-date guidance, still exclude correct FDIC figures and still lead with a dearer design; without the second fact check the stage-2 premise would have stood.

9.5 What the models did well, and badly

They were fast and broad, and the first review covered empirical sources, liquidity and capital rules, receivership law and securities law in one pass. Cross-model challenge worked, since the original outside draft supplied the framing that survived every version, two risks in one balance and the point that unfunded guarantees fail together with the banks they cover, one outside version contributed the passive-versus-operating cash split and the pooled vehicle, and the other rebuilt the pooled model faithfully and contributed four ideas the work adopted. And in stage 4 a rule stated before the run produced a champion I would not have chosen, and the stage-4 draft reported it.

They also failed in ways the log makes specific, and Section 9.3 has most of them. Attribution drifted in both directions, a parameter a model chose written up as my judgment and a methods note that overstated how much I had checked by hand, so that left alone the models would have made me look both more responsible for their choices and more thorough than the record shows. Every style pass also needed a scoped fact check behind it, because sentence splits moved a citation, a figure swap left its caption reversed, and an added “so” turned a description into a false cause.

9.6 Lessons

1. Use more than one model family and make them review one another, because they fail in different directions and each catches what the other cannot see.
2. Make the primary source the arbiter, never another model, and check the library already held before trusting a search extract, because the figures barred as unverifiable sat in my own files.
3. Keep a written facts sheet with an explicit status for each item, verified, secondary or barred, and reopen the barred items when a primary source becomes available, since a do-not-reuse list is itself a set of claims that can be wrong.
4. Insist that every number comes from one reproducible engine, rerun it in an empty folder before publishing, and run a parity check against the figures recorded at the earlier stage’s close before any new result is read.
5. Set reliability standards and other targets before the run, in round numbers, because a target read off the design’s own result is not a standard, and state an exhaustion point as a standard rather than as a count of members.
6. Treat every parameter without an evidenced source as a sweep, never as a base case, because the invented parameters in the log were all stated as facts.
7. Date-stamp anything that moves, rates, guidance, thresholds, and recheck it on the day of publication, because the stale items in the log were all of that kind.
8. Write posture rules and disclaimers down and check them every round, because models regress on them.
9. Watch attribution in both directions, because the log shows it drifting each way (Section 9.5).
10. Ask what the problem left out, because the models answer the problem they are given, thoroughly, and the gap a practitioner notices first came from a practitioner’s question.

11. Put a scoped fact check behind every style pass, and a cap on the loop, because gate fixes create tells of their own and fresh graders each round are the protection.
12. Send the finished work to a model family that built none of it, and ask it what each number measures rather than whether it reproduces, because the stage-7 review reproduced every figure it recomputed and found that five of them measured the wrong thing.

9.7 Disclosure

Every empirical fact here traces to the public source cited. The fact checks of each stage and of this manual verified the named figures against the primary source where it could be reached, and the items they could not reach, the California collateral percentages quoted from a secondary reproduction of the statute among them, are listed as unverified in the process record. Every model number traces to one of the four engines in the repository, and Table 1 to the panel script; the routes engine checks the fourteen headlines it imports against the values recorded at the close of each earlier stage before any new result is read, and the settlement engine checks the twelve values it imports from the routes engine the same way (Appendix A). I used AI-assisted drafting, modeling and checking tools under my direction throughout, as this chapter describes. Text produced with these tools carries Anthropic's statistical watermark, which I disclose as a matter of transparency. The analysis, the judgments and any errors are mine.

> **What the models did here.** Stage 5 produced this manual. Claude Fable drafted it from the four stage drafts and the process record with the result frozen, and a fresh Claude Opus instance made about 842 checks on the draft and failed 24. No engine number was misquoted, but three statements built on engine numbers were wrong, among them a claim that a frictions quotation of about 10 basis points would close most of the gap between the two champions, when a straight-line extension of the engine's sweep, which stops at 10, puts half the gap near 15; a figure marker pointed at the wrong chart. A scoped recheck of the revision made about 340 checks and found nine new slips the edits had introduced. A first style round by another instance failed 35 items, with announced counts and enumeration scaffolds the most common pattern, in nine of them; a second round failed 17 and a scoped third round 7, no earlier fix was reverted, the long-sentence count was left at the cap, and a confidentiality check passed on all five points. Reorganizing the code for release changed no number in the results files, about 11,000 of them compared before and after, and a fresh-clone test found the comparison engine still needed a folder outside the repository, since fixed. Stage 6, the settlement engine and the Chapter 6 it rewrote, is told in that chapter's box, and stage 7, the outside review, in Section 9.1.

Chapter 10. The ask

The experiment's last step is human expert review, and the questions below are the ones I most want challenged, each with the chapter that sets it up. Comments and corrections can be posted as issues in the repository, and the engines can be rerun with other conventions in a few minutes.

1. The deciding question: may a national bank make a 3 percent entry capital contribution to a state-licensed excess-deposit mutual insurer, under what authority (the incidental powers of 12 U.S.C. 24(Seventh), the non-controlling investment rules of 12 CFR 5.36 and the Comptroller's interpretive letters on bank investments in insurers are where counsel would start), and does a Texas state bank reach it through the pledging carve-out for excess deposit insurance in Finance Code 34.304(c) (Chapters 4 and 5)? A yes makes the mutual the champion at 21.03 basis points; a no leaves the vehicle at 37.94 with no challenger.
2. The rule and the screens: is "the cheapest all-in price among routes passing every screen" the right rule, or should it weigh fat-tailed reliability, the lendable share or the dollars at a given bank? Should an open legal question count as a failure or be carried at a probability? Are these the right five screens, with a lendable share of 0.9 and a Gaussian 1 percent standard the right bars when the same designs fall short one time in four or five under the fat-tailed copula (Chapters 2 and 4)?
3. The mutual's form and frictions: should a bank mutual be an admitted insurer in the Ohio form, a risk retention group or a fund under enabling legislation, and how does each treat the capital contribution and the special assessment for solvency? What would a licensed carrier, a rating review and a tail investor charge against the 5 basis points assumed, given that the engine's sweep, which stops at 10, moves the price by about a basis point for each basis point of frictions, so that a credible quotation near 15 basis points would close half the gap between the two champions and one near 20 most of it (Chapters 5 and 8)?
4. Rule M and the trigger mechanics: would the FDIC run a purchase and assumption with a private fund paying the uninsured loss share for a $50 billion member rather than a $500 million one, and in a cluster, and is the receivership carry under the Massachusetts rule a timing cost rather than a subsidy? Can a vehicle pay par within stated business days against assignment of the claim and collect the advance dividend as assignee, or is the day-seven rule in which the depositor collects the dividend operationally cleaner (Chapters 3, 5 and 8)?
5. Tail calibration and the pricing measure: is an asset correlation of 0.20 to 0.30 defensible for bank failures, what evidence could choose between the Gaussian and the Student-t copula, and what observable would pin the risk-premium multiplier, given that a rating agency's capacity requirement sized to the fat-tailed copula would decide the capacity in practice (Chapters 2, 3 and 5)?
6. The weekend repo and the always-on share: does the qualified-financial-contract safe harbor work as read for a repo moved to a bridge bank over a weekend, does a multilateral netting contract among an all-bank consortium qualify under 12 U.S.C. 4402 to 4404 so that the agent's net position discharges the members' obligations, is 2 percent over three days enough haircut,

and is the 1.38 to 4.15 basis point bound the right way to score the facility's subsidy? Is a tenth of the balance the right ceiling for an always-on tier, and would depositors pay 75 to 240 basis points on the tier dollar for it (Chapter 6)?

7. The price as an internal charge, and the service yield: even if nobody builds any of this, is the self-insured cost of about 21 basis points, or the champion's 21.03, the right funds transfer pricing charge for a bank's uninsured corporate funding? Should that charge stack by clock, as Table 17 stacks it, so that an uninsured tokenized balance carries the run-clock charge and the settlement-clock charge together, and should the balance be credited as funding only for its lendable share? Would a published weekend secured rate give the desk a usable curve point? How should the earnings credit, the fees it offsets and the value of payment and credit services be measured, so that the depositor's advantage of Chapter 7 can be placed for a real balance (Chapters 2, 6 and 7)?
8. The replacement cost and the pilot: what would a regional bank pay to replace a large uninsured balance that moved to a fund or a stablecoin, and how much of the balance would in fact leave? Would a pilot of a dozen members in one state whose law accommodates the form, with the settlement rule agreed with the FDIC and the state regulator before the first member fails and the covered depositors' wire behavior in the next stress compared against uncovered cohorts at the same banks, answer the behavioral question (Chapters 1, 7 and 8)?
9. The settlement clock, for the people who can answer it. For the Reserve Banks: will the six-day schedule run on the announced timetable, what would decide the seven-day proposal, will collateralized intraday credit be there for a net debtor at a Sunday opening, and under seven-day operation will Reserve Bank credit carry weekend debits to Monday's securities opening, would a joint account for a weekend settlement arrangement among regional banks be approved under the 2017 guidelines and on what interest terms, and may a liquidity management transfer settle a netted weekend claim between banks that arose from tokenized deposits, or does the purpose rule confine it as I read it? For counsel: does the qualified-financial-contract stay run as read for the net creditors of a member closed on a Friday, does the consortium's netting contract qualify under the netting statute, and may a non-member reach the facility through a sponsoring member on the FICC pattern? For any bank that can observe one: what are the median and dispersion of the weekend peak net debit on its tokenized deposits, read from timestamped attempted and completed payments with their receivers and opening balances rather than from settled totals, the numbers that decide whether the facility can be nine tenths lendable with a member's cap that binds in 1 percent of weekends (Chapter 6)? At the assumed dispersion of 0.8 the median has to be at or below 1.56 percent of the balance, 3.12 percent at a dispersion of 0.5 and 0.61 at 1.2.

A counsel's letter against the capital contribution would move the champion to the vehicle, and I would then expect the mutual to be rebuilt in a form a national bank can join rather than abandoned. A credible quotation for the mutual's frictions near 20 basis points, or a rating agency's capacity requirement sized to the fat-tailed copula, would close most of the gap between the two champions, since the mutual's alternative design already costs 31.01. And a pilot in which covered

depositors wired anyway would say that the full-count reading of the replacement cost is wrong for the balances that matter, in which case every route on the list is a cost to the bank and the right answer is the cheapest route to a stable insured base, the reciprocal layer the market has already chosen. On the settlement clock, a bank's weekend record with a median peak net debit at or below 1.56 percent of its tokenized balance, and a dispersion no wider than the 0.8 assumed, would make the facility a product that keeps the balance nine tenths lendable with a cap that binds in 1 percent of weekends, provided the Sunday settlement runs on Reserve Bank credit against loans the bank already holds, and a record near the 5 percent assumed would leave it a product for a tenth of the balance until the public rail runs seven days and the Reserve Banks lend over the weekend. My hypothesis is that passive corporate cash, which the regionals now hold on costlier insured terms, would stay as a plain deposit if that deposit at its own bank were made as safe as a government fund, and that the price of making it so, on these numbers and if one legal question resolves, is about half of what a vehicle that does not use the members' own capital would charge.

Appendix A. Reproducing the numbers

Four engines produce every model number in this manual, and a fifth script produces Table 1 from FDIC bank-level data, all in the public repository that accompanies the manual. The clearing-cost engine (Chapter 3) prices the single-name and pooled vehicles, runs both copulas and computes reliability in the year the depositor's own bank fails. The mutual engine (Chapter 5) extends it to a member pool with unequal weights, a mutual layer, a re-attached tail and a settlement draw, and imports the clearing-cost engine rather than copying it; the panel script beside it pulls the FDIC BankFind data for Table 1 (Chapter 1) and caches the pull so that offline runs are byte-identical. The routes engine (Chapters 4 to 7) imports both earlier engines by path, copies no code from either, prices the nine routes, scores the screens, runs the rule and draws Figures 1, 4 to 9, 11, 15 and 16. The settlement engine (Chapter 6) imports the routes engine by path, copies no code from it, and runs the same-weekend factor model over an open debtor cohort, the median sensitivity, the closed-window calendar with its elapsed-hours check, the buffer sensitivity, the sponsored non-member price, the Monday cash gap on both intervals and the Sunday cash source, checking the twelve values it imports against the routes engine's results file before any new result is read, and draws Figures 12 to 14.

From the repository root (github.com/chihchen22/cash-that-never-sleeps), after `pip install -r requirements.txt` under Python 3.12, the commands are these:

```
python -I -B model/clearing_cost.py
python -I -B model/mutual_tail.py
python -I -B model/uninsured_panel.py
python -I -B model/routes.py
python -I -B model/settlement.py
python -I -B tests/fixtures.py
```

`node tests/parity.mjs` then checks the browser dashboard in `docs/` against the Python. All were run from a fresh copy of the repository, and the first five reproduce every results file: `model/results_clearing_cost.md`, `results_mutual_tail.md`, `results_panel.md`, `results_routes.md` and `results_settlement.md`.

The reference run used Python 3.12.13 and numpy 2.5.2 with matplotlib, takes a few minutes, and writes each engine's results file and the figures. Seeds are 20261110 for the weekend outflow Monte Carlo check, a million draws that agree with the closed-form cap-binding figures of Chapter 6 within three standard errors, 20261111 for the illustrative weekend path of Figure 11, and 20261120 for the settlement engine's same-weekend Monte Carlo of 500,000 weekends; the imported engines carry their own. Before any new result is read, the routes engine reads fourteen headline figures from the imported engines in memory, with no number typed in, and compares each with the value recorded at the close of each earlier stage at the precision recorded; it then checks its own sizing wrappers against the two engines' iso-reliability rows and that the bearer splits of Figure 6 sum to the self-insured cost. The parity table and the map from each chapter of this manual to the sections of the results files are in the repository README.

Appendix B. Parameters

Parameter	Value	Label
Balance above the insured limit	$100 million	author assumption
lambda_P; mu; LGD; a; D	0.25%; 3; 25%; 50% (linked to LGD, a = (2/3)(1 - LGD)); 2 years	author assumption (the clearing-cost engine's base)
r; s_b	4.3%; 1.0%	author assumption (r rounded from a reported 3-month bill rate, 15 October 2026)
rho	0.20 for prices, 0.30 for reliability; Gaussian copula, Student-t(5) disclosed	author assumption
Screen S2 threshold; screen S5 standard	lendable share at least 0.9; P(not paid in full given own bank fails) at most 1%, Gaussian, rho 0.30	author assumption
R1 over-collateralization	110% of the deposit	reported figure, Cal. Gov. Code 53651 and 53652 (via a secondary source)
R1 cost, pledged from securities held; if bought	10.0 to 30.0; 150.0 to 300.0 loan-to-security spread on 110%	author assumption (Chapter 4)
R2 public-unit letter fee by district	10.5 to 18.0; to 35.0 variable-balance at one district and 20.0 other purposes at another; activity stock 10.0 bps of the letter where required (not priced)	reported figure, FHLBank Des Moines 2026; FHLBank San Francisco 2025; FHLBank Boston 2022; FHLBank Topeka 2023
R3 placement fee; rate uplift	12.5; 0.0, 10.0, 25.0 (base 10.0)	reported figure, Chain Bridge Bancorp 2026 (one bank's disclosure); author assumption
R4 replacement spread	50.0 to 300.0	author assumption (Chapter 3)
R7 always-on share; haircut	20%; 2% on bills	author assumption (Chapter 6)
R7 terms: credit; access; encumbrance spread; frictions	0.0 to 1.0; 2.0 to 5.0; 25.0 to 100.0 on the bill slice; 3.0 to 8.0	author assumption (Chapter 6)
R7 weekend net outflow / always-on tier	lognormal, median 25%, sigma 0.8 (sweep 0.5, 0.8, 1.2); 52 weekends a year, independent	author assumption (illustrative)
R8 shared-infrastructure saving	50% of the smaller structure's access term plus 1.0 bp of frictions; swept from none to the full access term	author assumption

Parameter	Value	Label
R9 prefunded share; joint-account balance earns IORB	20% (by share 5% to 50%); yes in the base, no as a sensitivity	author assumption; reported figure for the live network's joint account (Board order effective 7 June 2023), no general rule for other joint accounts
R9 IORB; 3-month bill (coupon-equivalent)	3.90%; 4.13%	reported figures, Board of Governors (IORB since 17 September 2026) and U.S. Department of the Treasury, 3 November 2026
R9 terms: credit; access; loan spread over bills; frictions	0.0 to 0.5; 2.0 to 6.0; 25.0 to 100.0 on the position; 2.0 to 6.0	author assumption (Chapter 4)
R9 per-payment fee; per-payment limit; liquidity management transfer limit	$0.045 per payment sent, nothing to receive; $10 million; $500 million	reported figures, The Clearing House RTP fee schedule 2026; FedNow operating procedures v3.7, 13 October 2026
Mutual tail: rule-M assessment advance	advanced at closure, repaid within 7 days at r + s_b (5.3%); funded commitment T plus the advance capacity, 7.00% of covered excess for the reference design	author assumption (the mutual engine's)
Banks: assets; liabilities / assets; covered excess / liabilities	$50bn and $5bn; 90%; 20%	author assumption
Reciprocal allowance (non-brokered)	50% of the first $1bn of liabilities, 40% of the next $9bn, 30% above, at most $30bn; agent institution well capitalized, CAMELS 1 to 3	reported figure, 12 CFR 337.6(e); FDIC 2026a
Replacement spread s_rep; retention cells	20.0, 40.0, 60.0, 80.0; none, V 100 with a run cut of 1%, V 150 with a run cut of 5%	author assumption (the mutual engine's)
Mutual design (R6)	N 150, theta_m 3% paid in, special assessment cap 2 x the annual assessment, reserve opportunity cost 100.0 bps, f_m 5.0 bps, q_acq 0.5	author assumption (the mutual engine's)
Pooled vehicle (R5)	50 banks at 2% each; buffer 20% (reference) or 22% (sized); lock-up premium 10 bps; frictions 15 bps; state-dependent LGD 15% to 40%	author assumption (the clearing-cost engine's)

Parameter	Value	Label
Settlement engine (Chapter 6): members; correlation; copulas; bind targets; hour scalings; buffer; sponsor terms; bridging rate	20 (also 10, 40), equal size, an open debtor cohort; 0.00 to 0.50; Gaussian and Student-t(5); 1% of weekends and 1 in 520; linear and square root of hours / 50; 0% to 40%; 2.0 to 6.0 per tier dollar plus 1.0 to 3.0; 4.3% plus 10.0 to 50.0	author assumption (illustrative); the calendar constants, the announced schedule and the QFC stay are reported figures, sourced in the settlement engine's results, Section 1
Settlement engine: weekend peak net debit; median set; Sunday cash source; Fedwire Securities hours	lognormal peak debit to all counterparties, median 25% of the tier (5% of the balance), sigma 0.8; medians 1% to 5% of the balance by sigma 0.5, 0.8, 1.2; daylight overdraft fee 0 bps collateralized (50 uncollateralized as a sensitivity), repaid Monday 12:00; 8:30 a.m. opening and 3:15 p.m. secondary-market cutoff	author assumption (illustrative); the overdraft fees and the securities hours are reported figures (Board of Governors 2023d; Federal Reserve Financial Services 2026g)

Table 18. Parameters of the engines, with their labels; reported figures are not model numbers. The R2 row's two-district wording follows the fee schedules cited; the routes engine's Section 1 label still reads "20.0 other purposes at one district". Source: the routes engine's results, Section 1; the clearing-cost engine's results, Section 1; the mutual engine's results, Section 1; the settlement engine's results, Section 1.

S2 bar	S2 verdicts that change (R7, R8, R9)	Settlement problem, open items resolved, S2 applied	Both problems together, open items resolved	Cap-test boundary at a dispersion of 0.8, grossed-up bills
0.90 (rule of record)	R7 fail, R8 fail, R9 fail	none	none	1.52% (fails at the 5.0% median)
0.85	R7 fail, R8 fail, R9 fail	none	none	2.29% (fails)
0.80	R7 pass, R8 fail, R9 pass	R7 at 22.00; challenger R9 at 25.35	none	3.05% (fails)
0.75	R7 pass, R8 pass, R9 pass	R7 at 22.00; challenger R9 at 25.35	R8 at 41.84	3.81% (fails)

Table 19. The lendable bar as a sensitivity: the S2 verdicts, the rule and the cap-test boundary at each bar, at the published route parameters with the facility's Sunday cash as Reserve Bank credit against loans already pledged (Section 4.9). The verdicts of the other variants and the run-clock champions (R5 sized strictly, R6 with the open items resolved) do not move at any bar. The boundary is the largest median weekend peak net debit, as a share of the balance, at which a member's cap binds in at most 1 percent of weekends, counting the grossed-up bills. Source: the routes engine's results, Section 9b; the settlement engine's results, Table A6.

Disclaimer

The views expressed are my own and do not represent those of any employer or organization with which I am affiliated. The parameters and results are illustrative and are not estimates for any bank, mutual, vehicle, facility or investor, nor are they a forecast; the designs are indicative and for discussion only. Nothing here is legal, investment or regulatory advice; readers should consult qualified professionals before relying on it. The private protection and the private arrangements described are not deposit insurance and are not insured by any government agency. This manual is shared for discussion and challenge and may change in later versions.

References

American Mutual Share Insurance Corporation and Excess Share Insurance Corporation. 2026. Annual Statements for the year ended 31 December 2025, filed with the Ohio Department of Insurance (NAIC company codes 12700 and 10003).

Axos Bank. 2023. “Axos Bank Provides FDIC Deposit Insurance Coverage Up to $150MM with Insured Cash Sweep.” Press release, 13 March, as reproduced by InsuranceNewsNet. https://insurancenewsnet.com/oarticle/axos-bank-provides-fdic-deposit-insurance-coverage-up-to-150mm-with-insured-cash-sweep

Battey, P. 2013. “Competitor Distorts Our Insured Cash Sweep Service.” American Banker, BankThink, 6 February. https://www.americanbanker.com/opinion/competitor-distorts-our-insured-cash-sweep-service

Berger, A. N., N. H. Miller, M. A. Petersen, R. G. Rajan and J. C. Stein. 2005. “Does Function Follow Organizational Form? Evidence from the Lending Practices of Large and Small Banks.” Journal of Financial Economics 76(2): 237-269. https://doi.org/10.1016/j.jfineco.2004.06.003

BlackRock. 2024. “BlackRock Launches Its First Tokenized Fund, BUIDL, on the Ethereum Network.” Press release, 20 March. https://www.nasdaq.com/press-release/blackrock-launches-its-first-tokenized-fund-buidl-on-the-ethereum-network-2024-03-20

BNY. 2025. “DTCC’s FICC and BNY Launch Collateral-in-Lieu Service via BNY’s Global Collateral Platform.” Press release, 23 December. https://www.bny.com/corporate/global/en/about-us/newsroom/press-release/dtccs-ficc-and-bny-launch-collateral-in-lieu-service-via-bnys-global-collateral-platform.html

BNY. 2026. “BNY Completes Tokenized U.S. Treasury Triparty Repo Workflow with Citadel Securities.” Press release, 13 October. https://www.bny.com/corporate/global/en/about-us/newsroom/press-release/bny-completes-tokenized-us-treasury-triparty-repo-workflow-with-citadel-securities.html

Board of Governors of the Federal Reserve System. 2017. “Guidelines for Evaluating Joint Account Requests.” Final guidance, 82 FR 41951, 5 September. https://www.federalreserve.gov/paymentsystems/joint_requests.htm

Board of Governors of the Federal Reserve System. 2023a. Financial Stability Report, May. https://www.federalreserve.gov/publications/files/financial-stability-report-20230508.pdf

Board of Governors of the Federal Reserve System. 2023b. Financial Stability Report, October. https://www.federalreserve.gov/publications/files/financial-stability-report-20231020.pdf

Board of Governors of the Federal Reserve System. 2023c. “Order Concerning the Payment of Interest on RTP Joint Account Balances.” Effective 7 June. https://www.federalreserve.gov/paymentsystems/files/interest_on_rtp_joint_balances.pdf

Board of Governors of the Federal Reserve System. 2023d. Federal Reserve Policy on Payment System Risk, as amended effective 20 July. https://www.federalreserve.gov/paymentsystems/files/psr_policy.pdf

Board of Governors of the Federal Reserve System. 2025a. “Federal Reserve Board announces expanded operating days of two large-value payments services, Fedwire Funds Service and the National Settlement Service (NSS), to include Sundays and weekday holidays.” Press release, 9 October. https://www.federalreserve.gov/newsevents/pressreleases/other20251009a.htm

Board of Governors of the Federal Reserve System. 2025b. “Federal Reserve Action To Expand Fedwire Funds Service and National Settlement Service Operating Hours.” Notice, 90 FR 51356, 17 November (document 2025-19942); text as published by the Board. https://www.federalreserve.gov/newsevents/pressreleases/files/other20251009a1.pdf

Board of Governors of the Federal Reserve System. 2026a. Assets and Liabilities of Commercial Banks in the United States, H.8 release; deposits at large and small domestically chartered commercial banks (FRED series DPSLCBW027SBOG and DPSSCBW027SBOG), observations through 21 October 2026; H.8 notice of 31 March 2023 on the treatment of FDIC bridge banks. https://fred.stlouisfed.org/series/DPSLCBW027SBOG and https://fred.stlouisfed.org/series/DPSSCBW027SBOG

Board of Governors of the Federal Reserve System. 2026b. Financial Stability Report, May. https://www.federalreserve.gov/publications/files/financial-stability-report-20260508.pdf

Board of Governors of the Federal Reserve System. 2026c. Financial Accounts of the United States, Z.1, money market fund shares held by households and nonprofit organizations (FRED series MMFSABSHNO) and by nonfinancial corporate business (FRED series BOGZ1FL103034000Q), 2022 Q4 to 2025 Q4. https://fred.stlouisfed.org/series/MMFSABSHNO and https://fred.stlouisfed.org/series/BOGZ1FL103034000Q

Board of Governors of the Federal Reserve System. 2026d. Interest on Reserve Balances (IORB) rate, 3.90 percent effective 17 September 2026, policy rates data feed, accessed 3 November 2026. https://www.federalreserve.gov/monetarypolicy/reserve-balances.htm

Board of Governors of the Federal Reserve System, Office of the Comptroller of the Currency and Federal Deposit Insurance Corporation. 2026. Revised Guidance on Model Risk Management. SR 26-2, 17 April (superseding SR 11-7 and SR 21-8). https://www.federalreserve.gov/supervisionreg/srletters/SR2602.htm

Caglio, C., J. Dlugosz and M. Rezende. 2024. “Flight to Safety in the Regional Bank Crisis of 2023.” Working paper, 8 January. SSRN 4457140. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4457140

Chain Bridge Bancorp, Inc. 2026. Annual Report on Form 10-K for the fiscal year ended 31 December 2025 (reciprocal deposit network fee). U.S. Securities and Exchange Commission, CIK 0001392272. https://www.sec.gov/Archives/edgar/data/1392272/000162828026020177/chnbrdg-20251231.htm

Chu, K., N. Frazier, C. Martin, N. T. Agarwal, J. Kuo, J. Northwood, T. Oey, C. Reed, A. Rodrigue and A. Starr. 2026. “Dissecting Depositor Flight: An Analysis of the Spring 2023 Bank Failures.” FDIC Center for Financial Research Staff Studies, May. https://www.fdic.gov/center-financial-research/staff-studies-may-2026.pdf

Cipriani, M., T. M. Eisenbach and A. Kovner. 2024. “Tracing Bank Runs in Real Time.” Federal Reserve Bank of New York Staff Report No. 1104, May (revised December 2024 and June 2026). https://doi.org/10.59576/sr.1104. See also “Anatomy of the Bank Runs in March 2023,” Liberty Street Economics, 20 December 2024.

Circle Internet Financial. 2023. Statement on USDC reserves held at Silicon Valley Bank, 10 March. USDC prices for 10 to 13 March 2023 from public market data.

Colorado Revised Statutes 11-105-404(1)(c)(II), as amended by HB 09-1053 (effective 5 August 2009); Colorado Revised Statutes 2024, official uncertified printout.

Correia, S., S. Luck and E. Verner. 2026. “Bank Runs With and Without Bank Failure.” Federal Reserve Bank of New York Staff Report No. 1198, July. Also arXiv:2601.20285, v3, 13 July 2026.

Cortés, K. R. and P. E. Strahan. 2017. “Tracing Out Capital Flows: How Financially Integrated Banks Respond to Natural Disasters.” Journal of Financial Economics 125(1): 182-199. https://doi.org/10.1016/j.jfineco.2017.04.011

Cross, M. 2023. “Appetite for expanded FDIC insurance is strong. Will it last?” National Mortgage News, 11 April. https://www.nationalmortgagenews.com/news/appetite-for-expanded-fdic-insurance-is-strong-will-it-last

Depositors Insurance Fund. 2025. Annual Report for the fiscal year ended 31 October 2025. Massachusetts. https://difxs.com/wp-content/uploads/2026/03/DIF_Annual-2025.pdf

Dime Community Bank. 2026. “FDIC Extended Insurance” (per-client limits for ICS and CDARS placements). Accessed 4 November 2026. https://www.dime.com/extended-deposit-insurance

Federal Deposit Insurance Corporation. 1984. “FDIC to Pay Off Depositors in Heritage Bank, Anaheim, California.” Press release PR-20-84, 19 March. https://archive.fdic.gov/view/fdic/11336/fdic_11336_DS1.pdf

Federal Deposit Insurance Corporation. 1989. “FDIC Transfers Insured Deposits of First Service Bank for Savings, Leominster, Massachusetts.” Press release PR-68-89, 31 March. https://archive.fdic.gov/view/fdic/10170/fdic_10170_DS1.pdf

Federal Deposit Insurance Corporation. 1990. “FDIC Transfers Deposits of New England Allbank for Savings, Gardner, Massachusetts.” Press release PR-233-90, 12 December. https://archive.fdic.gov/view/fdic/10078/fdic_10078_DS1.pdf

Federal Deposit Insurance Corporation. 1991. “FDIC Approves Assumption of Deposits of Bank Five for Savings, Arlington, Massachusetts.” Press release PR-138-91, 20 September. https://archive.fdic.gov/view/fdic/9711/fdic_9711_DS1.pdf

Federal Deposit Insurance Corporation. 1992a. “FDIC Approves Assumption of Deposits of Attleboro Pawtucket Savings Bank, Pawtucket, Rhode Island.” Press release PR-117-92, 21 August. https://archive.fdic.gov/view/fdic/9448/fdic_9448_DS1.pdf

Federal Deposit Insurance Corporation. 1992b. “FDIC Approves Assumption of Deposits of Plymouth Five Cents Savings Bank, Plymouth, Massachusetts.” Press release PR-131-92, 18 September. https://archive.fdic.gov/view/fdic/9462/fdic_9462_DS1.pdf

Federal Deposit Insurance Corporation. 1994. “FDIC Approves Assumption of Insured Deposits of Ludlow Savings Bank, Ludlow, Massachusetts.” Press release, 21 October (PR-67-94 in the archive record). https://archive.fdic.gov/view/fdic/465/fdic_465_DS1.pdf

Federal Deposit Insurance Corporation. 1997. “The Mutual Savings Bank Crisis.” In History of the Eighties: Lessons for the Future, Vol. 1, Ch. 6. https://www.fdic.gov/resources/publications/history-eighties/volume-1/history-80s-volume-1-part2-06.pdf

Federal Deposit Insurance Corporation. 2008. “Question and Answer Guide: IndyMac Bank, F.S.B., Pasadena, CA.” July. https://www.fdic.gov/bank-failures/question-and-answer-guide-indymac-bank-fsb-pasadena-ca

Federal Deposit Insurance Corporation. 2018. 2018 Small Business Lending Survey. October. https://www.fdic.gov/publications/small-business-lending-survey

Federal Deposit Insurance Corporation. 2023a. Options for Deposit Insurance Reform. 1 May. https://www.fdic.gov/analysis/options-deposit-insurance-reforms/report/options-deposit-insurance-reform-full.pdf

Federal Deposit Insurance Corporation. 2023b. “FDIC Creates a Deposit Insurance National Bank of Santa Clara to Protect Insured Depositors of Silicon Valley Bank.” Press release PR-16-2023, 10 March. https://www.fdic.gov/news/press-releases/2023/pr23016.html

Federal Deposit Insurance Corporation. 2023c. “FDIC’s Supervision of Signature Bank.” 28 April. https://www.fdic.gov/news/press-releases/2023/pr23033a.pdf

Federal Deposit Insurance Corporation. 2024. 2024 FDIC Small Business Lending Survey. October. https://www.fdic.gov/publications/2024-report-small-business-lending-survey

Federal Deposit Insurance Corporation. 2026a. “Reciprocal Deposits: Implementing the 21st Century ROAD to Housing Act.” Interim final rule, 91 FR 56022, 1 September; Financial Institution Letter FIL-52-2026, 27 August. https://www.fdic.gov/news/financial-institution-letters/2026/interim-final-rule-reciprocal-deposits

Federal Deposit Insurance Corporation. 2026b. Quarterly Banking Profile, second quarter 2026, time-series spreadsheet (estimated insured deposits as a percent of domestic deposits, by asset size group and for community banks), 2022 Q4 to 2026 Q2. https://www.fdic.gov/quarterly-banking-profile

Federal Deposit Insurance Corporation. 2026c. BankFind Suite, financial data API (financials, failures and history endpoints), bank-level data at 31 December 2022, 30 June 2023, 31 December 2024 and 30 June 2026, pulled 2 November 2026 (UTC). https://api.fdic.gov/banks/financials (formerly https://banks.data.fdic.gov/api/financials)

Federal Deposit Insurance Corporation. 2026d. BankFind Suite, financial data API (financials endpoint), gross loans and leases and total deposits by bank at 31 December 2007, 31 December 2019 and 31 December 2025, accessed 6 November 2026. https://api.fdic.gov/banks/financials

Federal Deposit Insurance Corporation. n.d. “Dividends from Failed Banks.” https://closedbanks.fdic.gov/dividends/

Federal Home Loan Banks Office of Finance. 2023. “Office of Finance Announces First Quarter 2023 Combined Operating Highlights for the Federal Home Loan Banks.” 28 April (advances at 31 March 2023). https://www.fhlb-of.com

Federal Home Loan Banks Office of Finance. 2026. “Federal Home Loan Banks Selected Financial Highlights” (advances at 31 December 2022, 2023 and 2025 and 30 June 2026). https://www.fhlb-of.com

Federal Reserve Bank of New York. 2026. Discount Window, New York district: access hours for depository institutions. Accessed 3 November 2026. https://www.frbdiscountwindow.org/pages/select-your-district/new-york

Federal Reserve Bank of Cleveland. 1986. “Unfoldings in Ohio: 1985.” 1985 Annual Report. https://www.clevelandfed.org/-/media/project/clevelandfedtenant/clevelandfedsite/publications/annual-reports/ar-1985-unfoldings-in-ohio-pdf.pdf

Federal Reserve Bank of St. Louis. 2026a. Market Yield on U.S. Treasury Securities at 3-Month Constant Maturity, Quoted on an Investment Basis (FRED series DGS3MO), observation for 15 October 2026.

Federal Reserve Bank of St. Louis. 2026b. Market Yield on U.S. Treasury Securities at 1-Month, 3-Year and 5-Year Constant Maturity, Quoted on an Investment Basis (FRED series DGS1MO, DGS3 and DGS5), observations for 28 October 2026.

Federal Reserve Bank of St. Louis. 2026c. Term Premium on a 3 Year and on a 5 Year Zero Coupon Bond (FRED series THREEFYTP3 and THREEFYTP5, Kim and Wright model estimates), observations for 23 October 2026.

Federal Reserve Banks. 2025. 2025 Report on Employer Firms: Findings from the 2024 Small Business Credit Survey. March. https://doi.org/10.55350/sbcs-20250327

Federal Reserve Financial Services. 2025. “FedNow Service will raise transaction limit to $10 million to meet increased demand, unlocking higher-value use cases.” Press release, 5 September. https://www.frbservices.org/news/press-releases/090525-fednow-transaction-limit-increase

Federal Reserve Financial Services. 2026a. “Fedwire Funds Service and National Settlement Service operating hours.” Accessed 31 October 2026. https://www.frbservices.org/resources/financial-services/wires/operating-hours.html

Federal Reserve Financial Services. 2026b. FedNow Service Operating Procedures, version 3.7, October. https://www.frbservices.org/wp-content/uploads/101326-fednow-service-operating-procedures.pdf

Federal Reserve Financial Services. 2026c. “Soaring volume, participation mark three years of the FedNow Service.” Fed360, 15 July. https://www.frbservices.org/news/fed360/issues/071526/fednow-service-soaring-volume-three-years

Federal Reserve Financial Services. 2026d. Operating Circular 8, “Funds Transfers Through the FedNow Service,” effective 1 April 2026 (Sections 2.24, 2.25, 9.7.1 and 15.3 on joint accounts and liquidity management transfers). https://www.frbservices.org/wp-content/uploads/040126-operating-circular-8.pdf

Federal Reserve Financial Services. 2026e. “FedNow Service operating hours.” Accessed 31 October 2026. https://www.frbservices.org/resources/financial-services/fednow-service-resources/fednow-service-operating-hours/

Federal Reserve Financial Services. 2026f. “Holiday Schedules.” Accessed 3 November 2026. https://www.frbservices.org/about/holiday-schedules/

Federal Reserve Financial Services. 2026g. “Fedwire Securities Service Schedule.” Accessed 3 November 2026. https://www.frbservices.org/resources/financial-services/fedwire-securities-resources/fedwire-securities-service-schedule-fedpayments-manager-securities-hours-availability/

FHLBank Boston. 2022. “Letters of Credit Fee Schedule.” Effective July 2022. https://www.fhlbboston.com/wp-content/uploads/2025/05/products-letters-of-credit-fee-schedule.pdf

FHLBank Des Moines. 2026. “Letters of Credit.” Product brochure, updated 2026. https://www.fhlbdm.com/webres/File/products-services/LOC_Product_Brochure_2026.pdf

FHLBank San Francisco. 2025. “Letters of Credit.” Product sheet (issuance and maintenance fees), December. https://assets.fhlbsf.com/x/ef83bb390c/fhlbsf_letters_of_credit.pdf

FHLBank Topeka. 2023. “Letters of Credit.” Member presentation. https://www.fhlbtopeka.com/doccenter/e73f150fd3734a46bbc6d4b3d16a9406

Gallagher, M. L. 2020. “Approval of the Merger of The Co-operative Central Bank with and into the Depositors Insurance Fund.” Massachusetts Division of Banks, letter to D. Elliott, 12 March. https://malegislature.gov/Reports/9307/SD2875%20–%20DOB%20Merger%20Cover%20Letter.pdf

Garratt, R. and H. S. Shin. 2023. “Stablecoins versus Tokenised Deposits: Implications for the Singleness of Money.” BIS Bulletin No. 73, 11 April.

Gilje, E. P., E. Loutskina and P. E. Strahan. 2016. “Exporting Liquidity: Branch Banking and Financial Integration.” Journal of Finance 71(3): 1159-1184. https://doi.org/10.1111/jofi.12387. Earlier as NBER Working Paper 19403. https://www.nber.org/papers/w19403

Gruenberg, M. J. 2022. “Remarks to the 10th Community Banking Research Conference.” Federal Deposit Insurance Corporation, 28 September. https://www.fdic.gov/news/speeches/2022/spsep2822.html

Huberdeau-Reid, D. A. and G. G. Pennacchi. 2025. “Adverse Selection in Deposit Insurance Following the 2023 Banking Crises.” Poster, American Economic Association Annual Meeting. https://topcat.aeaweb.org/conference/2025/program/paper/Q99Qy6H6

Hughes, H. 1985. “Proclamation of May 14, 1985.” Laws of Maryland 1985, Archives of Maryland vol. 760, p. 4221. https://www.msa.maryland.gov/megafile/msa/speccol/sc2900/sc2908/000001/000760/html/am760–4221.html

IntraFi. 2022. ICS Deposit Placement Agreement (version 2022-11), as published by a network member bank. https://westernalliancebank.com/sites/default/files/2023-06/ICS-Deposit-Placement-Agreement-230612.pdf

IntraFi. n.d. “Build Large Dollar Relationships” (coverage described as multi-million-dollar, no dollar figure given). Accessed 4 November 2026. https://www.intrafi.com/build-large-dollar-relationships

Investment Company Institute. 2026. “Money Market Fund Assets.” Weekly release for the week ended 28 October. https://www.ici.org/research/stats/mmf

Kim, D. H. and J. H. Wright. 2005. “An Arbitrage-Free Three-Factor Term Structure Model and the Recent Behavior of Long-Term Yields and Distant-Horizon Forward Rates.” Board of Governors of the Federal Reserve System, Finance and Economics Discussion Series 2005-33. https://www.federalreserve.gov/pubs/feds/2005/200533/200533abs.html

Kim, E. T., S. Kundu and A. Purnanandam. 2024. “The Economics of Market-Based Deposit Insurance” (later retitled “The Economics of Network-Based Deposit Insurance”). Working paper, September. https://anderson-review.ucla.edu/wp-content/uploads/2025/01/Deposit-Network-Kundu.pdf

Lu, L. and D. Scida. 2026. “Reciprocal Deposit Networks and Funding Stability.” Federal Reserve Bank of Boston, Supervisory Research and Analysis Working Paper 26-02, August. https://www.bostonfed.org/publications/risk-and-policy-analysis/2026/reciprocal-deposit-networks-and-funding-stability.aspx

Luck, S., M. Plosser and J. Younger. 2023. “Bank Funding during the Current Monetary Policy Tightening Cycle.” Liberty Street Economics, Federal Reserve Bank of New York, 11 May. https://libertystreeteconomics.newyorkfed.org/2023/05/bank-funding-during-the-current-monetary-policy-tightening-cycle/

Maryland Comptroller. 1990. Annual Report of the Comptroller, 1990. Archives of Maryland vol. 354, p. 54. https://msa.maryland.gov/megafile/msa/speccol/sc2900/sc2908/000001/000354/html/am354–54.html

Morningstar DBRS. 2025. “Global Methodology for Rating Government Related Entities.” 2 June (superseding the version of 15 April 2024; extended to deposit insurance and guarantee funds). https://dbrs.morningstar.com/research/455464/global-methodology-for-rating-government-related-entities; announcement of 2 June 2025, https://dbrs.morningstar.com/research/455465

National Credit Union Administration. 2002. Legal opinion 02-0269A, May (private excess share insurance). https://ncua.gov/regulation-supervision/legal-opinions/2002/guaranteeing-deposits

Office of Financial Research. 2016. “A Map of Collateral Uses and Flows.” OFR Working Paper 16-06. https://www.financialresearch.gov/working-papers/files/OFRwp-2016-06_Map-of-Collateral-Uses.pdf

Office of Financial Research. 2025. “Treasury Tri-party Repo Pricing.” OFR Working Paper 25-07, 30 September. https://www.financialresearch.gov/working-papers/files/OFRwp25-07_treasury-tri-party-repo-pricing.pdf

Nguyen, H.-L. Q. 2019. “Are Credit Markets Still Local? Evidence from Bank Branch Closings.” American Economic Journal: Applied Economics 11(1): 1-32. https://doi.org/10.1257/app.20170543

Petersen, M. A. and R. G. Rajan. 1994. “The Benefits of Lending Relationships: Evidence from Small Business Data.” Journal of Finance 49(1): 3-37. https://doi.org/10.1111/j.1540-6261.1994.tb04418.x

Pulkkinen, T. E. and E. S. Rosengren. 1993. “Lessons from the Rhode Island Banking Crisis.” New England Economic Review, Federal Reserve Bank of Boston, May/June: 3-12. https://www.bostonfed.org/economic/neer/neer1993/neer393a.pdf

Rose, J. 2024. “The Speed of Discount Window Lending: A Look Back at 1985.” Federal Reserve Bank of St. Louis, On the Economy, 23 May. https://www.stlouisfed.org/on-the-economy/2024/may/speed-discount-window-lending-look-back-1985

rwa.xyz. 2026. Tokenized Treasuries, distributed value, July. https://app.rwa.xyz/treasuries

Ryfe, D. and A. Saretto. 2023. “Reciprocal deposit networks provide means to exceed FDIC’s $250,000 account cap.” Dallas Fed Economics, Federal Reserve Bank of Dallas, 28 November (updated 6 September 2024). https://www.dallasfed.org/research/economics/2023/1128

Stein, J. C. 2002. “Information Production and Capital Allocation: Decentralized versus Hierarchical Firms.” Journal of Finance 57(5): 1891-1921. https://doi.org/10.1111/0022-1082.00483

The Clearing House. 2024. “Higher $10 Million RTP Network Transaction Limit Empowers New Uses.” Press release, 4 December (limit effective 9 February 2025). https://www.theclearinghouse.org/payment-systems/Articles/2024/12/Higher_10_Million_RTP_Network_Transaction_Limit_Empowers_New_Uses_12-04-2024

The Clearing House. 2026. RTP Operating Rules, effective 1 November 2026 (Rules I.E, II.C.2, VI.C to VI.G); RTP Participant Fee Schedule, effective 1 January 2026. https://www.theclearinghouse.org/-/media/New/TCH/Documents/Payment-Systems/RTP/RTP-Operating-Rules-Effective-November-1-2026.pdf and https://www.theclearinghouse.org/-/media/New/TCH/Documents/Payment-Systems/RTP/RTP_Participant_Fee_Schedule_01-01-26.pdf

Todd, W. F. 1994. “Lessons from the Collapse of Three State-Chartered Private Deposit Insurance Funds.” Federal Reserve Bank of Cleveland, Economic Commentary, 1 May. https://www.clevelandfed.org/publications/economic-commentary/1994/ec-19940501-lessons-from-the-collapse-of-three-state-chartered-private-deposit-insurance-funds; also https://fraser.stlouisfed.org/files/docs/historical/frbclev/econcomm/econcomm_19940501.pdf

U.S. Department of the Treasury. 2026. Daily Treasury Bill Rates, 13-week bill, 3 November 2026 (bank discount 4.03 percent; coupon equivalent 4.13 percent). https://home.treasury.gov/resource-center/data-chart-center/interest-rates/

U.S. Securities and Exchange Commission. 2025. Order approving a proposed rule change by the Fixed Income Clearing Corporation, as modified by Partial Amendment No. 1, to establish a collateral-in-lieu offering within the Sponsored GC Service (File No. SR-FICC-2025-019), describing Government Securities Division Rule 3A on sponsored membership. Release No. 34-104374, 12 December. https://www.sec.gov/files/rules/sro/ficc/2025/34-104374.pdf

U.S. Securities and Exchange Commission. 2026. Order Extending the Phase-In Termination Date for the Security-Based Swap Dealer De Minimis Exception. Release No. 34-104573, 9 January.

Statutes and regulations cited: 12 U.S.C. 24(Seventh); 12 U.S.C. 90 and 92a(d); 12 U.S.C. 1430(e); 12 U.S.C. 1821(d)(11) and 1821(e)(8), (9) and (10), including (e)(8)(G)(ii), (e)(10)(B)(i) and (e)(10)(D); 12 U.S.C. 1823(c)(4)(G); 12 U.S.C. 1831f(i); 12 U.S.C. 4402 to 4404; Federal Deposit Insurance Act Section 18(a)(4); 12 CFR 5.36; 12 CFR 7.1016; 12 CFR 249.3 and 249.22, including 249.22(b)(1); 12 CFR 249.32(e) and (h), including (h)(1) and (h)(4); 12 CFR 327.3(b), 327.5(a) and 327.16(e)(3); 12 CFR Part 327, Appendix A to Subpart A; 12 CFR Part 328 Subpart B; 12 CFR 337.6, including 337.6(e); 12 CFR 337.7; 12 CFR 360.8; 12 CFR 740.3; 12 CFR 1269.2; Federal Register notices 86 FR 6742 (brokered deposits final rule), 89 FR 68244 (brokered deposits proposal of 2024), 90 FR 12115 (its withdrawal), 82 FR 41951 (joint account guidelines), 90 FR 51356 (expanded Fedwire Funds and National Settlement Service operating days, announced) and 91 FR 56022 (reciprocal deposits interim final rule, effective 1 September 2026); New York Insurance Law Article 69; Ohio Revised Code Chapter 1761; Colorado Revised Statutes 11-105-404(1)(c)(II); Texas Finance Code 34.304, as amended by Acts 2007, 80th Leg., ch. 110; Illinois Banking Act, 205 ILCS 5/5(7)(h), as amended by Public Act 104-0310 (HB 3467), effective 15 August 2025; California Government Code 53651 and 53652 (as reproduced at california.public.law); Texas and Pacific Railway Co. v. Pottorff, 291 U.S. 245 (1934).